GEK-83923

# UNIVERSAL STATIC VOLTAGE REGULATOR FOR AC GENERATORS USING ROTATING EXCITERS

DS3820SVR

SAFETY SYMBOL LEGEND

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* *
* WARNING *
* *
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THIS EQUIPMENT CONTAINS A POTENTIAL HAZARD OF ELECTRICAL SHOCK OR BURN. ONLY PERSONNEL WHO ARE ADEQUATELY TRAINED AND THOROUGHLY FAMILIAR WITH THE EQUIPMENT AND THESE INSTRUCTIONS SHOULD INSTALL, OPERATE, OR MAINTAIN THIS EQUIPMENT.

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* *
* CAUTION *
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THIS EQUIPMENT IS DESIGNED TO WORK IN A NORMAL INDUSTRIAL ATMOSPHERE. OPERATING RADIO TRANSMITTERS NEARBY MAY CAUSE THE CONTROL EQUIPMENT TO MALFUNCTION. IT IS RECOMMENDED THAT RADIO TRANSMITTERS NOT BE OPERATED WITHIN THE VICINITY OF THE CONTROL EQUIPMENT WHEN THE PANEL DOORS ARE OPEN, NOR AT ANY TIME WITHIN THE CONFINES OF A POWER CONTROL ROOM.

NOTE:

Commands attention to an essential operating or maintenance procedure, condition, or statement which must be highlighted.

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## 1. Introduction:

GE has introduced a universal Static Voltage Regulator, SVR, for excitation control of industrial and utility generators. This regulator can be applied to most generators using a rotating exciter, dc commutator or brushless type. The regulator can be easily applied to retrofit old generators or for new generator applications.

The performance of generators with dc rotating exciters and mechanical or Amplidyne* regulators can be improved by retrofiting with a static regulator. Some new generators with rotating exciters are brushless. In these applications the static voltage regulator will provide optimum performance.

The SVR power source can be single phase, 98 - 140 Vac, three phase, 98 - 124 Vac, or 100 - 143 Vdc. The maximum output is 30 amps for 20 seconds at 155 vdc. The rated output is 20 amps continuous at 100 Vdc.

The versatility of this regulator is the result of its basic controlling power amplifier - a switching transistor. The power source can be an auxiliary bus, the generator terminals, a Permanent Magnet Generator (PMG) of most frequencies, or a station battery.

## 2. ABBREVIATIONS

| | |
|---|---|
| AC Regulator | Automatic Regulator, regulates Generator AC terminal Voltage. |
| BERG jumper | Used to make connections on the regulator boards. |
| DCOM | Common voltage reference for DC power supplies. |
| DC Regulator | Manual Regulator, regulates Exciter Field Voltage. |
| Droop | Used to parallel generators. |
| DVRC | URAL board on the regulator. |
| EFARL | Exciter Field Amps Rated Load. |
| EFVRL | Exciter Field Volts Rated Load. |
| KI | Kiloamps. |
| KV | Kilovolts |
| KVA | Kilovoltamps. |
| LDC | Line Drop Compensation. |
| NEPF | Board that contains power components on the regulator. |

| | |
|---|---|
| NVRC | Board that has AC and DC regulator |
| PMG | Permanent Magnet Generator. |
| PPT | Power Potential Transformer. |
| PT | Potential Transformer. |
| pu | per unit. |
| SVR | Static Voltage Regulator. |
| SVRA | Regulator without the URAL option. |
| SVRB | Regulator with the URAL option. |
| TVM | Transfer voltmeter. |
| URAL | Underexcited Reactive Ampere Limit. |
| VAR | Voltamps Reactive. |
| V/HZ | Volts per Hertz. |

## 2.1 Function Diagram

A functional diagram of this regulator is illustrated in Figure 1. The pulse width modulated switching of the power transistor provides control of the exciter field voltage by varying the "on" period. Thus, the average dc voltage applied to the exciter field is the shaded area in Figure 1.

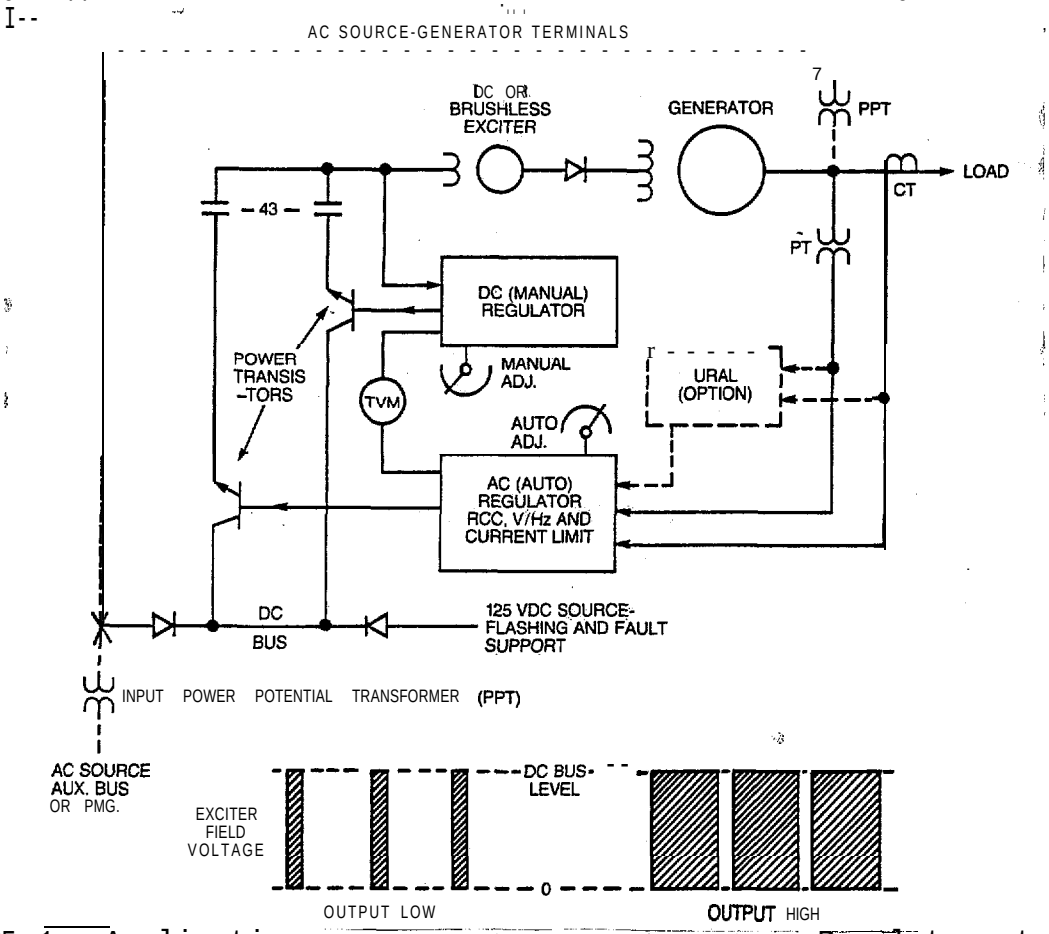



FIGURE 1. Application of Pulse Width Static Voltage Regulator--to a DC or Brushless Exciter.

## 2.2 AC/DC Regulator

The SVR has both an Automatic (AC) and Manual (DC) regulator. The ac regulator regulates the generator ac voltage, the DC regulator regulates the generator dc field voltage. Both regulators are pulse width modulators that control the SVR output voltage. The DC regulator controls the SVR output voltage to regulate the field current of the synchronous machine. The ac regulator controls the SVR output voltage to regulate the ac voltage at the terminals of the synchronous machine. These regulators are independent so that if one fails, the other regulator will operate. Selection of ac or dc regulator control is by the operator, or protective relaying.

The ac regulator provides operator adjustment range from 85% to 110% of rated generator voltage with a 120 volt PT circuit.

The dc regulator provides an operator adjustment range from 10 volts dc to 110 volts dc.

## 2.3 Flashing

Flashing is usually required because the residual voltage of the generator is not high enough to power the regulator and provide voltage build-up. If the regulator is powered from an auxiliary bus or PMG, then flashing is not required. Flashing current is typically 5 to 10 amperes for 1 to 5 seconds.

## 2.4 Fault Current Support

When generator fault current support is required excitation will be sustained automatically from a dc source (such as a station battery). See Figure 1. The rectifier between the dc bus and dc source is normally biased off by the higher ac source voltage. Should this ac source voltage dip, as during a fault, the dc source will take over during the interruption. Such interruptions are seldom and usually last for only 5 to 10 cycles which is a small load on a battery. An auxiliary bus ac source should not be relied on for fault support because these sources usually follow generator terminal voltage dips. Fault current support is not required if the ac power source is a PMG.

## 2.5 Protective Functions

The SVR has a volts per hertz limit. When the volts per hertz of the synchronous machine drops past a pre-set value, the volts per hertz function will cause the AC regulator to reduce field current of the synchronous machine so that a pre-set volts per hertz limit is not exceeded. This function does not work under DC regulator control.

The SVR will regulate generator terminal voltage under ac regulator control up to a pre-set field current limit value. It will not increase the field current past this limit under steady state conditions. In dc regulator control the operator cannot increase field current beyond this field current limit value. The field current limit is instantaneous.

Underexcited Reactive Ampere Limit (URAL) is an optional function. The URAL limits the Leading Vars that the synchronous machine can produce. The URAL has a circle limit function.

Droop or Line Drop Compensation (LDC) is standard. Droop tends to stabilize the VAR load when the machine is paralleled with other machines. LDC is used to regulate voltage at some point in the power system. The regulator can be set for either Droop, or LDC, but not both.

## 2.6 Other Functions and Features

1. This regulator can be used with dc exciters or with brushless exciters, with or without a PMG.

2. The ac (or automatic) regulator controls generator terminal voltage. The dc (or manual) regulator controls the exciter field voltage. Completely redundant circuits and power transistors are used for the ac and dc regulators.

3. Bumpless transfer is obtained by using a transfer voltmeter, TVM, for switching between the ac and dc regulators. When the transfer voltmeter reads "0", the voltage out of the ac regulator is the same as the voltage out of the dc regulator, and switching from dc to ac regulator will not affect excitation or cause a generator terminal voltage change. Likewise when the voltage out of the dc regulator is the same as the voltage out of the ac regulator, the TVM will read "0", and switching from ac to dc regulator will not affect excitation or cause a generator terminal voltage change.

4. The ac power source for the regulator can be a PMG, an auxiliary bus, or the generator terminals at any frequency from 50 to 360 hertz. Although a three phase power source is preferred, a single phase power source can be used.

5. The regulator can provide "black start" capability by taking ac power from the generator terminals through a power potential transformer (PPT), and flashing from the station battery. A PMG source also provides "black start" capability.

6. Fault Current Support can be provided by applying a dc power source to sustain excitation during interruptions in the ac power source. The dc power source must be 125V; 48V or 250V power sources cannot be used.

7. The regulator is self powered from the ac power source. DC power is required only for flashing and fault current support, or associated relaying for customized applications.

8. The regulator has an adjustable exciter field current limit which protects the exciter field windings and the power transistors.

9. The regulator has an adjustable volts/hertz limit. This protects the generator and line transformers from low frequency operation.

10. The regulator has adjustable gain and stabilizing settings.

11. The regulator also has a field current stabilizing feedback for fast regulator response.

12. The ac regulator will regulate generator terminal voltage to within ±1%, no load to full load.

13. The dc regulator will regulate the exciter field voltage to within ±5% of maximum steady state exciter field voltage.

14. Three phase, or single phase sensing can be used.

15. The regulator current ratings are 20 amps continuous, 30 amps for 30 seconds, or 36 amps for 20 seconds. The ceiling voltage to the exciter field is typically 155 volts dc, with a 115 volt, three phase power source. Regulator output voltage at generator rated load should be between 50 and 100 volts dc. If this voltage is less than 50 volts a series resistor can be added. If this voltage is more than 100 volts, the exciter field windings should be reconnected for lower voltage and higher current.

16. Line filters are provided for both ac and dc power source inputs.

17. A rectifier bridge is provided to convert ac power to dc power for the switching transistors.

18. The regulator is designed to operate, convection cooled, up to a maximum ambient temperature of 50 degrees Celcius.

19. The basic regulator described above can be furnished as a wall mounted module with required PPT, adjusting potentiometers, switches, and meters for remote mounting, or as a complete system with optional accessories and features, mounted in a free standing enclosure with customized system diagrams. See Application section of this instruction book.

## 3. RECEIVING AND STORAGE

As soon as the equipment is received, it should be examined for any damage that might have been sustained in transit and all shipping material removed. If injury or rough handling is evident, a damage claim should be filed immediately with the carrier and the nearest GE sales office should be notified promptly.

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected from damage. Particular care should be exercised to avoid storing the equipment in locations where construction work is in progress. Do not store the regulator in a damp place!

## 4. EQUIPMENT DESCRIPTION

### 4.1 WEIGHT AND SIZE

Weight: 29 pounds.
Size: 11.76"W X 16.10"H X 7.73"D.

### 4.2 REGULATOR OPTIONS

1. Underexcited Reactive Ampere Limit (URAL) to prevent the ac regulator, and/or operator, from reducing generator excitation to a level which would result in loss of synchronism.

2. Input Power Transformer (PPT) for an ac source.

3. Input Power Transformer primary fuses.

4. Input molded case ac breaker with overcurrent trip for protection and maintenance.

5. Input molded case dc breaker with overcurrent trip for protection and maintenance.

6. Motor operated voltage adjustors for remote control of the ac regulator or dc regulator voltage set points.

7. Exciter field over-excitation protective relay which transfers from the ac regulator to the dc regulator after 20 seconds of over-excitation, and trips the generator if over-excitation persists for 5 more seconds.

8. DC field circuit breaker with a discharge resistor for rapid de-excitation.

9. VAR-Power Factor controller. Provides "operator" type control of generator VAR's or Power Factor. This requires a motor operated voltage adjustor for the ac regulator.

10. 125 volt dc battery pack for flashing, fault forcing and relaying, where 125 volt station battery is not available.

11. PT and CT disconnect switches for maintenance and testing.

12. Exciter field ammeter.

13. Exciter field voltmeter.

14. Indicating lights and special relaying.

15. Free standing NEMA 1 enclosure.

16. Enclosure lights and convenience outlets. (Floor mounted enclosure only)

17. Operators control panel insert.

## 4.3 TWO AC REGULATOR OPTION

The "two ac regulator option" consists of two SVR regulators configured so that each ac regulator has the same sensing inputs. Note that each SVR has both an ac and dc regulator, but the dc regulator is not used.

Each ac regulator has a voltage reference setpoint potentiometer. These potentiometers are tandem connected to the shaft of a motor operated voltage adjuster. When the shaft turns, each potentiometer is adjusted the same amount. Thus the voltage reference to each ac regulator is identical. Since each regulator is sensing the same generator voltage and current, and the voltage references are identical, all the inputs to each regulator are the same. Certain adjustments are required to insure that the output of each regulator will be the same under all load conditions. These adjustments are outlined below. Once the adjustments are made, transfer from one regulator to the other regulator simply requires that an operator use the 43CS switch. There is no need to "null" the reference setpoints.

A. DS3800NVAC sh. 4BA, measure voltage at R40 or pin 10, location D37. Use R5 to adjust the voltage.

B. DS3800NVRC sh. 4BA, measure voltage at TP4, location G33. Use R3 to adjust the voltage.

C. DS3800NEPF sh. 4AA, measure output voltage (TB2-3 to TB2-4), and adjust R2, DS3800NVRC sh. 4AA location G13.

3. Adjust Current Limit.

To adjust the current limit, attach to the output of the regulator, TB2-3 to TB2-4, a resistor with the approximate resistance of the field and correct wattage. Adjust R2, sheet 4AA location E25, for desired field current limit. Reconnect exciter field when current limit has been adjusted for both primary and backup ac regulators.

4. Adjust V/hz, Droop or LDC.

A. V/hz. Adjust primary regulator for the desired compensation. Adjust the output of the backup regulator to be the same as the output of the primary regulator using R8, sheet 4CA NVRC. Switch from primary regulator to backup regulator and fine tune, if necessary, using R8. Note that the machine should be off-line and the speed of the prime mover at the speed V/hz takes over.

B. Droop or LDC. Adjust primary regulator for the desired compensation. Measure the output of the primary and backup regulator and adjust one output of the backup regulator, using R7, sheet 4CA NVRC, to be equal to the output of the primary regulator. Measure the output of each regulator from TB2-3 to TB2-4. Switch from primary to backup regulator, if necessary fine tune, by adjusting R7 on the backup regulator, so that the machine is providing the same VARS. The machine must be on-line for this adjustment.

5. URAL

Follow the URAL procedure outlined in this manual to set up URAL on the primary regulator, repeat the procedure to URAL on the backup regulator. To be sure each URAL is set the same, each URAL function must be tested at load conditions. Switching from primary to backup regulator should not affect the operation VARS of the machine when operating within the URAL limit.

The URAL cards should not be switched from primary regulator to backup regulator. Tag the cards so that they can be replaced on the regulator from which they are removed.

The elementary sheets of 237B4355 pertaining to the two ac regulator option are found on sheets labeled "TWO AC REGULATOR OPTION".

The inputs to the summing junction of the ac regulator shown on sheet 4AA, DS3800NVRC, are as follows:

1. Voltage reference.
2. Machine voltage - recitified average.
3. Volts/Hz.
4. Droop or Line Droop Compensation (LDC).
5. Underexciter Rective Ampere Limit (URAL).
6. Feedback from field current.
7. Current limit.

Parameters which affect the output of the regulator:

1. Gain of the ac regulator.
2. Input voltage to the bridge.

All of the above items must be the same for the output of each ac regulator to be identical at all machine load conditions. If a protective function is added, the function must affect each regulator equally. If it does not, transfer from primary to backup regulator may result in a change in machine field current, and thus a change in VARS.

The following procedure may be used to set up the regulators for equivalent outputs.

1. Remove the URAL card. Adjust V/hz, droop or LDC, and current limit so that these functions have no effect on regulator output. Adjust the lead, lag and gain for stable operation on the primary regulator.

2. Set the lead, lag and gain on the backup regulator equal to the settings on the primary regulator. Make sure BJ1 and BJ2 are in the same position. Measure the input voltage to each bridge. The voltages should be the same. Measure the voltages on the primary regulator, at the points described below, and adjust the backup regulator to have the same voltage at these points. The voltages measured when the primary regulator is in control should be recorded. Switch control to the backup regulator and adjust the voltages.

## 5. PRINCIPLES OF OPERATION

Refer to Section 9 for a more detailed description of the elementary.

### 5.1 Description

The SVR acts as a pilot exciter with voltage regulation functions. One basic module design will serve for all identified ac and dc input voltages. The power circuit for the regulator is a switching transistor dc to dc converter. This configuration allows the SVR to flash the field of the generator during startup, and to sustain the field excitation during a fault on the ac input by drawing power from a station battery in parallel with the rectified ac input. Refer to figure 2 for a block diagram of the SVR.

The SVR has an ac and dc regulator. The ac regulator, which regulates the generator ac terminal voltage, has its own power transistor to control the regulator output current. The dc regulator, which regulates the field voltage of the synchronous machine, also has its own power transistor to control the regulator output current. Selection of either ac or dc regulator is made by relay contacts which connect either the ac or dc power transistor to the regulator power supply. A transfer voltmeter provides bumpless transfer from one regulator to another. The transfer voltmeter output is zero volts when the ac and dc base drives have the same average "on" time. The operator can switch from dc to ac regulator , or ac to dc regulator, without the generator terminal voltage changing. Rate feedback stabilization from exciter field current is used to stabilize the ac regulator. Both power transistors have independent base drives and control circuits, so a failed transistor on one regulator will not make the module inoperable. In the event of a failed transistor, the "43" relay is used to isolate the failed device, and the regulator will operate using the functioning power transistor.

### 5.2 Sensing

The SVR senses generator voltage from either a single phase or three phase voltage transformer. The current sensing is in phase B. For either single or three phase sensing the phase relationships necessary for Droop or LDC compensation, as well as URAL regulation are maintained. A BERG jumper, BJ2, on the regulator board must be in either the three phase or single phase position for the appropriate sensing. For single phase sensing TB3-12 and TB3-13 must be shorted together.

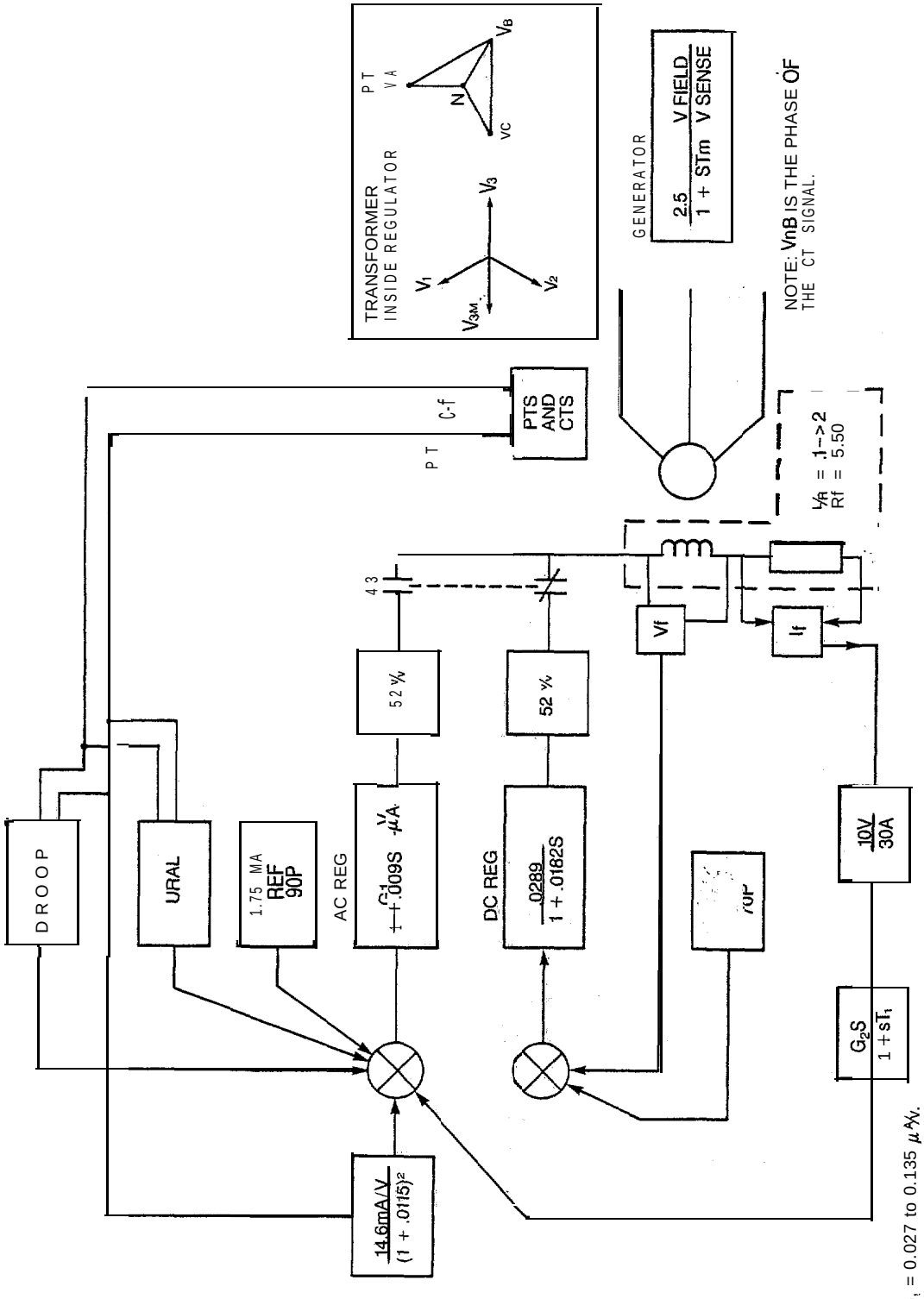



FIGURE 2. Block Diagram of the SVR.

## 5.3 REGULATOR POWER SOURCE

Input power is either from a PMG, the generator, station ac power, or a battery. The ac source must be 98 to 124 Volts, three phase, or 98 to 140 Volts single phase. The dc source must be 125 Volts nominal, with a 100 to 143 Volt range.

A power transistor in the switching mode is used to control the voltage applied to exciter field. The transistor is switched on and off at a constant frequency, approximately 250 Hz. The unregulated dc supply voltage is converted to a regulated dc voltage by controlling the ratio of the transistor on time to off time (see figure 3). This is done by use of a pulse width modulating chip on the NVRC card which responds to an error signal by modulating the pulse width applied to the base drive transistors and to the power transistors. The transfer function of the dc/dc converter is:

Vout = Vin * (ton/(ton + toff)).

Vout in this case is the average exciter field voltage and Vin is the unregulated dc bus voltage, normally 155 volts dc with a 120 volt ac line to line three phase input. The free wheeling diode (CR5) provides a path for the exciter field current to flow during the power transistor off time. The pulse width modulating chip also provides a current limit function to protect the power transistors and exciter field windings from overcurrent. Figure 3 illustrates how the field voltage will look for two different levels of excitation.

Because separate pulse width modulating chips are used for the ac and dc regulator, their switching frequencies will not be the same.

## 5.4 Power Supply

The regulator requires no external power supply to operate control circuits on NVRC and DVRC. The internal power supplies are t15 and -15 volts with respect to COM which operate over an input range of 50 to 175 volts dc. The input dc to these supplies is the unregulated dc bus voltage of the power circuit, which can vary from 50 to 230 volts dc. A clipper circuit on NEPF limits the voltage to the supply inputs to a maximum of 180 volts dc. Internally the power supply chops the input dc voltage, thus changing it to ac voltage.

Isolation transformers are used to convert this ac voltage to a lower voltage. The transformer secondary voltage is then rectified and regulated, and is adequate for power requirements of the regulator.

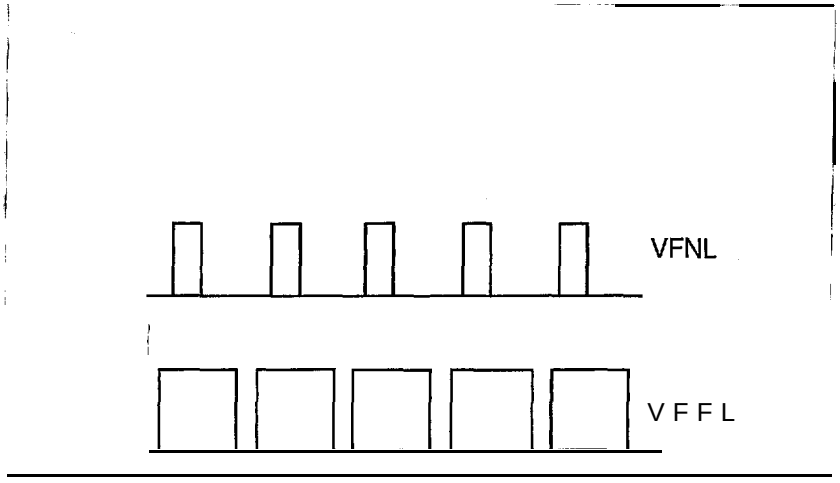



FIGURE 3. VFNL and VFFL.

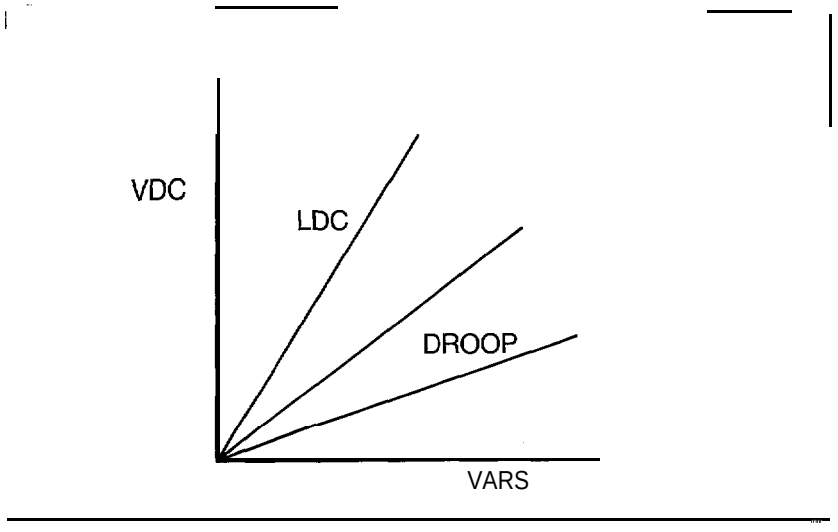



FIGURE 4. LDC and Droop Modification to Field Voltage With Increase in VARS out of Generator.

## 5.5 Current Limit

This function allows a maximum limit to be set on the field current to the rotating exciter. There are two ranges allowing for a high limit in the range of 8.3 to 36 amps, and a low limit from about 2 amps to 8.3 amps. Operation is not recommended with the current limit set above 20 amps.

The current limit function is achieved by applying a voltage which is adjusted by R2 to the current limit input of both the ac and dc pulse width modulating chip. The current limit should be set when operating in the ac regulator, and should then be checked when operating in the dc regulator. It should be within ±10% of the value set in the ac regulator.

## 5.6 Volts/Hz Limit

The Volts/Hz Limit takes control of the ac regulator to insure that the synchronous machine will operate at constant volts per hertz below a pre-set volts per hertz limit. The Volts/Hz Limit does not cause an alarm or trip the unit. The range of Volts/Hz Limit adjustment takeover is from 43.5 to 57 Hz, which allows the system to be used with either a 50 or 60 Hz generator.

The Volts/Hz Limit uses a frequency to voltage converter. The voltage output becomes negative when the frequency drops below the "takeover" frequency, and causes the ac regulator to reduce field voltage. It does not operate in dc regulator control.

## 5.7 DROOP/LDC

The Droop changes the amount which the generator voltage will be reduced when the generator load is inductive. The present circuit will droop the generator voltage 20% when 4 amps lagging phase B-N current is flowing into the CT burden resistors on the NEPF board. This adjustment is over a range of 0 to 20%.

The LDC increases the generator voltage for inductive load. BERG jumpers BJ3 and BJ4 select the regulator mode, either Droop or LDC. See figure 4.

## 5.8 URAL

The Underexcited Reactive Ampere Limit (URAL) option is available in the SVRB board. The URAL is a takeover regulator. When the generator operation penetrates the limit line the URAL will take over control from the ac regulator on NVRC and drive excitation up in order to operate on the limit line. The URAL limits operation in the underexcited (leading power factor) region of the generator capability curve.

The URAL Center (C) and Radius (R) are used to establish the limit line. When the generator armature current increases, the Center (C) and the armature current (I) are added vectorally. When the vector sum of (C) + (I) is greater than or equal to (R), the URAL regulator will control the field voltage. The URAL center and radius are ac voltages. The radius is a percentage of phase 2 voltage, and the center is a percentage of phase 2 voltage, but smaller than the radius. From the phasor diagram of the sensing voltages, figure 2, it is seen that the voltage V2 is 90 degrees lagging the voltage from the CT, phase 2-N. The phasor addition can be understood if we consider the generator operating as a synchronous condenser underexcited (the machine current leads the machine voltage by 90 degrees). In this case the phasor addition of the center voltage plus the voltage from the CT would be in phase. That is if VTP1 = 2 volts rms, and the voltage from the CT = 1 volt rms, then their addition would result in 3 volts rms. If the machine was operating as a synchronous reactor over-excitited the phasor addition would have resulted in 1 volt rms. When the machine puts out power at unity power factor the addition must take into account the 90 degree phase shift resulting from the real component of current out of the machine. The result of this addition of voltage and current is rectified by the URAL board and available for measurement at TP3. The radius voltage is rectified and compared with VTP3. If VTP3 is greater than the rectified radius voltage, VTP7, the URAL limit is reached, and the URAL should control the ac regulator. At this time VTP6 should be -15 volts, and VTP5 is the voltage that modifies the ac regulator.

## 6. APPLICATION

If the SVR is purchased with accessory functions assembled in an enclosure, there will be custom diagrams to indicate interconnections between the regulator, generator, and operator controls.

If the SVR is purchased "off the shelf" and user plans to design their own system, the following outline is important to assure that the SVR is applied correctly. Refer to typical elementary diagram 237B4355 in the back of this book. This diagram illustrates many ways the SVR can be applied.

### 6.1 Power Input - see sheet 03A.

a. The primary power source is ac into AC1, AC2, AC3, on TB1. Three phase, 115 Volt is recommended. The frequency can be from 50 to 360 hertz. An isolation transformer (PPT) is recommended:

KVA(PPT) = 115 x 1.5 x EFARL.

EFARL = exciter field amps required at generator rated load.

The PPT primary can connect to an auxiliary bus, a PMG, or the generator terminals - see sheet 02A. Using a PMG or generator terminals will provide "black start" capability.

A single phase source can be used - 140 Volts maximum steady state.

b. The secondary power source is 125 Vdc from a station battery - required for flashing the machine at start-up if the ac source is the generator terminals and for fault current support if the ac source is not a PMG. The maximum current required from the 125 Vdc source is about 2 x EFARL for 5 seconds during faults.

6.2 Exciter field output is F1 and F2 at TB2-3 and TB2-4. F1 is positive during operation. Polarity of the exciter field winding is not important except to provide correct meter readings of exciter field voltage and current.

The exciter field characteristic must match the regulator. This means the exciter field voltage must be between 50 and 100 Vdc when the generator is at rated load and temperature. The following describes methods to determine this voltage - EFVRL (Exciter Field Voltage Rated Load).

The exciter field voltage requirement is the exciter field current requirement times the field winding resistance. The field winding resistance will typically vary 20% to 30% from room temperature (25°C) to rated temperature (75°C to 100°C). Example:

$$R(75°C) = R(25°C)(1 + .00393 \times 50°C \text{ rise}).$$

$$= R(25°C)(1.1965)$$

The exciter field voltage requirements will be:

$$EFVRL = EFARL \times R(75°C \text{ or } 100°C)$$

(This is the maximum steady state voltage required from the regulator.)

For a dc commutator exciter the above information can be obtained from generator data sheets (or saturation curve) and determining the exciter output voltage required for a particular load condition with the generator at rated temperature. Next refer to the exciter data (or saturation curve) to find the exciter field current required and multiply this value by the exciter field resistance to find the required exciter field voltage.

For a brushless exciter the method may be the same as above, using first the generator saturation curve, then the exciter saturation curve. For some brushless exciters, the exciter field requirements (or saturation curve) may be in terms of generator loading and can be read directly.

If no data or saturation curves exist, then the simplest method is to measure exciter field current and voltage required for the various loading conditions, with the generator at rated temperature.

If EFVRL is less than 50 volts, then adding a resistance in series with the exciter field winding circuit is required. If EFVRL is more than 100 volts, the exciter field winding circuit must be reconnected. See figure 5.

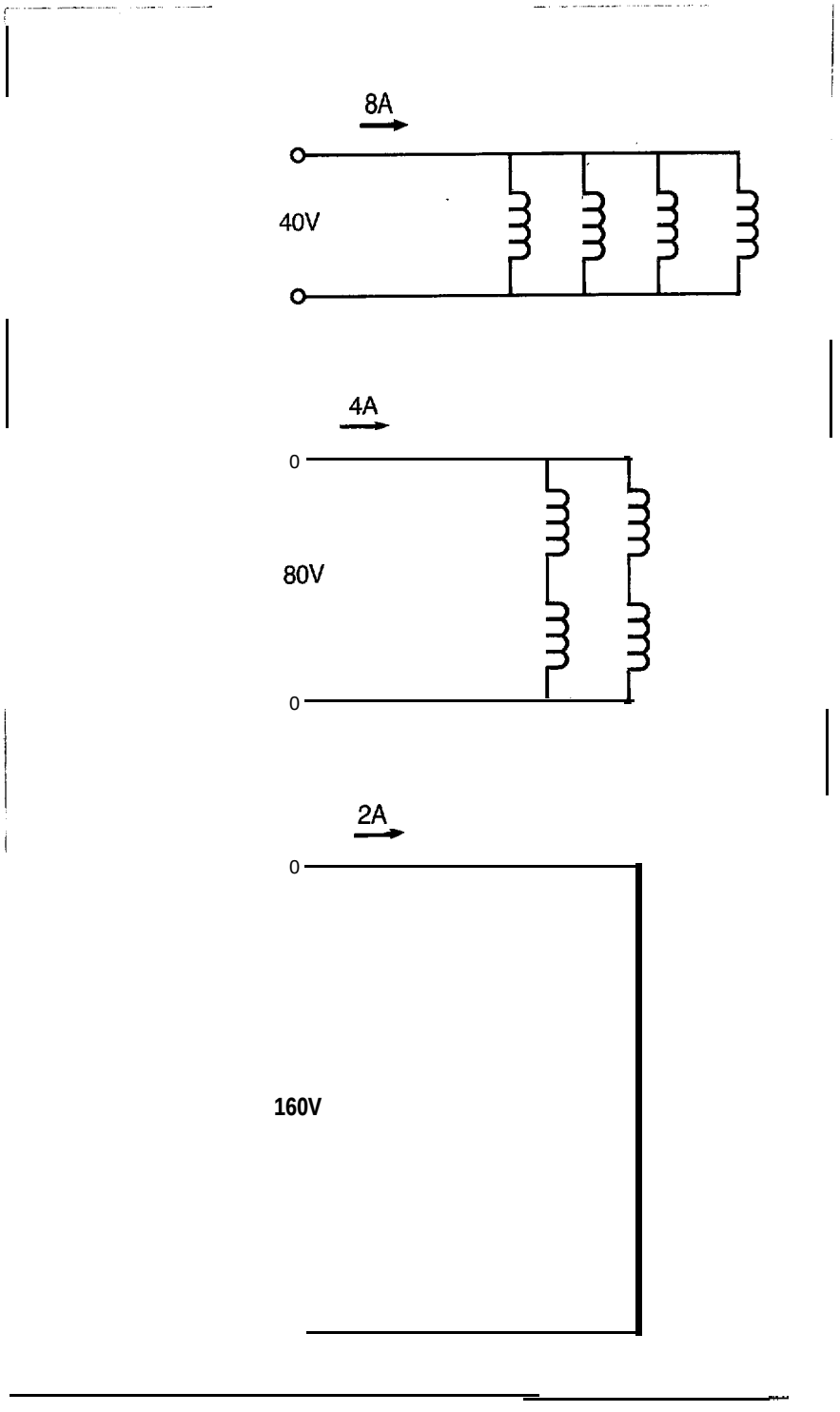



FIGURE 5.

Exciter field circuits are typically 4 pole windings. If all 4 poles are connected in series, the exciter field current will be low, but the voltage high. By reconnecting these poles in parallel, the field voltage will be lower and the field current will be higher.

6.3 Regulator Mode Transfer - sheet 03A. A switch or contactor should be connected to 43C3, 43C2, 43C1 on TB2 to transfer regulator mode between ac regulator and dc regulator. An ON-OFF contact may be included - described in section 6.4 below. Contacts do not need to overlap and should be rated to carry maximum exciter field current and to interrupt 3 x EFARL at 250 Vdc. An operator control switch may directly operate this function, or the function could be interfaced through a contactor with protective relaying as required.

6.4 De-excitation. Most users want to start and stop excitation when required. This can be done by closing and opening both ac and dc power switches - DC1, DC2, AC1, AC2, and AC3; Closing and opening power transistor circuit - 43C1, 43C2 and 43C3; or closing and opening the exciter field circuit, F1 and F2.

Molded case breakers can be used in the ac and dc source circuits. Shunt trip coils can be included if remote tripping is required.

A contact of the 43 switch or contactor could be used to close and open the 43C circuit.

A dc field circuit breaker and discharge resistor can be used to close and open the exciter field circuit.

6.5 Regulator Sensing Circuits - sheet 02A. PT and CT sensing from the generator output is required to connect to PT!, PT2, PT3, CT1, and CT2 as shown on sheet 02A. Polarity and phasing must be as shown.

For single phase sensing, connect as shown on sheet 02AA. See Installation for correct jumpers for single phase sensing.

CT and PT disconnect switches may be used if required.

6.6 Voltage Adjust Pots - sheet 05A. Connect the ac regulator adjuster - 90P and the dc regulator adjuster - 70P per sheet 05A. These pots must be connected using shielded leads as shown - unless they are motor operated pots in the same enclosure.

If motor operated pots are used, then raise - lower switches must be connected as typically shown on sheets 09A and 09B.

6.7 Transfer Voltmeter - sheet 05A. This meter is required to get bumpless transfer with the 43 function between the ac and dc regulators. This is a 10 volt zero center meter and is usually furnished with the SVR connected as shown using shielded leads.

6.8 The Underexcited Reactive Ampere Limit (URAL) is a board (DVRC) that requires no external connections. This board plugs directly into the regulator board (NVRC).

6.9 Refer to 237B4355 for interconnections for other accessories.

## 7. INSTALLATION AND START UP

The regulator should be mounted so that it is accessible. It should also be mounted in such a way as to protect personnel from exposed voltages. If mounted in an enclosure, the ambient temperature should not exceed 50 degrees Celsius.

The regulator must always be mounted vertically so that it will vent its heat properly. There must be at least a 4 inch space above and below the regulator. Figure 6 shows the voltage regulator with the clearances labeled.

### Regulator Connections, Jumpers and Pots

Before the generator is started - connections, jumpers and pots should be checked as follows.

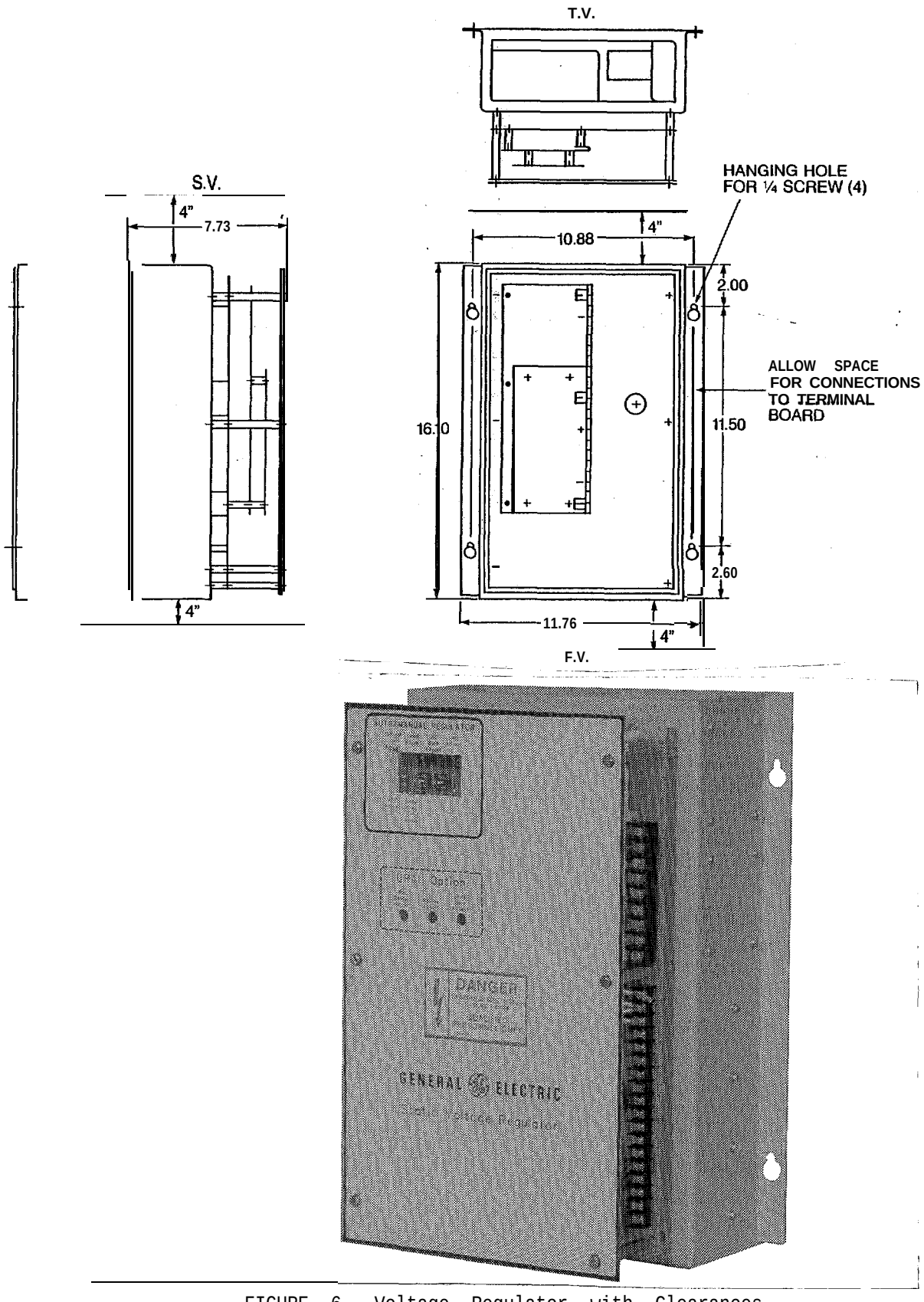



FIGURE 6. Voltage Regulator with Clearances.

A. Connections

While the generator is stationary the connections to the regulator should be checked to insure that the sensing connections, the power connections and the jumpers are correct. The table on page 22 summarizes the regulator connections at the terminal board.

Sensing Voltage:

Phase sequence must be A-B-C.

| Three phase | Single phase | Terminal Board |
|---|---|---|
| ØA | ØA | TB3-14 |
| ØB | PC | TB3-13 |
| ØC | short TB3-13 to TB3-12 | TB3-12 |

Sensing Current:

CT polarity must be per elementary diagram.

| | | |
|---|---|---|
| ØB+ | ØB+ (CT2) | TB3-16 |
| ØB- | ØB- (CT1) | TB3-15 |

AC Power:

| | | |
|---|---|---|
| ØA | +AC volts (AC1) | TB1-1 |
| ØB | -AC volts (AC2) | TB1-2 |
| ØC | leave open | TB1-3 |

Station Battery:

125 Vdc only.

| | | |
|---|---|---|
| | + DC (DC1) | TB2-2 |
| | - DC (DC2) | TB2-1 |

Field:

| | |
|---|---|
| + Field Voltage (F1) | TB2-3 |
| - Field Voltage.(F2) | TB2-4 |

AC or DC regulator select (41CS):

| | | |
|---|---|---|
| Common | (43C1) | TB2-5 |
| DC regulator | (43C2) | TB2-7 |
| AC regulator | (43C3) | TB2-6 |

To select the ac regulator, 41CS contacts should connect TB2-5 to TB2-6. To select the DC regulator, 41CS contacts should connect TB2-5 to TB2-7.

Transfer Voltmeter:

| | |
|---|---|
| TVM1 | TB3-11 |
| TVM2 | TB3-10 |

Power Supply Voltages:

These are internally connected to the terminal board points as indicated:

| | |
|---|---|
| N15 | TB3-1 |
| COM | TB3-2 |
| P15 | TB3-3 |

B. BERG jumpers

There are 4 BERG jumpers on the NVRC board (basic regulator), and 2 BERG jumpers on the DVRC board (URAL). Be sure to check that the jumpers are in the correct position as shown for your application.

NVRC (basic regulator)

BJ1 - High or Low Field Current Rate Feedback. The 6.81k resistor should be shorted to increase feedback. Place the BERG jumper on "H" for exciter field currents greater than 8 amps dc.

BJ2 - Three phase or single phase sensing. R101 should be shorted for three phase sensing. Place BERG jumper on 1 for single phase sensing, and 3 for three phase sensing.

BJ3/BJ4 - Droop or LDC. Place BERG jumper on D for Droop, on L for LDC.

DVRC Board (URAL)

BJ1 - URAL gain. Three positions for low, medium or high.

BJ2 - URAL integration time constant.

C. Pots.

There are 6 Pots on the basic NVRC board.

R2. Current Limit Ad just.
R3. Lead Adjust.
R4. AC Regulator Gain Adjust.
R5. Lag Adjust.
R7. Droop/LDC Adjust.
R8. Volts/Hz Limit Adjust.

INPUT/OUTPUT CONNECTIONS

| Terminal Board Connection | Designation | Connected to |
|---|---|---|
| TB1-1 | AC1 | ac Input, Phase A |
| TB1-2 | AC2 | ac Input, Phase B |
| TB1-3 | AC3 | ac Input, Phase C |
| TB2-1 | DC2 | dc Input, (-) |
| TB2-2 | DC1 | dc Input, (+) |
| TB2-3 | F1 | Field Output (+) |
| TB2-4 | F2 | Field Output (-) |
| TB2-5 | 43C1 | 43 Relay Center Pole |
| TB2-6 | 43C3 | 43 Relay (ac) |
| TB2-7 | 43C2 | 43 Relay (dc) |
| TB2-8 | PS1 | Power Supply Input (H) |
| TB2-9 | PS2 | Power Supply Input (L) |
| TB3-1 | P15 | -15 volts from Power Supply |
| TB3-2 | DCOM | Common from Power Supply |
| TB3-3 | N15 | Auto Regulator Potentiometer |
| TB3-4 | 90P1 | t15 from Power Supply |
| TB3-5 | 90P2 | Auto Regulator Potentiometer |
| TB3-6 | | Unused |
| TB3-7 | 70P1 | Manual Regulator Pot (+) |
| TB3-8 | 70P2 | Manual Regulator Pot Wiper |
| TB3-9 | 70P3 | Manual Regulator Pot (-) |
| TB3-10 | TVM2 | Transfer Voltmeter Output |
| TB3-11 | TVM1 | Transfer Voltmeter Output |
| TB3-12 | PT3 | Generator Voltage Sense Ph. C. |
| TB3-13 | PT2 | Generator Voltage Sense Ph. B. |
| TB3-14 | PT1 | Generator Voltage Sense Ph. A. |
| TB3-15 | CT2 | Generator Current Sense (-) |
| TB3-16 | CT1 | Generator Current Sense (+) |
| TB3-17 | TVMSH | TVM Shield |
| TB3-18 | GND | Earth Ground |

There are 3 Pots on the DVRC board.

R1. Center Adjust.
R2. Radius Adjust.
R3. Gain Adjust.

On all of these pots, clockwise increases the function and counterclockwise decreases the function.

## 7.2 Test points

Test points are provided for the purpose of setting up the regulator and trouble shooting.

### NVRC Test Points

| Test point | function | voltage signal |
|---|---|---|
| TP1 | Droop or LDC | dc. |
| TP2 | Volts/Hz | dc. |
| TP3 | AC regulator error signal | dc. |
| TP4 | Lead adjust | dc. |
| TP5 | Lag adjust | dc. |
| TP6 | DC voltage sense input | dc. |
| TP7 | Phase 2 or B ac voltage | ac. |
| TP8 | AC sensing voltage | dc. |
| TP9 | AC current sensing | ac. |

### DVRC Test Points

| Test point | function | type of signal |
|---|---|---|
| TP1 | URAL CENTER | ac, phase B or phase 2. |
| TP2 | URAL RADIUS | ac, phase B or phase 2. |
| TP3 | CENTER + IMAGINARY CURRENT | dc, rectified V2 + Im(I2). |
| TP4 | URAL SIGNAL | dc. |
| TP5 | URAL OUT | dc. |
| TP6 | URAL ACTIVE | dc, -15Vdc for URAL control. |
| TP7 | URAL RADIUS | dc, rectified V2. |

## 7.3 Off Line Checks

Before starting the turbine, check exciter field current limit with a dummy load. Use the 125Vdc as the power source for the regulator if the ac power input is the generator ac voltage. A recommended limit is twice EFARL. This is a compromise in protecting the field of the exciter and generator.

Set the 43CS function for dc regulator.

Set the dc regulator set point, 70P, for minimum voltage.

Start turbine and apply excitation by closing the exciter field breaker, ac source breaker, or initiate flashing as required. If generator ac voltage goes above rated, trip immediately and trouble shoot.

Exercise 70P to verify smooth control of excitation.

**********
* CAUTION *
**********

Do Not exceed generator rated voltage at any time:

Using 70P, adjust generator voltage to rated.

Adjust the ac regulator set point, 90P, to zero the transfer voltmeter.

While watching generator ac voltage, switch to ac regulator using 43CS. If the generator voltage raises above rated, switch back to dc regulator immediately and trouble shoot. The ac voltage should not change, watch for instability such as oscillations of exciter field voltage or generator ac voltage.

To adjust the ac regulator for optimum response, in may be necessary to adjust the Lead pot R3 and the Lag pot R5. The Gain pot R4 has been factory set for 45 V/% and should not be changed unless absolutely necessary.

Connect an oscilloscope or strip chart recorder to TP8. The oscilloscope should be set at a sweep rate of 5 seconds. The recorder should be set at a minimum of 1 second per inch of paper.

Adjust 90P for rated generator ac voltage. Switch to dc regulator and lower the generator ac voltage to 90% of rated with 70P.

Without zeroing the TVM, switch back to ac regulator and watch the generator ac voltage on the oscilloscope or recorder. The goal is to adjust the Lead and Lag to obtain the "critically damped response" shown in figure 7.

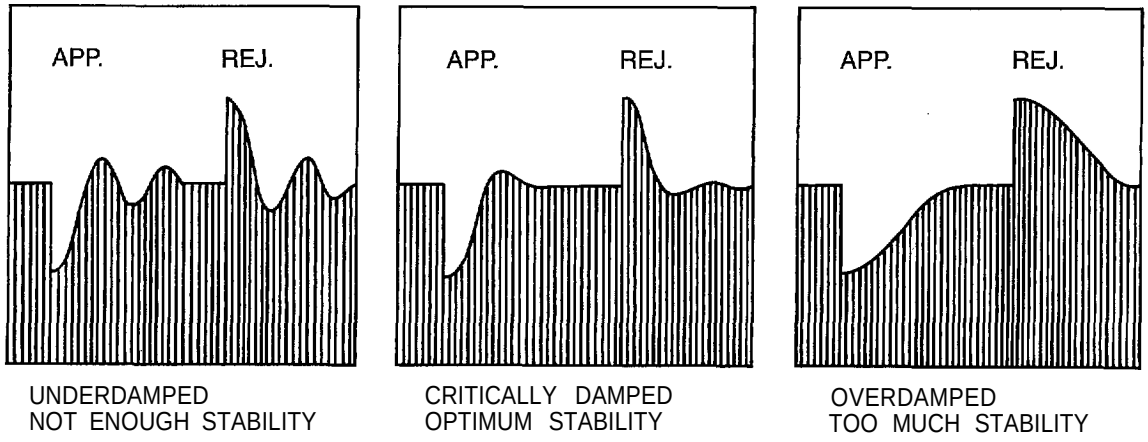



FIGURE 7.
Typical Oscillograms or Strip-chart Recordings looking at TP8. TP8 is Rectified and Filtered Generator Line Voltage.

If the response is "over damped" turn the Lag pot (R5) CCW, and the Lead pot (R3) CW. Adjust both pots 10% until response appears "critically damped".

If the response is "under damped", turn the Lag pot (R5) CW and the Lead pot (R3) CCW. Adjust both pots 10% until response appears "critically damped".

If adjustment of Lead and Lag pots will not result in "critically damped response", then adjust the Gain pot (R4). If the system is under damped, then the Gain pot should be adjusted CCW. If the system is over damped, then the Gain pot should be adjusted CW. (CW increases the gain)

The Volts/Hz Limit take over is factory set to take over at 57 hertz. To check take over, operate in ac regulator mode. Lower the generator speed to provide 57 hertz. The generator ac voltage should begin to drop at 57 hertz, if it does not, adjust the take over pot (R8), until generator voltage begins to drop. Return to rated speed.

## 7.4 ON LINE CHECKS

The following checks must be done with the generator on line.

Switch to dc regulator before going on line the first time. After this the generator may be synchronized in either dc or ac regulator control.

Synchronize the generator with the power system by user's established procedure.

Load the generator to at least 10% load.

Adjust 70P to raise excitation so that the generator is operating with lagging VARS and a power factor of 0.8 or less.

Zero the TVM using 90P, the ac regulator adjust pot.

A good method to check to see if connections are correct for DROOP is to monitor the voltage at TP1 on NVRC with a voltmeter. The generator is loaded and the dc regulator is in control. Increase excitation with the dc regulator. VTP1 should go more negative for DROOP as excitation is increased. VTP1 should go more positive for LDC as excitation is increased.

With BJ3/BJ4 in the "D" droop position - increase the droop pot R7 setting. Voltage at TP1 should go negative. If VTP1 goes positive, short the wires on the CT side of CT1 and CT2 and reverse the wires on the regulator side of CT1 and CT2. This now provides correct CT polarity for Droop operation.

Turn the Droop pot R7 to 50 on the dial. Zero the TVM with the ac voltage adjust pot 90P. While watching excitation or generator VAR's, switch to ac regulator. Nothing should change. Should excitation, VAR's, or power factor begin to move away from its initial value, switch back to dc regulator and recheck the CT polarity in the step above.

The Droop/LDC can be set to provide a drooping voltage characteristic as the generator is loaded with lagging power factor load, or a rising voltage characteristic as the generator is loaded with lagging power factor load (LDC).

If the generator is to be paralleled with other generators directly, without a step-up transformer or line reactor, the Droop mode must be used. This is the case for many small generators, industrial plants, or cross-compound turbine generators. Droop compensation gives the regulator the characteristics of trying to hold constant power factor or VAR's for wider excursions in system voltages. This simply means that the operator will not have to adjust the VAR's or power factor as often using 90P.

If the generator is to be paralleld with other generators through a step-up transformer or line reactor it is ofen better to connect this circuit for Line Drop Compensation (LDC), or reverse droop, to provide better voltage regulation on the system beyond the transformer or reactor. A typical step-up transformer or line reactor will have an impedance of 8% or .08 per unit. In this case the Droop/LDC should be set for LDC and about 4% reverse droop so that the net droop is only 4% to the system. This will provide adequate droop for stable parallel (on-line) operation and also reasonable voltage regulation to the power system.

The final setting is a compromise between stable on-line operation - the operator doesn't have to correct VAR's or power factor as often, and good system voltage regulation. Only operating experience will establish this compromise.

8. URAL SET-UP

In order to set up the URAL, a per unit CENTER and RADIUS must be calculated. 1 per unit voltage at TP3 must be calculated, or measured at generator nameplate data. This is done in the following manner:

Step 1 and Step 2 should be done before adjusting the URAL board.

STEP 1. (Note that pu = per unit.1

Let 1 pu KVA = generator rated KVA.
Let 1 pu KV = generator rated KV.
Let 1 pu KI = generator rated KI.

From the URAL limit line pick 2 points: (P1,Q1),(P2,Q2).

(P1,Q1) is one point on the generator limit line and (P2,Q2) is another point. Any two points may be chosen. P is the power out of the generator, and Q is the VARS out of the generator.

Calculate pu P1, pu Ql, pu P2, pu Q2.

P1 pu = P1/KVArated. Q1 pu = Q1/KVArated.
P2 pu = P2/KVArated. Q2 pu = Q2/KVArated.

STEP 2.

Calculate R and C in pu using the pu values of P and Q from step 1.

$$C = [(P2)(P2) - (P1)(P1) + (Q2)(Q2) - (Q1)(Q1)]/[2(Q1 - Q2)].$$

$$C = _____.$$

$$R = SQRT[(C)(C) + 2(C)(Q1) + (Q1)(Q1) + (P1)P1)].$$

Check your calculation of R by using the following.

R = SQRT[(C)(C) + 2(C)(Q2) + (Q2)(Q2) + (P2)P2)].

R = _____.

STEP 3.

When the generator line breaker is closed, and generator is at least 50% load, measure and record the KVA, KV, KI and VTP3. Make sure that R1 is turned to minimum and R2 is turned to maximum.

KVA(1) = _____. Also record the rated values.
KV(1) = _____.
KI(1) = _____.
VTP3(1) = _____.

KVA(rated) = _____.
KV(rated) = _____.
KI(rated) = _____.
VTP3(rated) calculate as shown below

VTP3(rated) = VTP3(1) * KI(1)/KI(rated) * KV(1)/KV(rated).

VTP3(rated) = _____.

STEP 4.

Short the CT input to the regulator and using pot R1 set VTP3 to the voltage for the Center, and VTP2 to the voltage for the radius. Set the Center before the Radius.

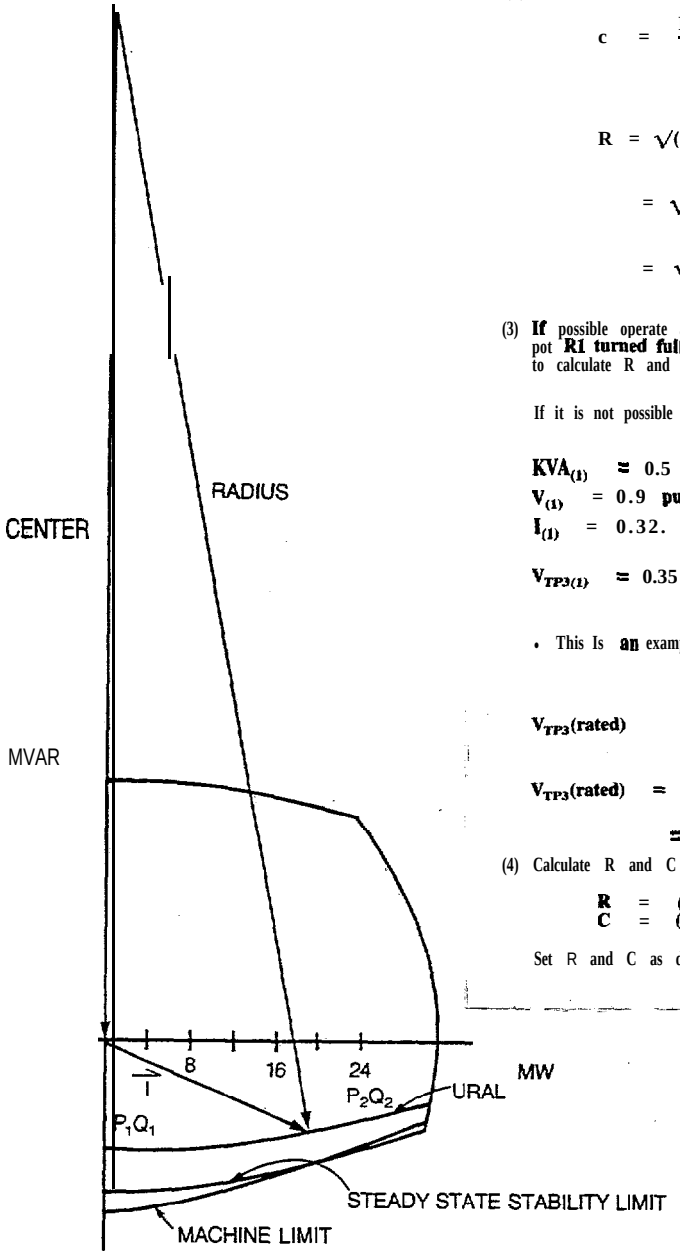
VTP3(center) = VTP3(rated) * C.
VTP3(center) = _____.

VTP7(radius) = VTP3(rated) * R.
VTP7(radius) = _____.

Note: VTP3 is negative, VTP7 is positive.

Remove the jumper across the CT input to the regulator. The URAL is now set. The URAL can be checked at other than rated conditions by setting R and C for different values and following the above procedure.

See the example in figure 8 for setting the URAL.




**MACHINE' 31 MVA = lpu**

**(1) From Limit Line**

$P_1$ = 0 MW = 0 pu
$Q_1$ = 10 MVARS = 0.3226 pu
Pa = 28 MW = 0.9032 pu
$Q_2$ = 6.8 MVARS = 0.2194 pu

**(2) Calculate R and C in pu**

$$c = \frac{Pa' \cdot P_1^2 + Q_2^2 \cdot Q_1^2 = 0.9032^2 \cdot 0^2 + 0.2194^2 \cdot 0.3226^2}{2(Q_1 \cdot Q_2) \qquad 2(0.3226 \cdot 0.2164)}$$

$$= 3.681 pu$$

$$R = \sqrt{(C + Q_1)^2 + P_1^2} = \sqrt{(C + Q_2)^2 + p_2^2}$$

$$= \sqrt{(3.681 \text{ t } 0.3226)^2 + 0^2} = 4.004 \text{ pu}$$

$$= \sqrt{(3.681 \text{ t } 0.2194)^2 \text{ t } .9032^2} = 4.064 \text{ pu}$$

(3) If possible operate at rated load and measure the voltage signal at TP3 on URAL card with pot R1 turned fully CCW. This will give 1 pu voltage at TP 3. Record this voltage and use it to calculate R and C voltages to be set on URAL according to Section 8.

If it is not possible to operate at rated load, operate at as great a load as possible and record.

$KVA_{(1)}$ = 0.5 pu'
$V_{(1)}$ = 0.9 pu*
$I_{(1)}$ = 0.32.

$V_{TP3(1)}$ = 0.35 volts

* This Is an example, $V_{TP3(1)}$ may not be 2 volts at these conditions.

$V_{TP3}$(rated)

$$V_{TP3}(\text{rated}) = 0.35 \times \left(\frac{1.0}{0.9}\right) \times \left(\frac{1.0}{0.5}\right) \times \left(\frac{1.0}{0.32}\right)$$

$$= 0.35 \text{ x } 6.94 = 2.43 \text{ Volts}$$

(4) Calculate R and C in Volts

R = (2.43)(4.004) = 9.73
C = (2.43)(3.681) = 8.94

Set R and C as described in Section 8.

FIGURE 8. URAL.

Tuning the URAL:

The URAL has 2 Berg jumpers. One jumper sets the gain of the URAL, the other jumper the time constant. The URAL also has a Pot, R3, which sets the level of the signal that goes into the output stage.

It is suggested that initially R3 should be at 50%, Gain in the middle position, and the time constant in the short position.

If URAL is unstable, decrease the gain, decrease R3, and increase the time constant.

If URAL is too slow, increase gain, increase R3, and decrease the time constant.

## 9. MODULE ELEMENTARY

Elementary DS3800NEPF sh. 4A shows the ac (Q1) and dc (Q2) power transistors. These power transistors are used to regulate dc bus voltage to the exciter field. When a power transistor is turned on the dc bus voltage is applied thru the transistor to the field of the exciter. When the transistor is turned off, the free-wheeling diode (CR5) conducts the field current around the transistors. As explained in section 5.3, the ratio of "on time" to "total time" that a transistor, Q1 or Q2 is selected by 43CS, multiplied by the regulator dc bus voltage, determines the exciter field voltage.

Diode pairs, CR2, CR3, CR4 form a three phase power rectifier, if single phase power is applied one diode pair is not utilized.

The 125 Vdc from the station battery is applied to DC1 (+) and DC2 (-). One of the diodes in CR1 is used to protect the regulator dc bus against failure in the dc power supply. The power supply is also filtered by inductors L1 and L2.

The isolation operational amplifier shown in section K5, provides the current signal to the ac regulator. The rate feedback and current limit use the current signal from this device.

The regulated ± 15 volt power supplies for control circuits are dc to dc power supplies. Their source is the regulator dc bus at PS1 and PS2. The circuit in section L38 will not allow voltage spikes greater that 180 volts to enter the input to the 15 volt power supplies. The dc power supplies are only shown in a block diagram on DS3820SVR. These supplies may be inspected visually - if the LED is on, the power supply works.

The ac regulator, sensing , and TVM are shown on DS3800NVRC sh. 4AA. CR1, CR2 and CR3 rectify the 120 Vac PT signal. R98, R99, C19, C20 filter the rectified sensing voltage. VTP8 is the rectified filtered PT signal. For single phase sensing the jumper BJ2 is in the 1 position which means there is a large resistance across the sensing voltage. For three phase sensing BJ2 is in the 3 position and a smaller resistance is across the sensing voltage. The smaller resistance will have more current, and tends to drop the average voltage at TP8. This is necessary because 120 Vac rectified three phase voltage is greater in magnitude than 120 Vac rectified single phase voltage. The sensing circuit must work the same for either single phase or three phase voltages.

90P, the ac adjust pot, is connected across J1-3 and Jl-5. The milliamp current signal from the reference, from the sensing, from V/Hz and from Droop or LDC sum at the common point of R25 and R93. TP3 is the output of the ac regulator operational amplifier. This voltage is input to the LM3524 pulse width modulator. The more negative the voltage at TP3 becomes, the more time the LM3524 output is "on" , and if the ac regulator is in control, the higher the field voltage becomes.

The output of the LM3524 drives transistors Q1 and Q2 on DS3800NVRC sh. 4DA. The output of Q1, Q2 is BD1 which controls the power transistor Q2 on DS3800NEPF sh. 4A.

Going back to DS3899NVRC sh. 4AA section C25. The current signal from the isolation operational amplifier is input at Jl-6 and J1-12. The position of BJ1 determines how much of the signal will be available for current rate feedback and current limit. When BJ1 is in the "H" position less signal is available, thus a higher current limit , and less current rate feedback, for a given exciter field current.

DS3800NVRC sh. 4BA, sections 25 - 45, is the current rate feedback circuit. The Lag adjust determines how much of the signal will be available for the lead adjust. The input to the Lead adjust is filtered by a 100 microfarad capacitor, thus the signal is only ac. The Lead adjust determines how much of this ac signal will be input to the summing junction of the ac regulator at R93 and R25. The voltage at TP4 should not have a dc off-set. If it does this may indicate that C7 is installed with the wrong polarity, or that it is faulty.

Section Ill-25 of DS3800NVRC sh. 4BA is the dc regulator. The dc voltage from J1-14 and J1-16 is filtered and attenuated by the op amp in section C13. The dc reference, 70P is connected to J1-9, J1-13, J1-11. The op-amp in section El3 sums the reference with the feedback and controls the LM3524 just like the ac regulator. Similarily the dc LM3524 control transistors Q3 and Q4 on sheet 4DA control the power transistor Q1 on DS3800NEPF sh. 4AA.

Regardless of which regulator is controlling the exciter field voltage, both LM3524's continue to switch on and off. Thus the power transistor Q1 and Q2 continue to switch on and off. It is only the 43CS switch that selects which power transistor will control the exciter field voltage.

The Transfer Voltmeter circuit shown on DS3800NVRC sh. 4AA section F39 has the time modulated signals from both the ac and dc 3524's as input. These time modulated signals are filtered to dc, and when both signals have the same average value the output of the TVM is zero. If the ac regulator has control, then 70P must be adjusted so that output of the dc 3524 has the same ratio of "on time" to "total time" as the ac 3524. When this happens the output of the TVM is zero, and switching control from ac regulator to dc regulator will not change the exciter field voltage, consequently a bumpless transfer.

On DS3800NVRC sh. 4CA, the lower part (25-45), is the volts per hertz limit circuit. The V/Hz has only the voltage from the PT as input. The output of device U8 (E35) is input to C17. When the voltage at pin 8 is positive the output at pin 14 shorts the Cl7 to DCOM discharging it. When the voltage at pin 8 is negative Cl7 is charged through Q5 and R87 - a constant current source. The lower the frequency of the PT signal, the more time Cl7 has to charge, and so it will charge to a higher voltage during that time period.

When the frequency is low enough the voltage at C17 will be equal to the voltage on the takeover adjust pot. The output of LJ8 (H35) goes negative. The output of U3 (K33) also goes negative. This negative voltage at the summing junction causes current to flow into the V/Hz circuit, and the ac regulator senses that the generator ac voltage is rising. The ac regulator then reduces excitation and the generator terminal voltage drops. The lower the frequency the lower the generator terminal voltage will become.

The Droop/LDC circuit on DS3800NVRC sh.4CA, the upper portion (01-23), has as inputs a current signal from the CT, and a voltage signal. The phase relationship of the voltage and current is used to determine the amount of droop or LDC. The voltage signal input to the droop circuit is determined by the position of BJ3 and BJ4. The voltage is either 90° leading or 90° lagging the current at unity power factor. Note that the signal at BJ4 is 180° shifted from BJ3. The current signal into the static switch U7 is at pins 1 and 4. These signals are shifted by 180°. Device U7 is a switch that is opened and closed by the voltage signals at BJ3 and BJ4.

As an example, suppose that BJ3 and BJ4 are positioned so that the current is leading the voltage by 90° at unity power factor. Refering to figure 9, switch 1 will close for 180°, and the signal that is input to U3, section F09, has an average value of the integral of sin(u), integrating from u=90° to u=270°. Likewise the same thing happens with switch 2, since both the current and voltage are shifted by 180° as explained above. Thus at unity power factor the input to U3 is zero. As power factor increases the average voltage input to U3 is the integral of sin(u) from u=90°-(pf angle) to u=270°-(pf angle). The average voltage will increase. The output of U3 is inverted from the input and goes more negative. Thus the ac regulator will lower generator ac voltage. When increasing power factor causes a decrease in generator ac voltage, droop is the function.

By switching the phase of the voltage into switch 1 and 2 by 180° (using jumpers BJ3 and BJ4) the opposite will happen. When increasing power factor causes an increase in generator ac voltage, LDC is the function.

On DS3800DVRC sh. 4AA and sh. 4BA is the URAL circuit. On sh. 4AA the inputs to the URAL are current from the CT and voltage from the PT. At unity power factor these shifted by 90°. Op amp U1, section D15, adds a percentage of the voltage, determined by the position of R1, to the current signal. The phasor

addition occurs as previously described in section 5, this voltage (I + C) is rectified and filtered, and can be measured at TP3. A percentage of the voltage at J2 is determined by R2 as the Radius. This ac voltage is rectified and filtered, and can be measured at TP7. When the magnitude of TP3 is greater than TP7 the voltage at TP4 begins to go positive. The URAL will modify the ac regulator when the static switches U5 on sh. 4BA operate. Pot R3 determines how much of the URAL signal will pass into the ac regulator. A combination of the BJ1 and BJ2 positions determine the time constant and affect the gain of the URAL. The feedback from the ac regulator into the URAL is to charge C5 and C6, so that when switches U5 operate, the URAL control will not cause a "BUMP".

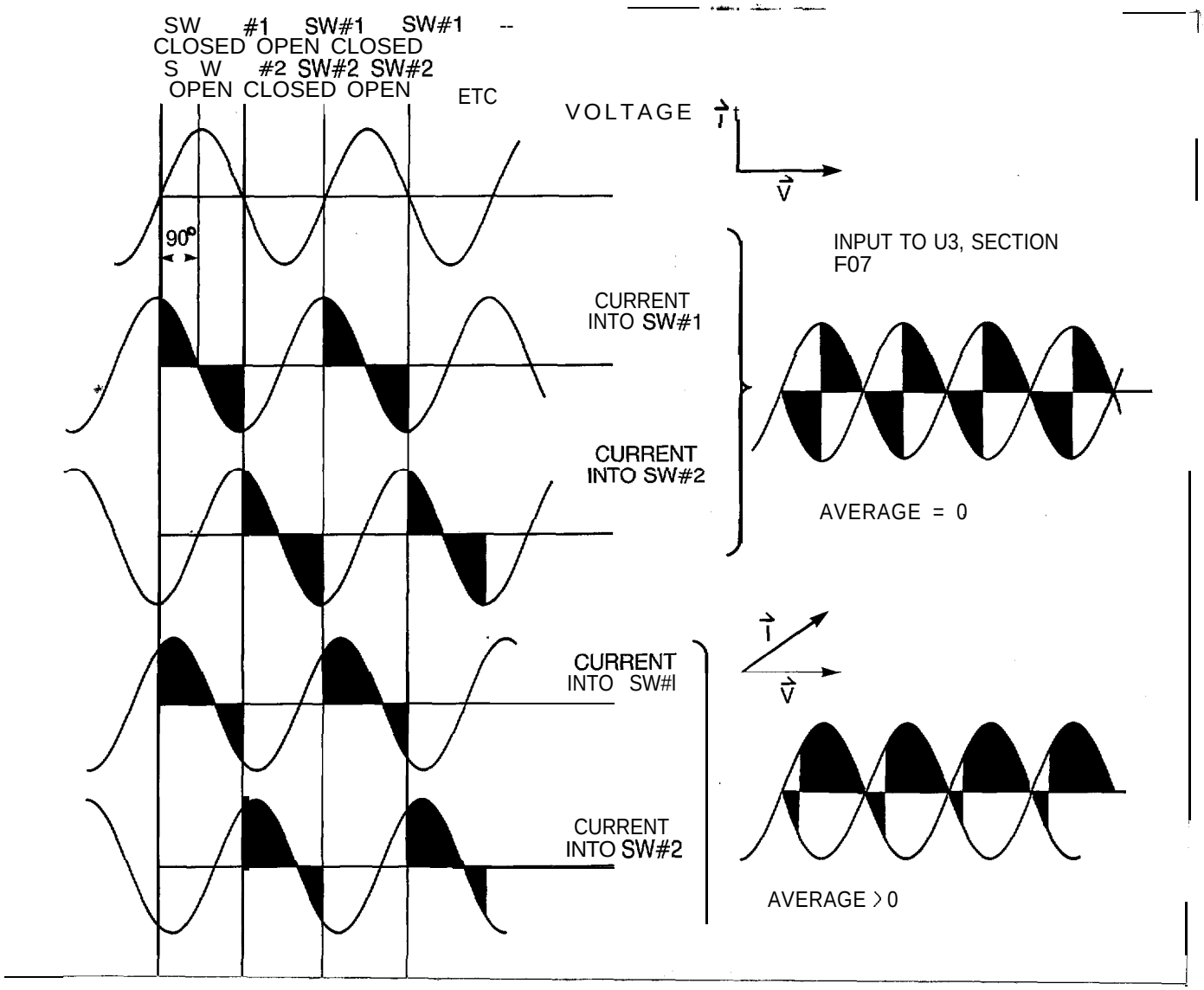



FIGURE 9.

RIG

# STATIC VOLTAGE REGULATOR

| | |
|---|---|
| CUSTOMER | |
| STATION | |
| REQUISITION | |
| EXCITER MODEL NUMBER | DS3820SVR |
| ML NUMBER | |
| EQUIPMENT INSTRUCTION BOOK | GEK-83923 |

ELEMENTARY DIAGRAM
TITLE SHEET
237B4355 A02
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM, VA. U.S.A.

237B4355 CONT ON SH 02A
SH NO A04B

FUNCTION INDICATOR --
CMC 4 LAMP •

| TERMINALS | NO. | INDICATOR LEGEND (BACK VIEW) UPPER | RATED | LOWER | | SH. | (INSERT COLOR) |
|---|---|---|---|---|---|---|---|
| C 1 | 1 | | X | | | | AMB |
| 2 12 | 2 | X | | | | | RED |
| (NOT USED) | 3 | | | | | | BLK |
| 4 14 | 4 | | | X | | | GRN |

RATED UPPER
LOWER
X = LAMP USEAGE

## △⊖ VOLTAGE ADJUSTED SWITCH FOR 2 AC REGULATORS

SBM SWITCH 90CS
PISTOL GRIP HANDLE

| HANDLE END | NO. | HANDLE POSITIONS (BACK VIEW) RAISE | | LOWER |
|---|---|---|---|---|
| 1 1C 2C 2 | 1 | X | | |
| | 2 | | | X |

SPRING RETURN TO NORMAL

90CS
LOWER RAISE
X = CLOSED
305A6752P1

## △ REGULATOR TRANSFER SWITCH

SBM SWITCH 55CS
PISTOL GRIP HANDLE

| CONTACTS HANDLE END | NO | HANDLE POSITIONS (BACK VIEW) ON | | OFF |
|---|---|---|---|---|
| 1 1C 2C 2 | 1 | X | | |
| | 2 | X | | |
| 3 3C 4C 4 | 3 | X | | |
| | 4 | | | X |

55CS
OFF ON
X = CLOSED

## △ REGULATOR TRANSFER SWITCH FOR 2 AC REGULATORS

SBM SWITCH 43CS
PISTOL GRIP HANDLE

| CONTACTS HANDLE END | NO | HANDLE POSITIONS (BACK VIEW) BACKUP | PRIMARY | OFF |
|---|---|---|---|---|
| 1 1C 2C 2 | 1 | | X | |
| | 2 | X | | |
| 3 3C 4C 4 | 3 | X | X | |
| | 4 | | X | |

MAINTAIN

43CS
PRIMARY
OFF BACKUP
X = CLOSED
278A2180LAP1

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM
SWITCH DEVELOPMENT
SH. NO. A04B
237B4355
02A
CONT ON SH

REVISED 05-16-88

ORIG

△ FIELD BREAKER CONTROL SWITCH ⊖
SBI SWITCH 1 STAGE
41CS STANDARD PISTOL GRIP HANDLE

| HANDLE END | NO. | HANDLE POSITIONS (BACK VIEW) ON | | OFF | SH. |
|---|---|---|---|---|---|
| 1 1C 2C 2 | 1 | | | X | 08A |
| | 2 | X | | | 08A |

41CS
OFF ON

X : CLOSED
16SB1AB300SUS2Y

△ AC VOLTAGE ADJUST ⊖
SBI SWITCH 2 STAGE
90CS STANDARD PISTOL GRIP HANDLE

| CONTACTS HANDLE END | NO | HANDLE POSITIONS (BACK VIEW) RAISE | | LOWER | SH. |
|---|---|---|---|---|---|
| 1 1C 2C 2 | 1 | X | | | 09A |
| | 2 | | | X | 09A |
| 3 3C 4C 4 | 3 | X | | | |
| | 4 | | | X | |

90CS
LOWER RAISE

X : CLOSED
16SB1BB320SUS2Y

AC REGULATOR TRANSFER SWITCH
△ C32 SWITCH 8 STAGE
43CS PISTOL GRIP HANDLE

| CONTACTS HANDLE END TOP BOTTOM | NO | HANDLE POSITIONS (BACK VIEW) AUTO | MAN | OFF | SH. |
|---|---|---|---|---|---|
| 1 2 3 4 | 1 | | X | | 09A |
| | 2 | | X | | 08A |
| 5 6 7 8 | 3 | | X | | 09A |
| | 4 | | X | | 08A |
| 9 10 11 12 | 5 | | X | | 08A |
| | 6 | | X | | 08A |
| 13 14 15 16 | 7 | X | | | 08A |
| | 8 | X | | | 08A |
| 17 18 19 20 | 9 | X | | | 08A |
| | 10 | X | | | 08A |
| 21 22 23 24 | 11 | X | | | 08A |
| | 12 | X | | | 08A |
| 25 26 27 28 | 13 | | X | | |
| | 14 | X | | | |
| 29 30 31 32 | 15 | | | X | |
| | 16 | X | X | | |

43CS
OFF MAN AUTO

X : CLOSED
*EVEN* NO. CONTACT ON BOTTOM
ODD NO. CONTACT ON TOP
C32US2234-700-KN1
(INCLUDES LEGEND AND HANDLE)

△ DC VOLTAGE ADJUSTOR SWITCH ⊖
SBI SWITCH 2 STAGE
70CS STANOARD PISTOL GRIP HANDLE

| CONTACTS HANDLE END | NO. | HANDLE POSITION (BACK VIEW) RAISE | | LOWER | SH. |
|---|---|---|---|---|---|
| 1 1C 2C 2 | 1 | X | | | 09B |
| | 2 | | | X | 09B |
| 3 3C 4C 4 | 3 | X | | | |
| | 4 | | | X | |

70CS
LOWER RAISE

X : CLOSED
16SB1BB320SUS2Y

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM
SWITCH DEVELOPMENT
237B4355 A04A
SH. NO A0
CONT ON SH

| REV | ISSUED | MADE BY |
|---|---|---|
| REV 1 | | |
| REV 2 | 05-16-88 | |
| REV 3 | | |
| REV 4 | | |
| REV 5 | | |

ENGINEER | PRINTS TO | FIRST MADE FOR NEXT | ITEM NO.

COPYRIGHT

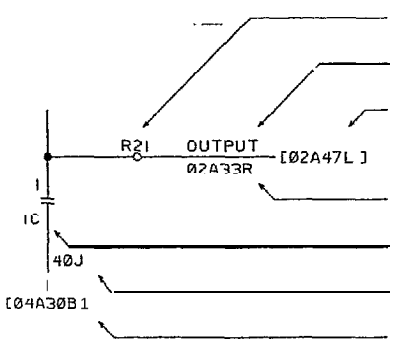
SH NO. A02 | 237B4355 CONT ON SH A04

## CUSTOMER NOTES

SYMBOLS USED.

- OPTION
- * PURCHASERS EQUIP
- A SUPPLIED BY GE DSD,MOUNTED BY CUSTOMER
- FACTORY TERMINAL BOARD CONNECTION
- CIRCUIT CONNECTION (DOES NOT DESIGNATE LOCATION)
- CUSTOMER TERMINAL WARD CONNECTION
- SPRING RETURN TO NORMAL
- CUSTOMER CABLE CONNECTION
- RELAY OR SWITCH CONTACT
- RELAY WITH BLOW OUT COIL
- FIXED RESISTOR
- AJUSTABLE RESISTOR
- INDUCTIVE, TRANSFORMER WINDING OR FIELD WINDING, (WITH POLARITY IF REQUIRED)
- FUSE
- DISCONNECT SWITCH
- SHORTING SWITCH
- BREAKER
- BREAKER WITH OVERCURRENT TRIP
- RECTIFIER
- THYRISTOR (SCR)
- ZENER DIODE
- POTENTIOMETER
- GROUNDEDRECEPTACLE
- LIGHT EMITTING DIODE (LED) INDICATOR
- THYRITE® RESISTOR
- RELAY COIL
- CAPACITOR
- H E A T E R
- 5 INDICATING LAMP
- SHUNT
- METER
- SHIELDED WIRE
- SHIELD
- ® REGISTERED TRADEMARK

| SHEET NO. | SUBJECT |
|---|---|
| A01 | TITLE |
| A02 | CUSTOMER NOTE5 |
| A04 | --- SVITCH DEVELOPHENT --- |
| 02A,02B | SENSING |
| 03A | POWER BRIDGE |
| 03B,03C | PRIMARY /BACKUP AC REG |
| 04A.07B | EXCITER FIELD |
| 05A | REGULATOR |
| 05B | URAL OPTION |
| 05H | VAR/PF OPTION |
| 0 8 A | OVERVOLTAGE OPTION |
| 08B | RELAYING, 2 A C R E G |
| 09A | 90P MOVA OPTION |
| 09a | 70P NOVA OPTION |
| 09C | 90PB,90PA MOVA |
| IIA | CUBICLE LIGHTS, TRANSDUCERS |
| I2A,I2B | CUSTOMER INTERFACE |




TERMINAL BOARD DESIGNATION

SIGNAL NAME

LOCATION OF SOURCE OF THE SIGNAL NAME ON ELEM SHEET 02A, VERT ZONE 47, HORIZ. ZONE L

WIRE NUMBER OF CIRCUIT (SHOWS CKT ORIGINATED ON SH 02A)

CONTACT DESIGNATIONS OR PIN NUMBERS

WIRE NUMBER OF CIRCUIT THAT ORIGINATED ON THIS SHEET.

DESTINATION OF CIRCUIT SHEET 04A, VERT ZONE 30, HORIZ. ZONE a

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA U.S.A.

CUSTOMER NOTES
237B4355 SH. NO. A02 A04

ENGINEER | FIRST MADE FOR REQN. | PRINTS TO | P.C. NO.
REV 1 | REV 2 | REV 3 | REV 4 | ISSUED 16-88

ORIG

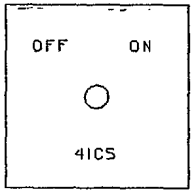
237B4355 A04B CONT ON SH

SH NO A04A

EXCITATION CONTROL SWITCH
CMC 1 BLOCK, 4 CIRCUIT, 2 LAMP
41CS

| TERMINALS | NO. | ON | OFF | SH. |
|---|---|---|---|---|
| 1 2 | 1 | | X | |
| 3 4 | 1 | X | | |
| 5 6 | 1 | | X | |
| 7 8 | 1 | X | | |
| C 1 | | | X | GRN |
| 2 12 | | X | | RED |

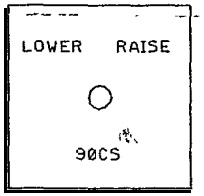
(INSERT COLOR)




X = CLOSED FOR CONTACTS AND LAMP USEAGE

AUTO VOLTAGE ADJUST
CMC 1 BLOCK, 4 CIRCUIT
90CS

| TERMINALS | NO. | RAISE | | LOWER | SH. |
|---|---|---|---|---|---|
| 1 2 | 1 | | | X | |
| 3 4 | 1 | X | | | |
| 5 6 | 1 | X | | | |
| 7 8 | 1 | | | X | |

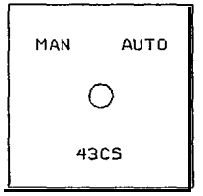



X = CLOSED SPRING RETURN TO CENTER

AC REGULATOR TRANSFER SWITCH
CMC 1 BLOCK, 4 CIRCUIT, 2 LAMP
43CS

| TERMINALS | NO. | AUTO | MAN | SH. |
|---|---|---|---|---|
| 1 2 | 1 | | X | |
| 3 4 | 1 | X | | |
| 5 6 | 1 | | X | |
| 7 8 | 1 | X | | |
| C 1 | | | X | GRN |
| 2 12 | | X | | RED |

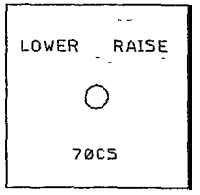
(INSERT COLOR)




X = CLOSED FOR CONTACTS AND LAMP USEAGE

MAN. VOLTAGE ADJUSTER SWITCH
CMC 1 BLOCK, 4 CIRCUIT
70CS

| TERMINALS | NO. | RAISE | | LOWER | SH. |
|---|---|---|---|---|---|
| 1 2 | 1 | | | X | |
| 3 4 | 1 | X | | | |
| 5 6 | 1 | X | | | |
| 7 8 | 1 | | | X | |




X = CLOSED SPRING RETURN TO CENTER

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.

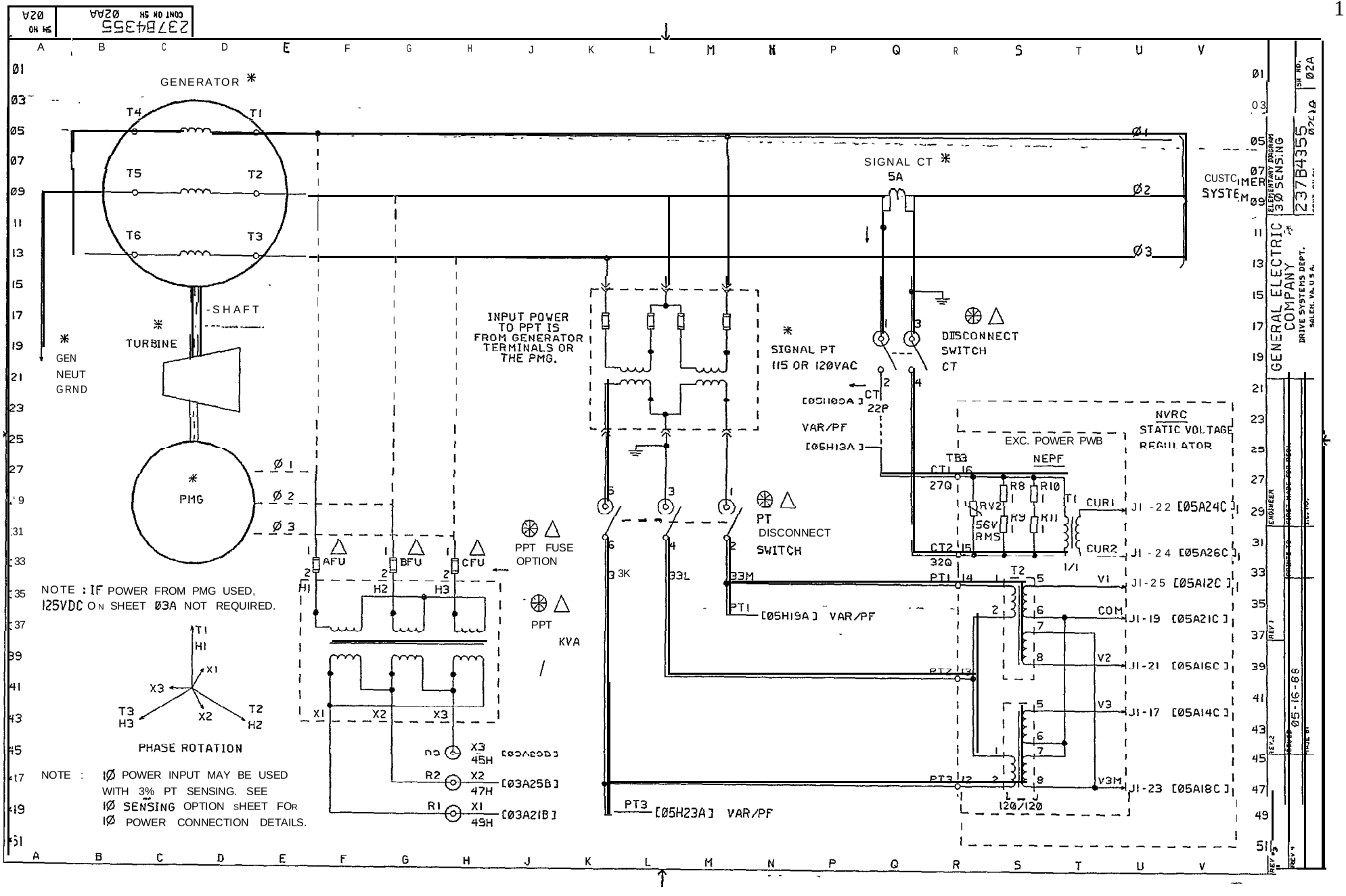
ELEMENTARY DIAGRAM
SWITCH DEVELOPMENT

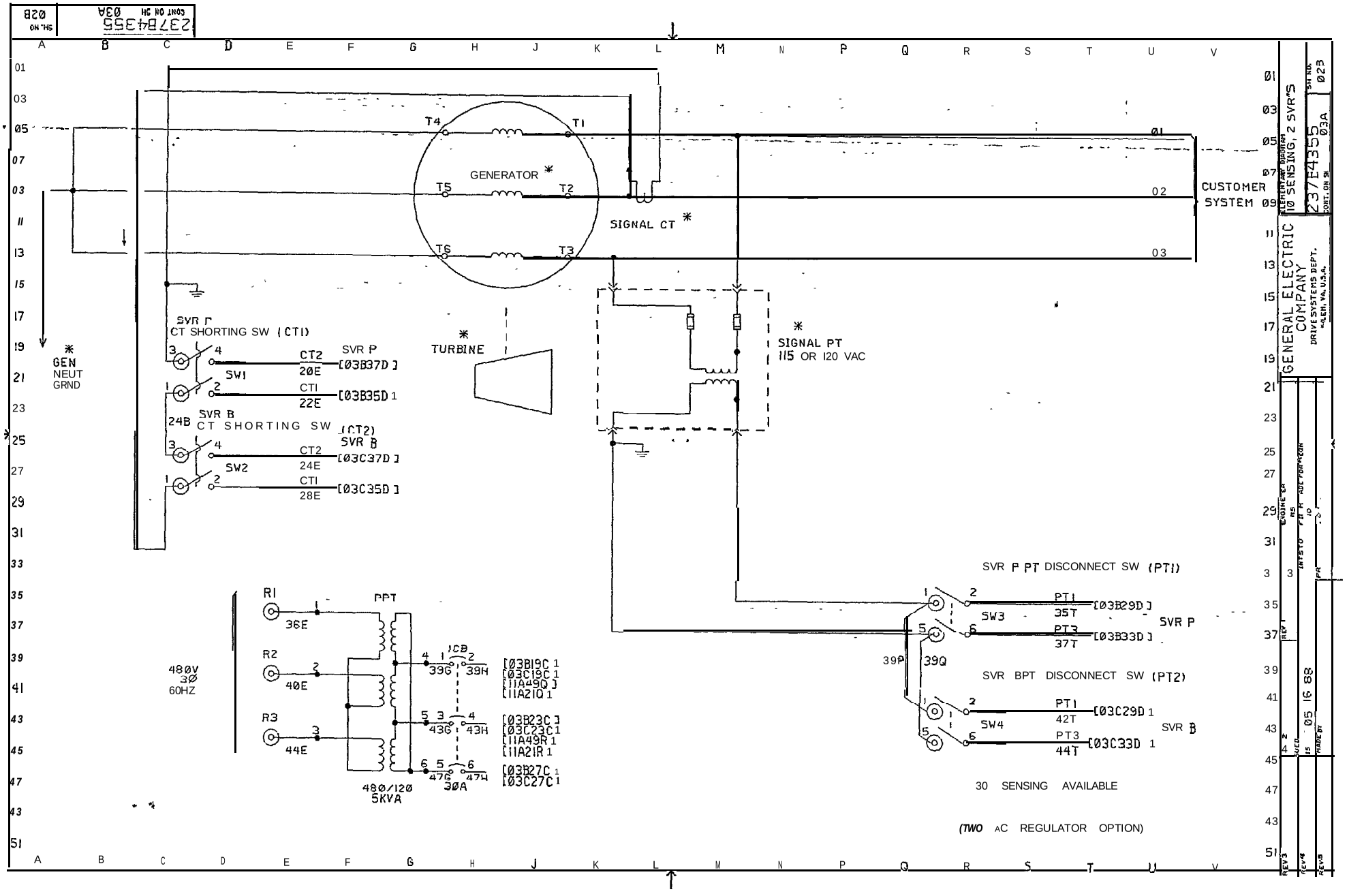
237B4355 A04B

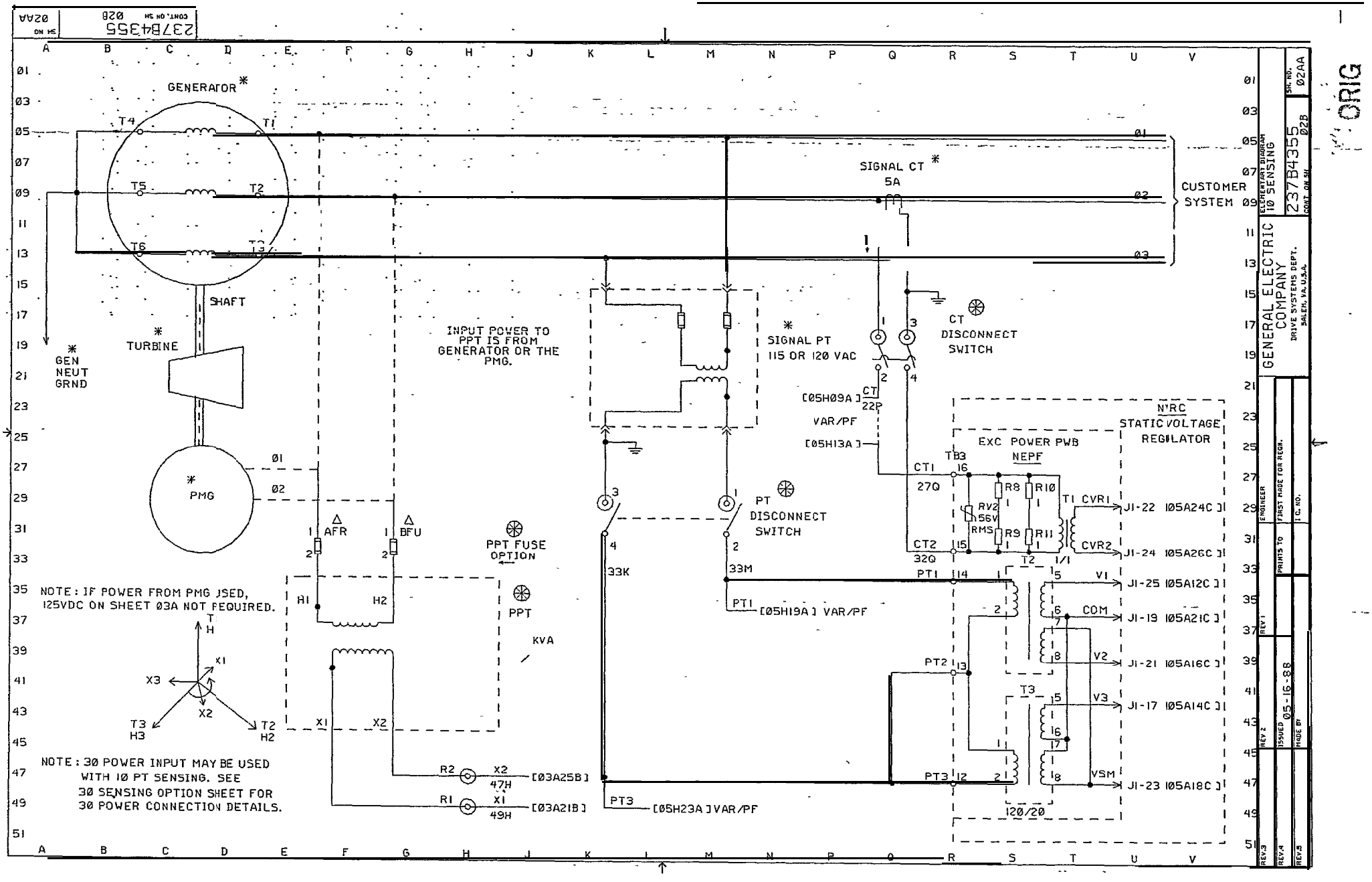
SH. NO. A04A

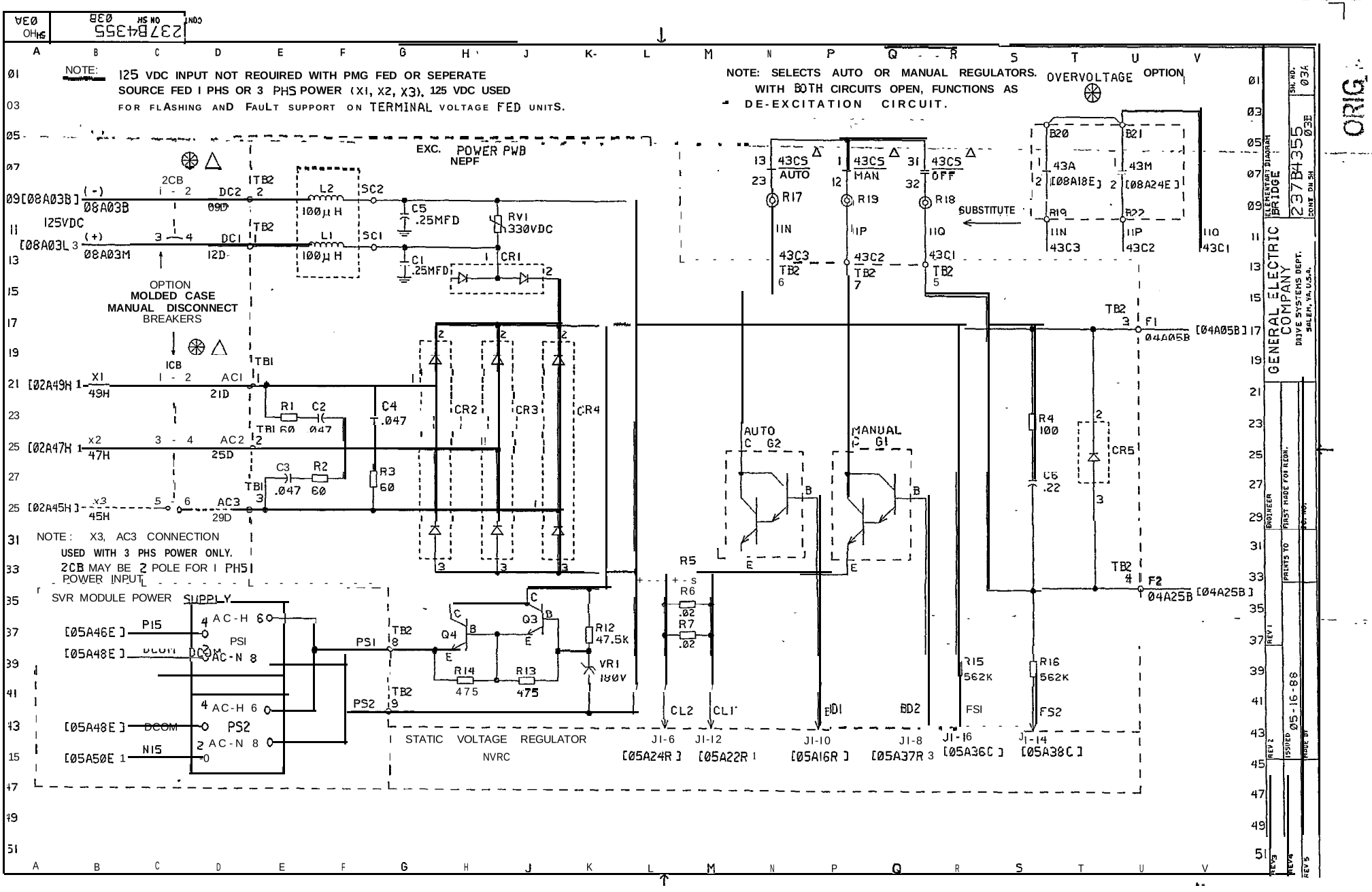
ENGINEER
FIRST MADE FOR REGN.
I.C. NO.
DRAWING TO
REV 1
REV 2
ISSUED 05-16-82
MADE BY
REV 3
REV 4
REV 5

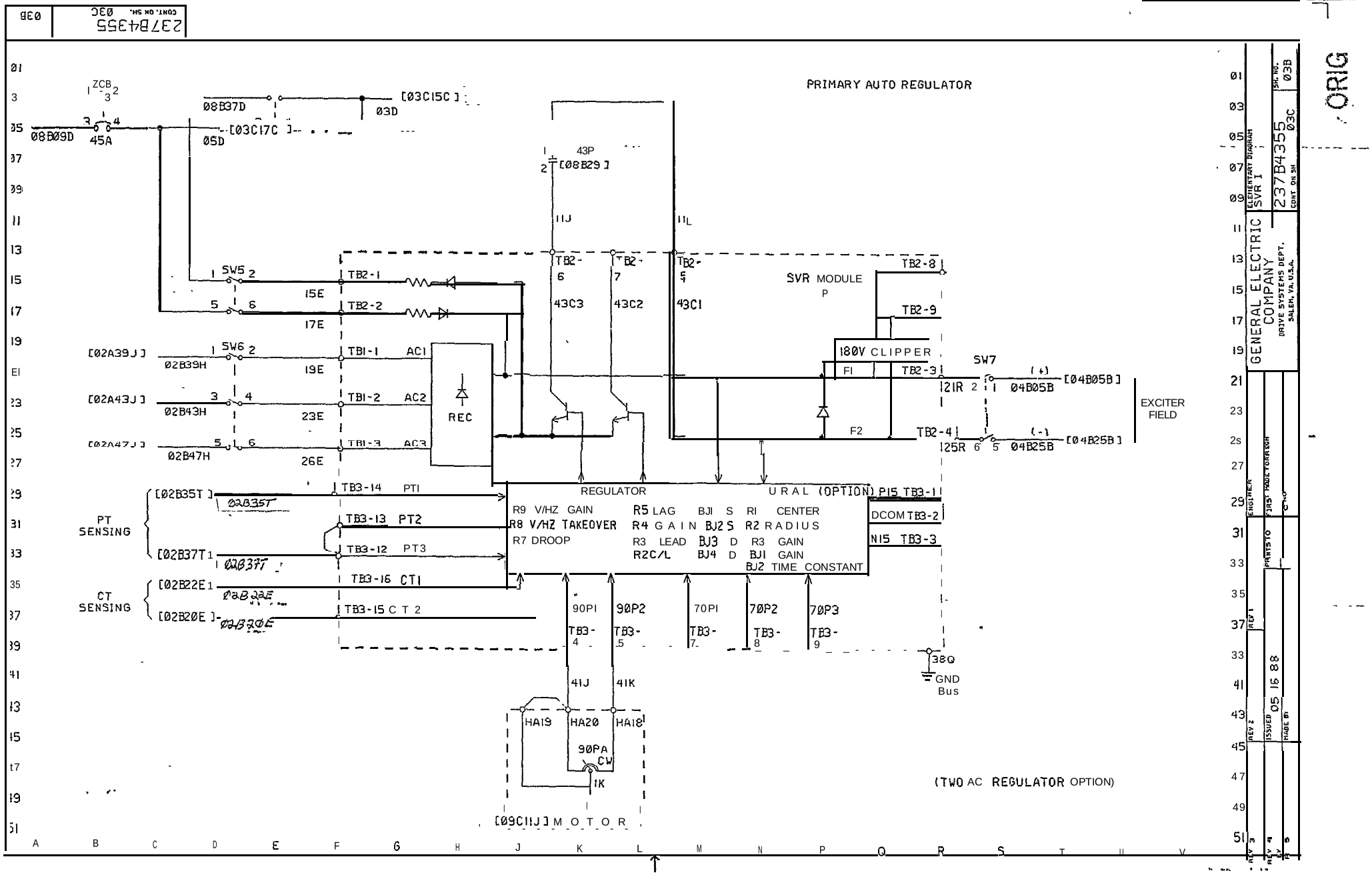
ORIG

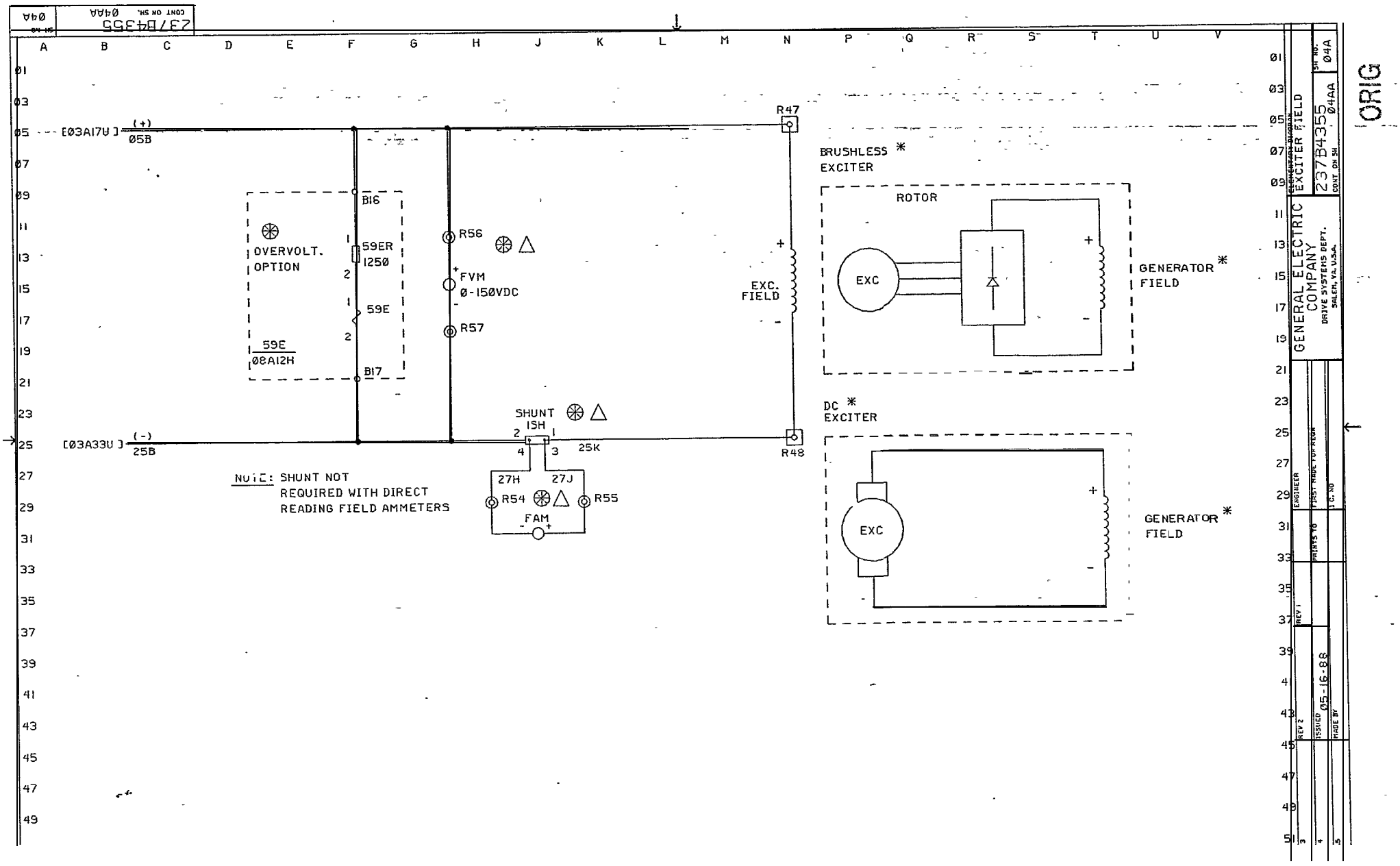

GENERATOR *
T4
T1
T5
T2
T6
T3
SHAFT
TURBINE
GEN NEUT GRND
PMG
Ø1
Ø2
Ø3
NOTE : IF POWER FROM PMG USED, 125VDC ON SHEET 03A NOT REQUIRED.
T1 H1 X1 X3 T3 H3 X2 T2 H2
PHASE ROTATION
NOTE : 1Ø POWER INPUT MAY BE USED WITH 3% PT SENSING. SEE 1Ø SENSING OPTION SHEET FOR 1Ø POWER CONNECTION DETAILS.
INPUT POWER TO PPT IS FROM GENERATOR TERMINALS OR THE PMG.
AFU
BFU
CFU
H1
H2
H3
X1
X2
X3
PPT FUSE OPTION
PPT
KVA
X3 45H [03A25B]
X2 47H [03A25B]
X1 49H [03A21B]
SIGNAL CT *
5A
SIGNAL PT
115 OR 120VAC
CT 22P
[05H19A] VAR/PF
DISCONNECT SWITCH CT
PT DISCONNECT SWITCH
3K
33L
33M
PT1 [05H19A] VAR/PF
PT3 [05H23A] VAR/PF
CUSTOMER SYSTEM
NVRC STATIC VOLTAGE REGULATOR
EXC. POWER PWB
TB3
NEPF
CT1
27Q
R8
R10
RV2
R9
R11
56V RMS
T1
CUR1
CUR2
CT2
32Q
PT1
T2
V1
COM
V2
V3
V3M
J1-22 [05A24C]
J1-24 [05A26C]
J1-25 [05A12C]
J1-19 [05A21C]
J1-21 [05A16C]
J1-17 [05A14C]
J1-23 [05A18C]
120/120
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA U.S.A.
ELEMENTARY DIAGRAM
130 SENSING
237B4355
SH NO. 02A
05-16-88

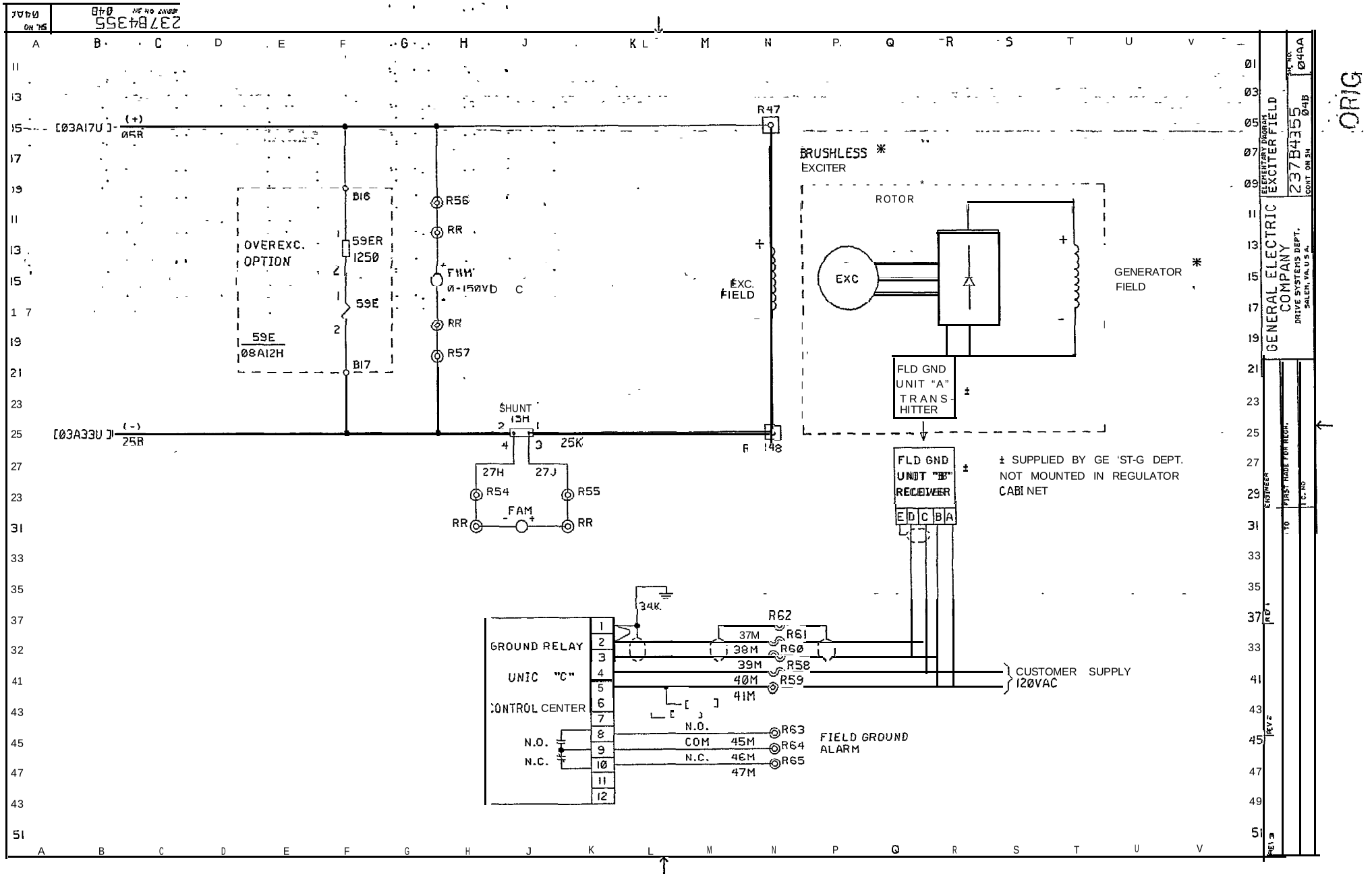

GENERATOR
T4
T1
T5
T2
T6
T3
SIGNAL CT
CUSTOMER SYSTEM
01
02
03
GEN
NEUT
GRND
TURBINE
SIGNAL PT
115 OR 120 VAC
SVR P
CT SHORTING SW (CT1)
SW1
CT2
20E
[03B37D ]
CTI
22E
[03B35D 1
SVR B
24B CT SHORTING SW (CT2)
SW2
CT2
24E
[03C37D ]
CTI
28E
[03C35D ]
R1
36E
R2
40E
R3
44E
PPT
480V
3Ø
60HZ
480/120
5KVA
ICB
39G
39H
43G
43H
47G
47H
30A
[03B19C 1
[03C19C 1
[11A49Q ]
[11A21Q 1
[03B23C ]
[03C23C1
[11A49R 1
[11A21R 1
[03B27C 1
[03C27C 1
SVR P PT DISCONNECT SW (PT1)
SW3
PT1
35T
[03B29D ]
PT3
37T
[03B33D ]
SVR P
39P
39Q
SVR BPT DISCONNECT SW (PT2)
SW4
PT1
42T
[03C29D 1
PT3
44T
[03C33D 1
SVR B
30 SENSING AVAILABLE
(TWO AC REGULATOR OPTION)
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
*LEM, VA, U.S.A.
10 SENSING, 2 SVR'S
2371E4355
03A
023
05 16 88

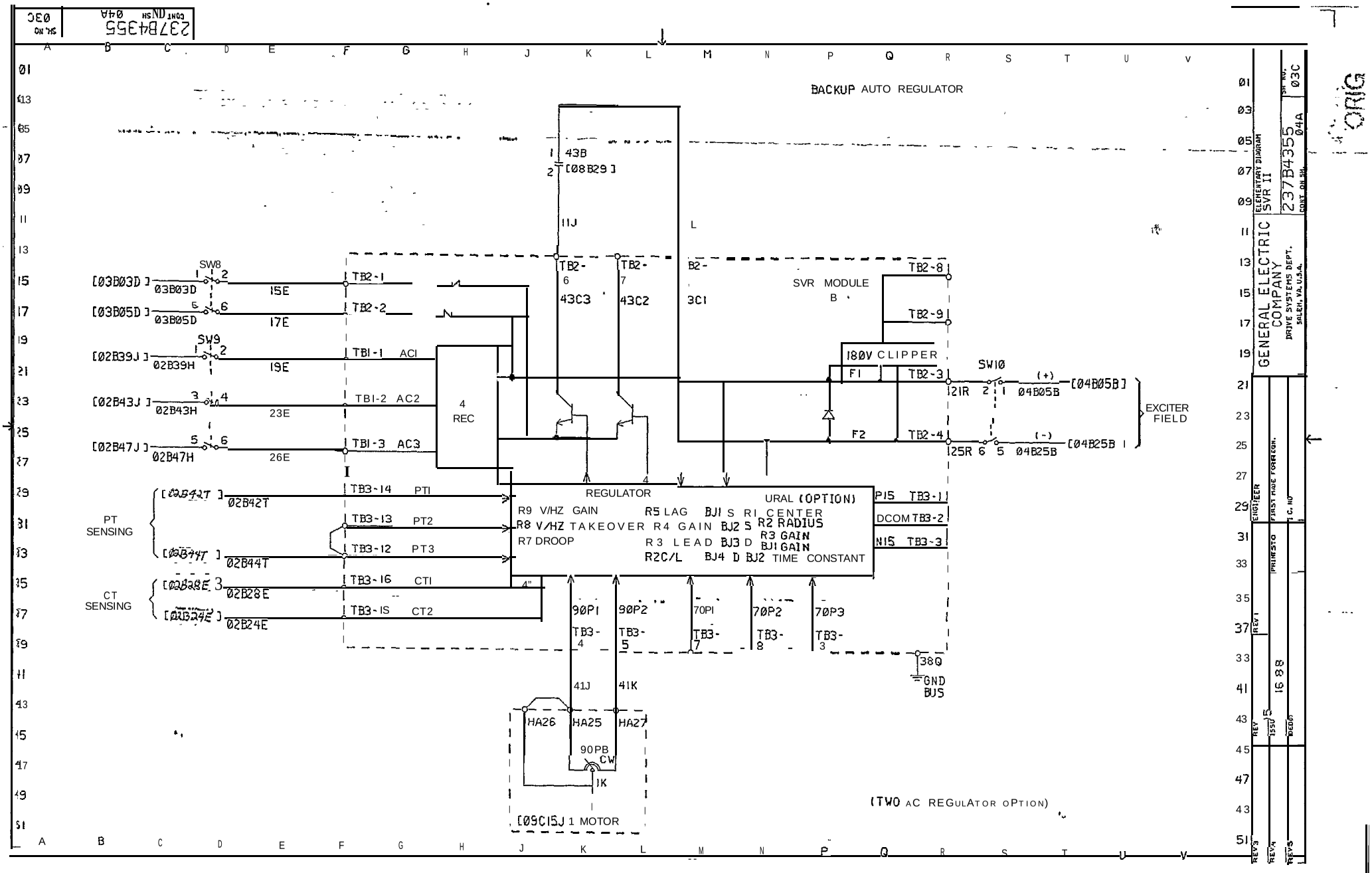

GENERATOR *
T4
T1
T5
T2
T6
T3
SHAFT
* TURBINE
* GEN NEUT GRND
* PMG
Ø1
Ø2
INPUT POWER TO PPT IS FROM GENERATOR OR THE PMG.
SIGNAL CT * 5A
CUSTOMER SYSTEM
* SIGNAL PT 115 OR 120 VAC
CT DISCONNECT SWITCH
[05H09A] CT 22P VAR/PF
[05H13A]
NRC STATIC VOLTAGE REGULATOR
EXC POWER PWB
NEPF
TB3
CT1 27Q
CT2 32Q
R8
R10
RV2 56V RMS
R9
R11
T1
CVR1
CVR2
T2
J1-22 105A24C
J1-24 105A26C
PT DISCONNECT SWITCH
33K
33M
PT1
[05H19A] VAR/PF
J1-25 105A12C
J1-19 105A21C
J1-21 105A16C
J1-17 105A14C
J1-23 105A18C
V1
COM
V2
V3
VSM
PT2
PT3
T3
120/20
AFR
BFU
PPT FUSE OPTION
PPT
KVA
A1
H2
X1
X2
R2
47H
[03A25B]
R1
49H
[03A21B]
PT3
[05H23A] VAR/PF
NOTE: IF POWER FROM PMG USED, 125VDC ON SHEET 03A NOT REQUIRED.
NOTE: 30 POWER INPUT MAY BE USED WITH 10 PT SENSING. SEE 30 SENSING OPTION SHEET FOR 30 POWER CONNECTION DETAILS.
H1
H3
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM 10 SENSING
237B4355 02B
SHT. NO. 02AA
CONT. ON SH
ENGINEER
PRINTS TO
DWG. NO.
REV 1
REV 2
ISSUED 05-16-88
MADE BY
REV 3
REV 4
REV 5
237B4355 CONT. ON SH 02B
SH NO 02AA
ORIG

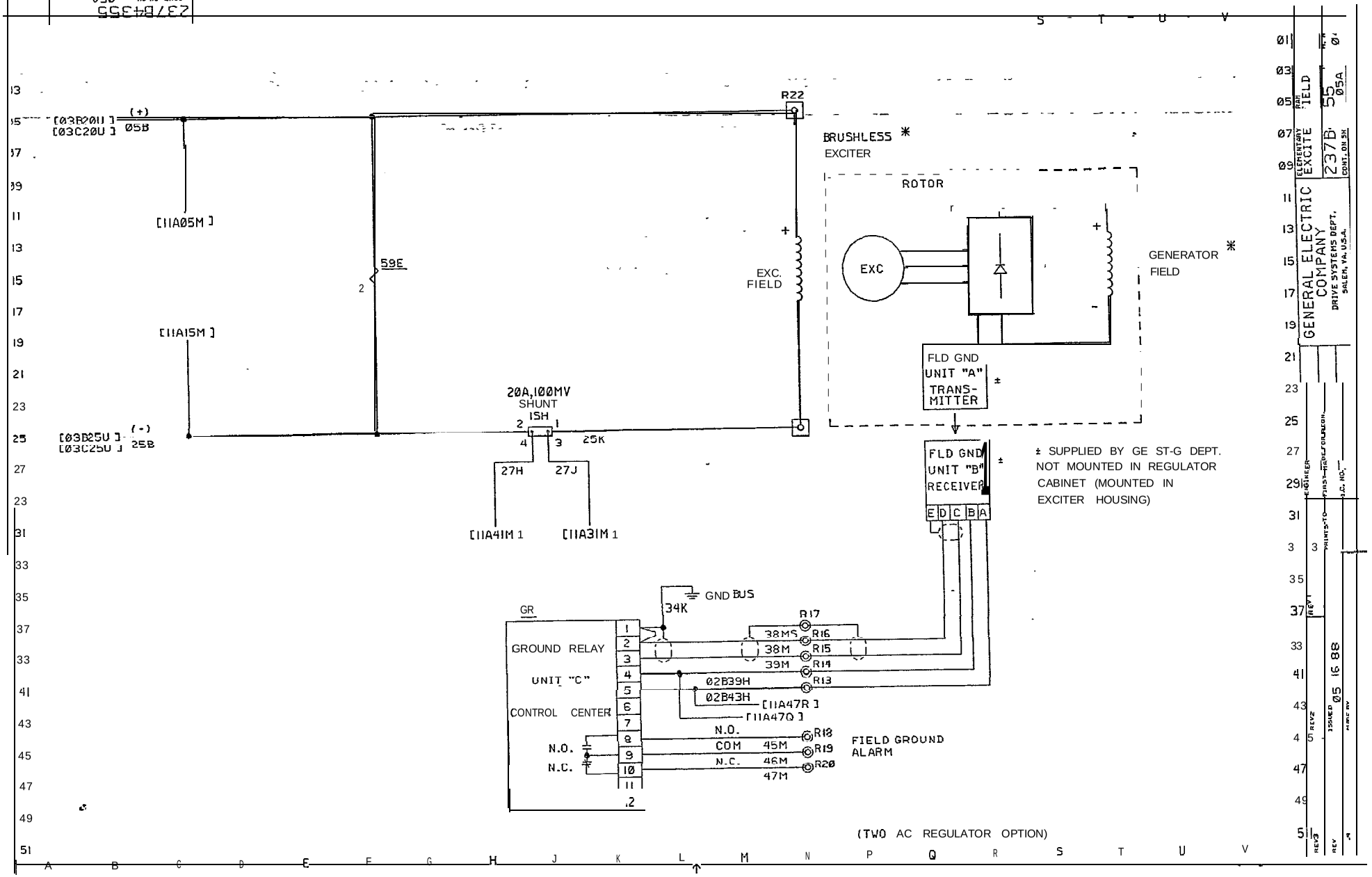

NOTE:
125 VDC INPUT NOT REQUIRED WITH PMG FED OR SEPERATE SOURCE FED I PHS OR 3 PHS POWER (X1, X2, X3). 125 VDC USED FOR FLASHING AND FAULT SUPPORT ON TERMINAL VOLTAGE FED UNITS.
NOTE: SELECTS AUTO OR MANUAL REGULATORS. WITH BOTH CIRCUITS OPEN, FUNCTIONS AS DE-EXCITATION CIRCUIT.
OVERVOLTAGE OPTION
EXC. POWER PWB NEPF
125VDC
OPTION MOLDED CASE MANUAL DISCONNECT BREAKERS
2CB
ICB
SUBSTITUTE
AUTO
MANUAL
OFF
NOTE: X3, AC3 CONNECTION USED WITH 3 PHS POWER ONLY.
2CB MAY BE 2 POLE FOR I PHS POWER INPUT
SVR MODULE POWER SUPPLY
STATIC VOLTAGE REGULATOR
NVRC
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
237B4355

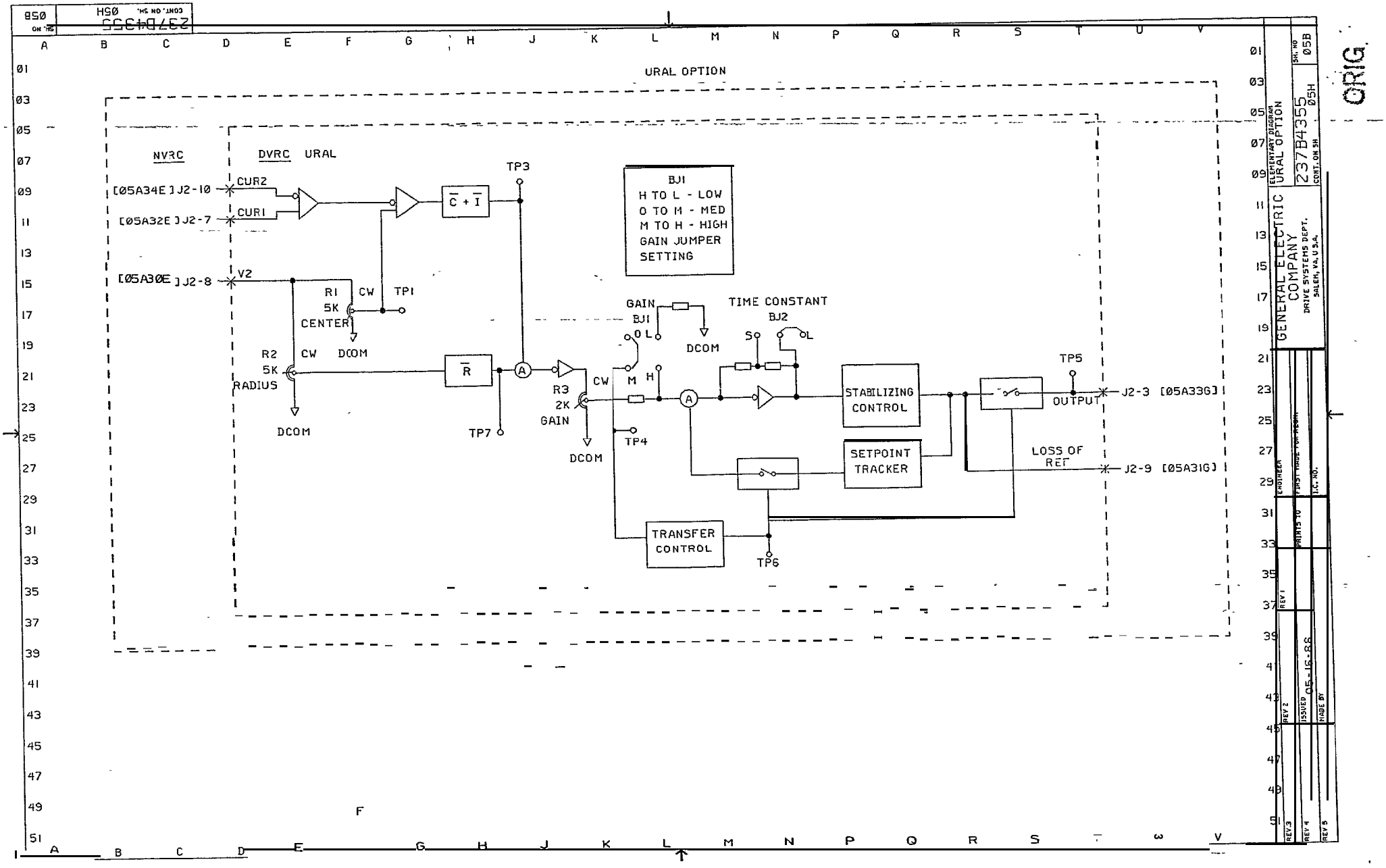

PRIMARY AUTO REGULATOR
ZCB
08B37D
[03C15C]
03D
08B09D
45A
[03C17C]
05D
43P
[08B29]
11J
11L
SW5
TB2-1
15E
TB2-2
17E
TB2-6
43C3
TB2-7
43C2
TB2-5
43C1
SVR MODULE
P
TB2-8
TB2-9
180V CLIPPER
F1
TB2-3
SW7
[04B05B]
04B05B
[02A39J]
02B39H
SW6
TBI-1
AC1
19E
[02A43J]
02B43H
TBI-2
AC2
23E
02B47H
TBI-3
AC3
26E
REC
F2
TB2-4
[04B25B]
04B25B
EXCITER FIELD
[02B35T]
TB3-14
PTI
PT SENSING
TB3-13
PT2
[02B37T]
TB3-12
PT3
[02B22E]
TB3-16
CT1
CT SENSING
[02B20E]
TB3-15
CT2
REGULATOR
URAL (OPTION)
P15 TB3-1
R9 V/HZ GAIN
R8 V/HZ TAKEOVER
R7 DROOP
R5 LAG BJ1 S R1 CENTER
R4 GAIN BJ2 S R2 RADIUS
R3 LEAD BJ3 D R3 GAIN
R2C/L BJ4 D BJ1 GAIN
BJ2 TIME CONSTANT
DCOM TB3-2
N15 TB3-3
90PI
TB3-4
90P2
TB3-5
70PI
TB3-7
70P2
TB3-8
70P3
TB3-9
GND Bus
41J
41K
HA19
HA20
HA18
90PA
CW
1K
[09C11J] MOTOR
(TWO AC REGULATOR OPTION)
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM
SVR 1
237B4355
SH. NO. 03B
CONT. ON SH. 03C
ISSUED 05 16 88
ORIG


237B4355 04AA
CONT ON SH.
04A
A B C D E F G H J K L M N P Q R S T U V
01 03 05 07 09 11 13 15 17 19 21 23 25 27 29 31 33 35 37 39 41 43 45 47 49
R47
[03A17U] (+) 05B
B16
OVERVOLT. OPTION
59ER 125Ø
59E
59E 08A12H
B17
R56
FVM Ø-150VDC
R57
EXC. FIELD
SHUNT 1SH
2 1 4 3 25K
[03A33U] (-) 25B
R48
NOTE: SHUNT NOT REQUIRED WITH DIRECT READING FIELD AMMETERS
27H 27J
R54 R55
FAM
BRUSHLESS * EXCITER
ROTOR
EXC
GENERATOR * FIELD
DC * EXCITER
EXC
GENERATOR * FIELD
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA, U.S.A.
EXCITER FIELD
237B4355
SH NO. 04A
CONT ON SH 04AA
ENGINEER
PRINTS TO
REV 1
REV 2
ISSUED 05-16-88
MADE BY
ORIG


[03A17U ] (+) 05R
[03A33U ] (-) 25R
R47
R48
BRUSHLESS *
EXCITER
ROTOR
EXC
GENERATOR FIELD *
EXC. FIELD
OVEREXC. OPTION
59ER 125Ø
59E
59E 08A12H
B16
B17
R56
RR
FHM Ø-15ØVDC
R57
SHUNT 15H
25K
27H
27J
R54
R55
FAM
FLD GND UNIT "A" TRANSMITTER ±
FLD GND UNIT "B" RECEIVER ±
E D C B A
± SUPPLIED BY GE 'ST-G DEPT. NOT MOUNTED IN REGULATOR CABINET
34K
GROUND RELAY UNIC "C" CONTROL CENTER
1 2 3 4 5 6 7 8 9 10 11 12
N.O.
N.C.
R62
37M R61
38M R60
39M R58
40M R59
41M
CUSTOMER SUPPLY 120VAC
N.O. R63
COM 45M R64
N.C. 46M R65
47M
FIELD GROUND ALARM
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM EXCITER FIELD
237B4355 Ø4B
CONT ON SH Ø4BA
DWG NO Ø4AA
ENGINEER
FIRST TRADE FOR RECM.
REV 1
REV 2
REV 3
ORIG


BACKUP AUTO REGULATOR
43B
[08B29]
11J
TB2-6
43C3
TB2-7
43C2
3C1
SVR MODULE B
TB2-8
TB2-9
180V CLIPPER
F1
F2
TB2-3
TB2-4
SW10
12IR
125R
04B05B
04B25B
EXCITER FIELD
SW8
SW9
[03B03D]
03B03D
15E
[03B05D]
03B05D
17E
[02B39J]
02B39H
19E
[02B43J]
02B43H
23E
[02B47J]
02B47H
26E
TB2-1
TB2-2
TB1-1 AC1
TB1-2 AC2
TB1-3 AC3
4 REC
REGULATOR
URAL (OPTION)
R9 V/HZ GAIN
R8 V/HZ TAKEOVER
R7 DROOP
R5 LAG
R4 GAIN
R3 LEAD
R2C/L
BJ1 S RI CENTER
BJ2 S R2 RADIUS
BJ3 D R3 GAIN
BJ4 D BJ2 TIME CONSTANT
P15 TB3-1
DCOM TB3-2
N15 TB3-3
PT SENSING
CT SENSING
02B42T
02B44T
02B28E
02B24E
TB3-14 PT1
TB3-13 PT2
TB3-12 PT3
TB3-16 CT1
TB3-15 CT2
90P1 TB3-4
90P2 TB3-5
70P1 TB3-7
70P2 TB3-8
70P3 TB3-3
38Q
GND BUS
41J
41K
HA26
HA25
HA27
90PB
CW
1K
[09C15J] 1 MOTOR
(TWO AC REGULATOR OPTION)
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM
SVR II
237B4355
04A
03C
PRIMSTO
1688


R22
BRUSHLESS *
EXCITER
ROTOR
EXC
GENERATOR *
FIELD
EXC.
FIELD
59E
[03B20U ] (+)
[03C20U ] 05B
[11A05M ]
[11A15M ]
[03B25U ] (-)
[03C25U ] 25B
20A,100MV
SHUNT
1SH
25K
27H
27J
[11A41M 1
[11A31M 1
FLD GND
UNIT "A"
TRANS-
MITTER
FLD GND
UNIT "B"
RECEIVER
E D C B A
± SUPPLIED BY GE ST-G DEPT.
NOT MOUNTED IN REGULATOR
CABINET (MOUNTED IN
EXCITER HOUSING)
GND BUS
34K
GR
GROUND RELAY
UNIT "C"
CONTROL CENTER
N.O.
N.C.
R17
38MS R16
38M R15
39M R14
02B39H R13
02B43H
[11A47R ]
[11A47Q ]
N.O. R18
COM 45M R19
N.C. 46M R20
47M
FIELD GROUND
ALARM
(TWO AC REGULATOR OPTION)
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA, U.S.A.
ELEMENTARY
EXCITE
237B
05A
CONT. ON SH
ISSUED 05 16 88




GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA, U.S.A.

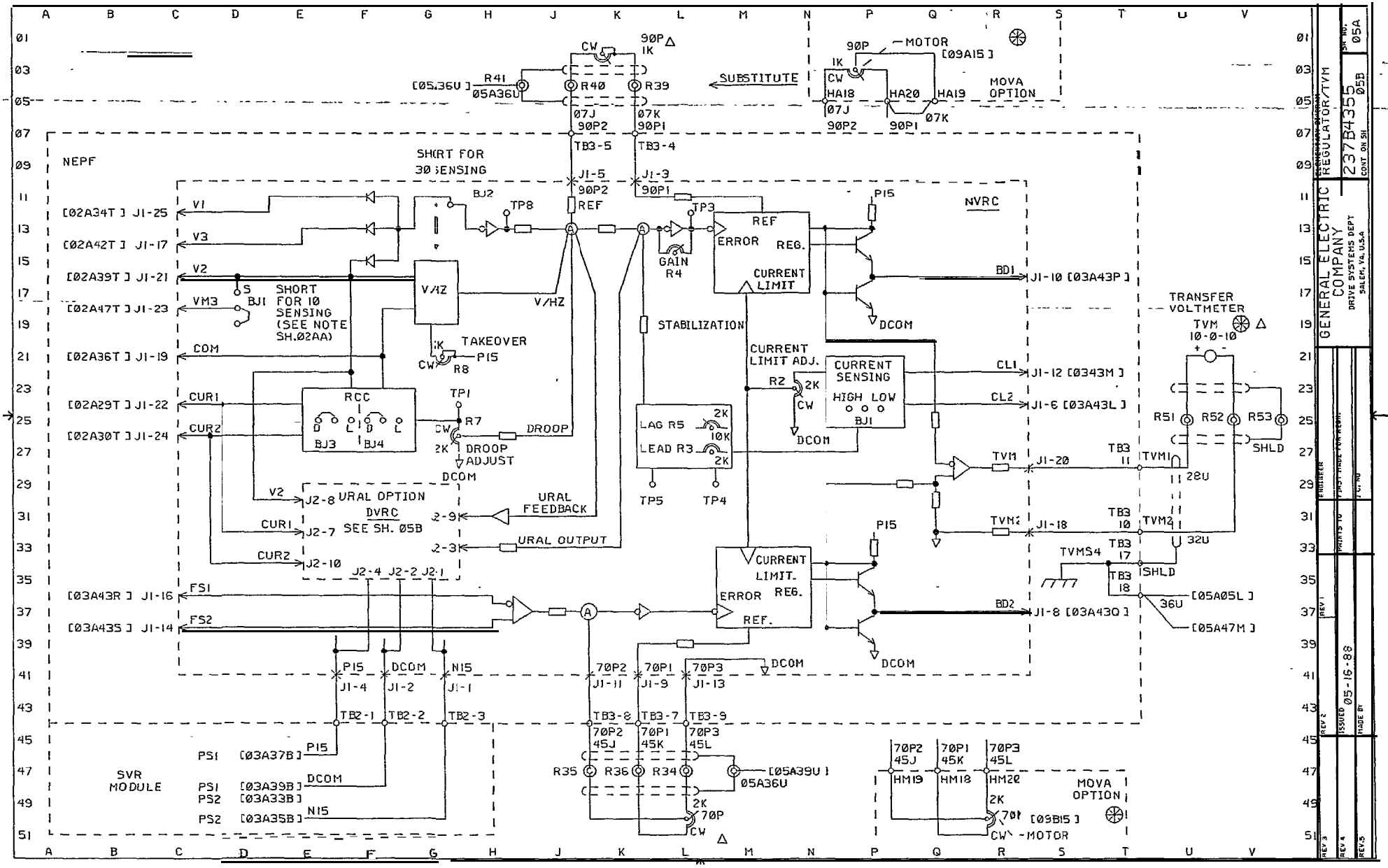
ELEMENTARY DIAGRAM
URAL OPTION

237B4355 05H

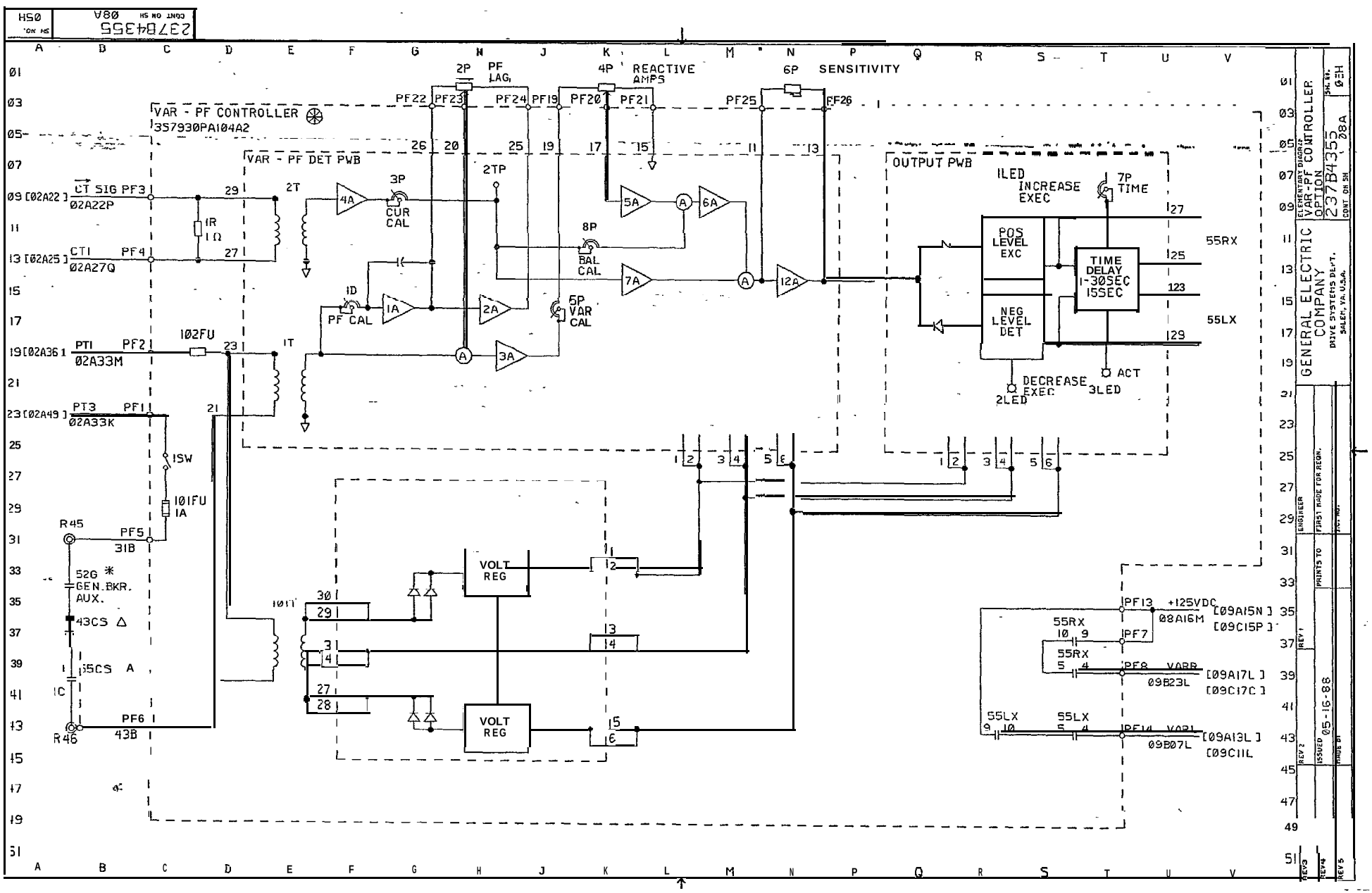
SH. NO 05B

CONT. ON SH 05H

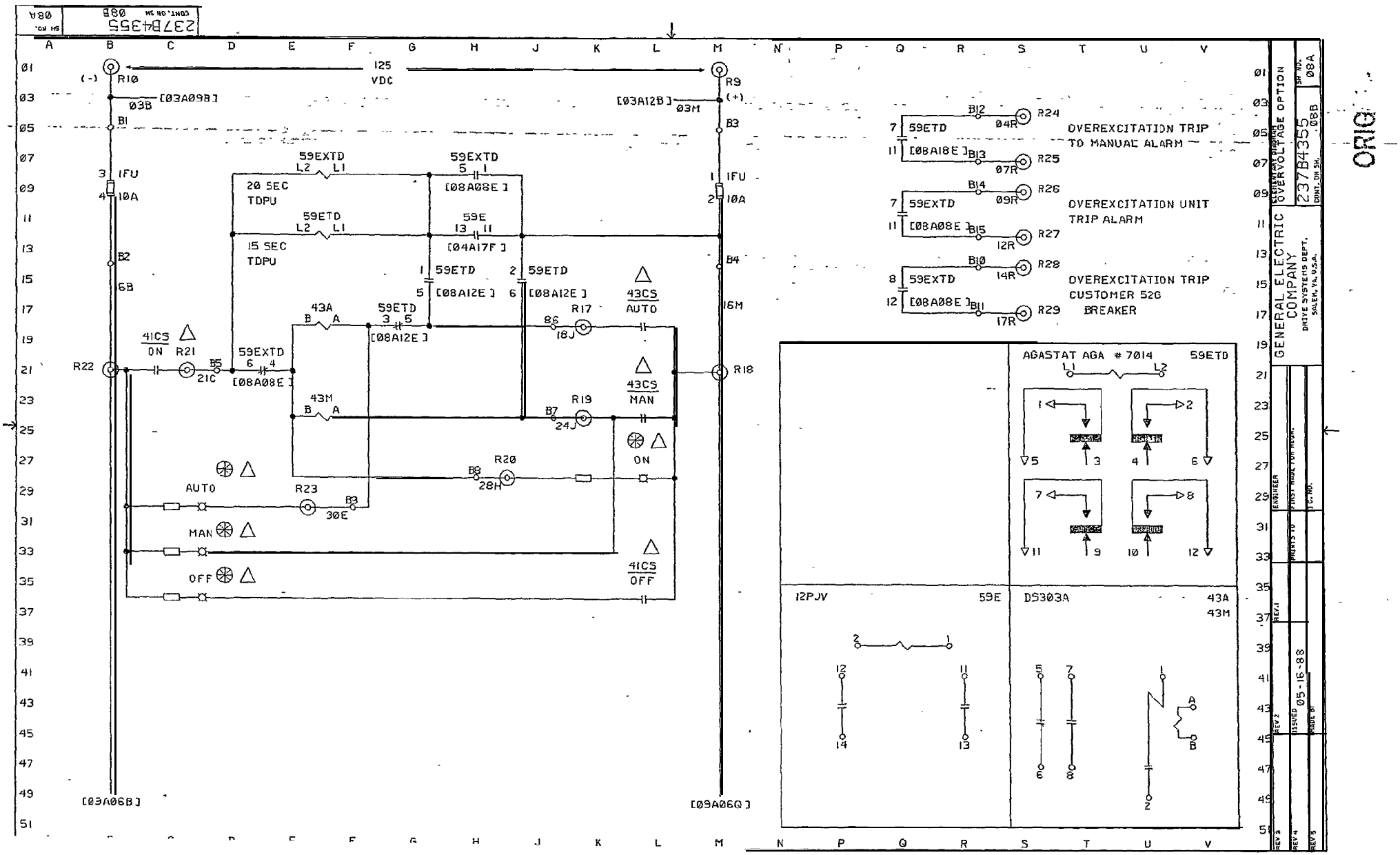
ORIG

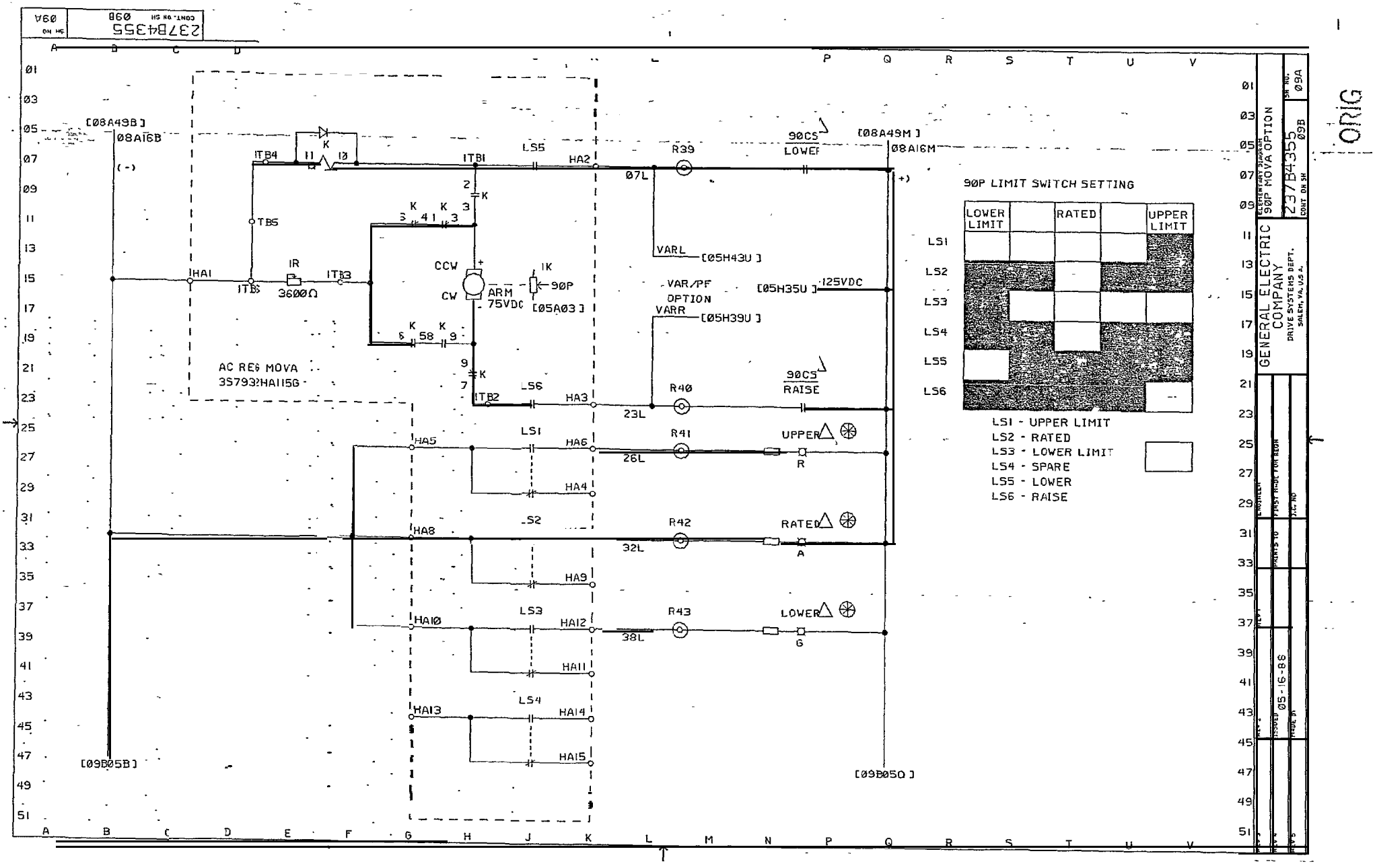

NEPF
SHORT FOR 3Ø SENSING
SUBSTITUTE
MOVA OPTION
MOTOR [09A15]
NVRC
REF ERROR REG.
CURRENT LIMIT
STABILIZATION
GAIN R4
CURRENT LIMIT ADJ.
CURRENT SENSING HIGH LOW
TRANSFER VOLTMETER
TVM 10-0-10
TAKEOVER
DROOP ADJUST
URAL OPTION
DVRC SEE SH. 05B
URAL FEEDBACK
URAL OUTPUT
LAG R5
LEAD R3
SHORT FOR 1Ø SENSING (SEE NOTE SH.02AA)
SVR MODULE
V1
V2
V3
VM3
COM
CUR1
CUR2
FS1
FS2
V/HZ
DROOP
DCOM
PI5
NI5
SHLD
36U
28U
32U
R1
R2
R3
R4
R5
R7
R8
R35
R36
R34
R39
R40
R41
R51
R52
R53
TP1
TP3
TP4
TP5
TP8
BJ1
BJ2
BJ3
BJ4
RCC
TB3-5
TB3-4
TB3-8
TB3-7
TB3-9
TB2-1
TB2-2
TB2-3
J1-5
J1-3
J1-4
J1-2
J1-1
J1-11
J1-9
J1-13
J2-8
J2-9
J2-3
J2-7
J2-10
J2-4
J2-2
J2-1
J1-25
J1-17
J1-21
J1-23
J1-19
J1-22
J1-24
J1-16
J1-14
J1-10 [03A43P]
J1-8 [03A43Q]
J1-12 [0343M]
J1-6 [03A43L]
J1-20
J1-18
[05A39U]
[05A36U]
[05A05L]
[05A47M]
[02A34T]
[02A42T]
[02A39T]
[02A47T]
[02A36T]
[02A29T]
[02A30T]
[03A43R]
[03A43S]
[03A37B]
[03A39B]
[03A33B]
[03A35B]
BD1
BD2
CL1
CL2
TVM
TVM1
TVM2
TVMS4
70P
90P
45J
45K
45L
HM19
HM18
HM20
HA18
HA19
HA20
07J
07K
90P1
90P2
IK
2K
10K
CW
REFERENCE DRAWING
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM, VA. U.S.A.
REGULATOR/TVM
237B4355
SHT NO. 05A
05B
CONT ON SH
ISSUED 05-16-88
MADE BY
ENGINEER
REV 1
REV 2
REV 3
REV 4
REV 5

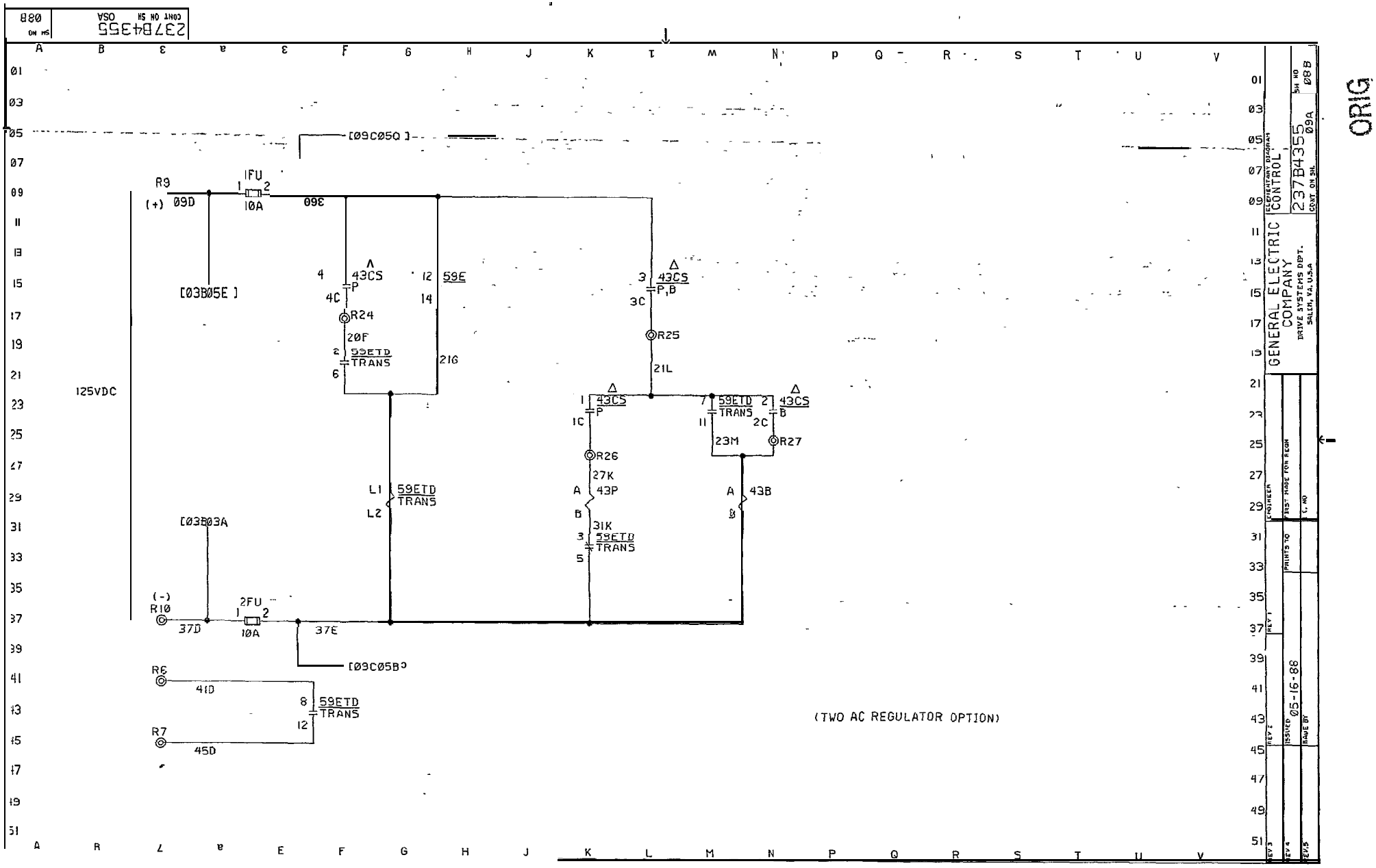
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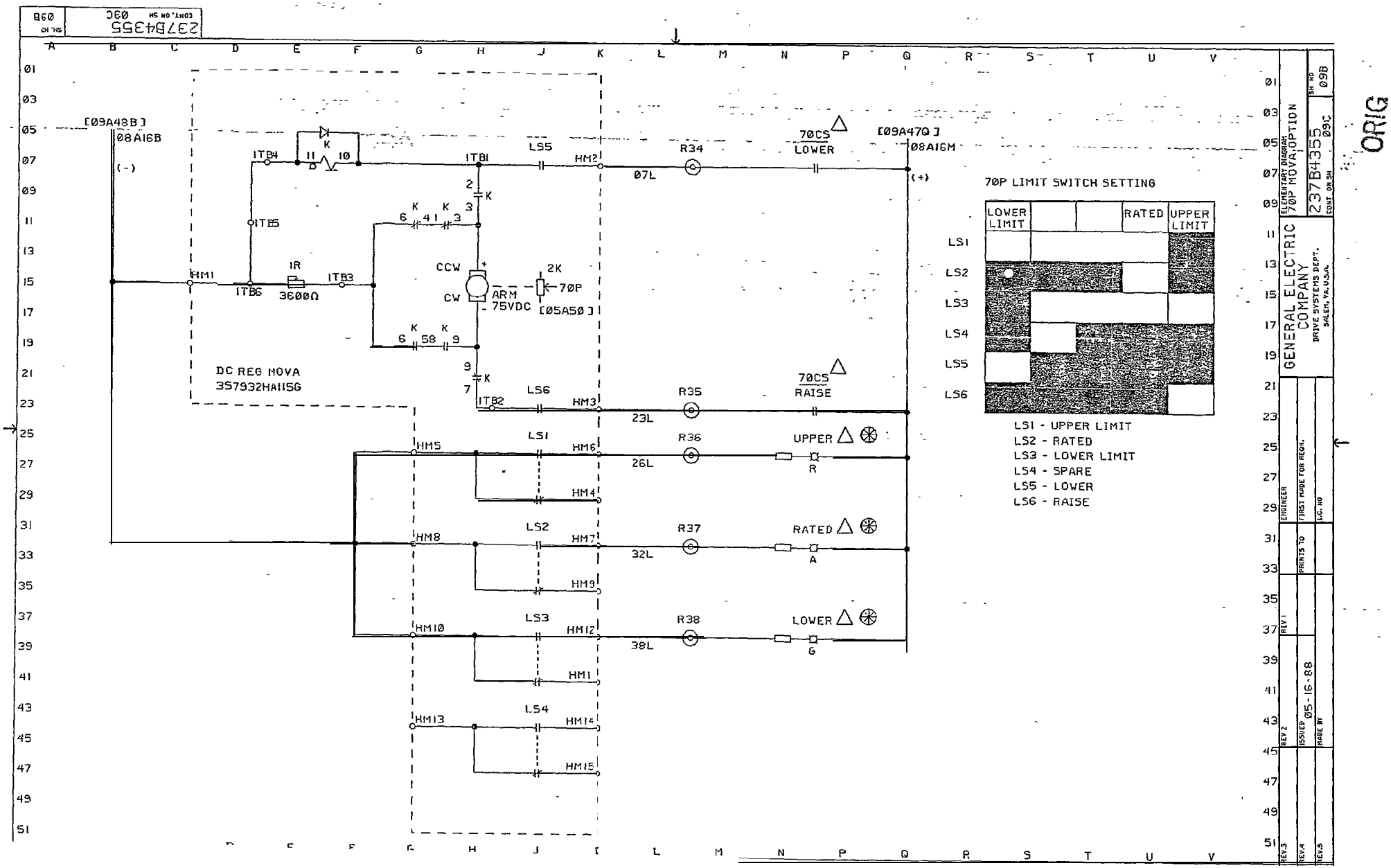

VAR - PF CONTROLLER
357930PA104A2
VAR - PF DET PWB
OUTPUT PWB
2P PF LAG
4P REACTIVE AMPS
6P SENSITIVITY
CT SIG PF3
02A22P
CT1 PF4
02A27Q
PT1 PF2
02A33M
PT3 PF1
02A33K
CUR CAL
PF CAL
BAL CAL
VAR CAL
ILED INCREASE EXEC
DECREASE EXEC
2LED
3LED
ACT
7P TIME
POS LEVEL EXC
NEG LEVEL DET
TIME DELAY 1-30SEC 15SEC
55RX
55LX
VOLT REG
VOLT REG
52G GEN.BKR. AUX.
43CS
65CS
R45
R46
PF5
PF6
31B
43B
PF13 +125VDC
08A16M
PF7
PF8 VARR
09B23L
PF14 VAR1
09B07L
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM
VAR-PF CONTROLLER OPTION
237B4355
05-16-88

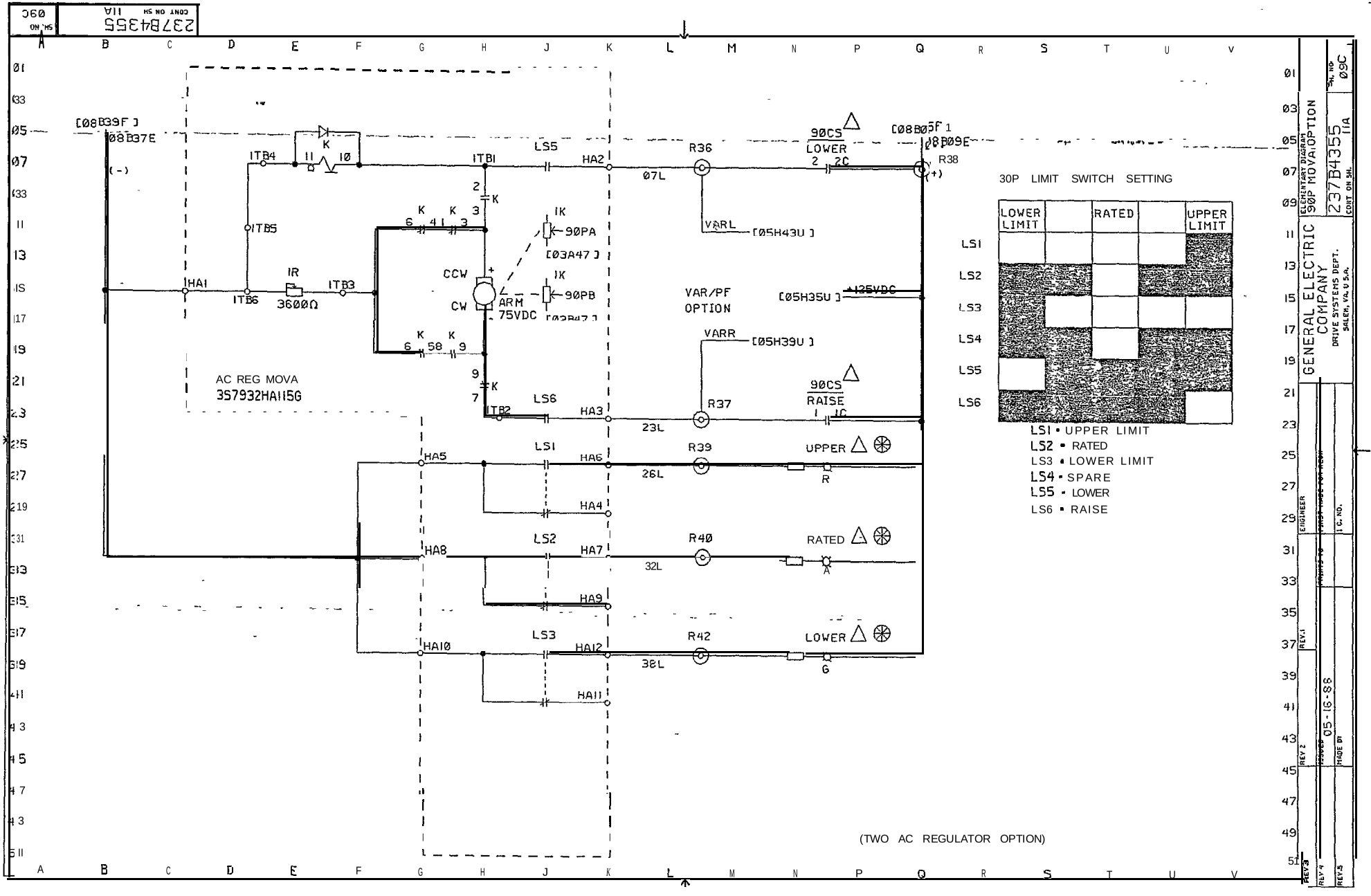
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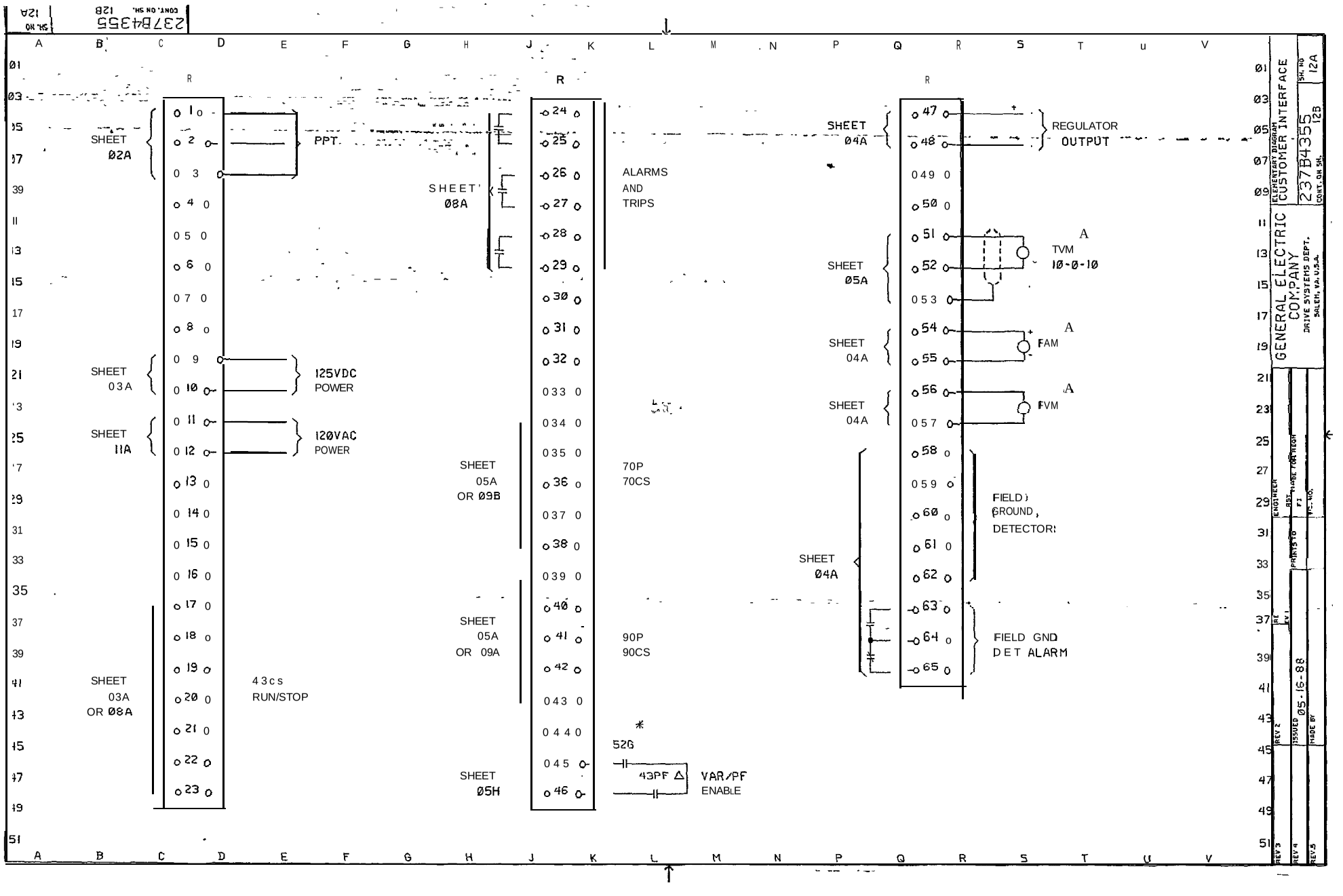

237B4355
08A
08B
125 VDC
(-)
(+)
R10
R9
[03A09B]
[03A12B]
03B
03M
B1
B3
1FU
10A
59EXTD
L2 L1
20 SEC TDPU
15 SEC TDPU
59ETD
59E
[08A08E]
[04A17F]
[08A12E]
[08A18E]
[08A08E]
B2
B4
6B
16M
43A
43M
43CS AUTO
43CS MAN
41CS ON
41CS OFF
R21
R22
R17
R18
R19
R20
R23
18J
24J
28H
30E
21C
AUTO
MAN
OFF
ON
[03A06B]
[09A06Q]
R24
R25
R26
R27
R28
R29
04R
07R
09R
12R
14R
17R
OVEREXCITATION TRIP TO MANUAL ALARM
OVEREXCITATION UNIT TRIP ALARM
OVEREXCITATION TRIP CUSTOMER 52G BREAKER
AGASTAT AGA # 7014
12PJV
DS303A
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA, U.S.A.
ELECTRICAL DIAGRAM OVERVOLTAGE OPTION
237B4355
ISSUED 05-16-88
ORIG

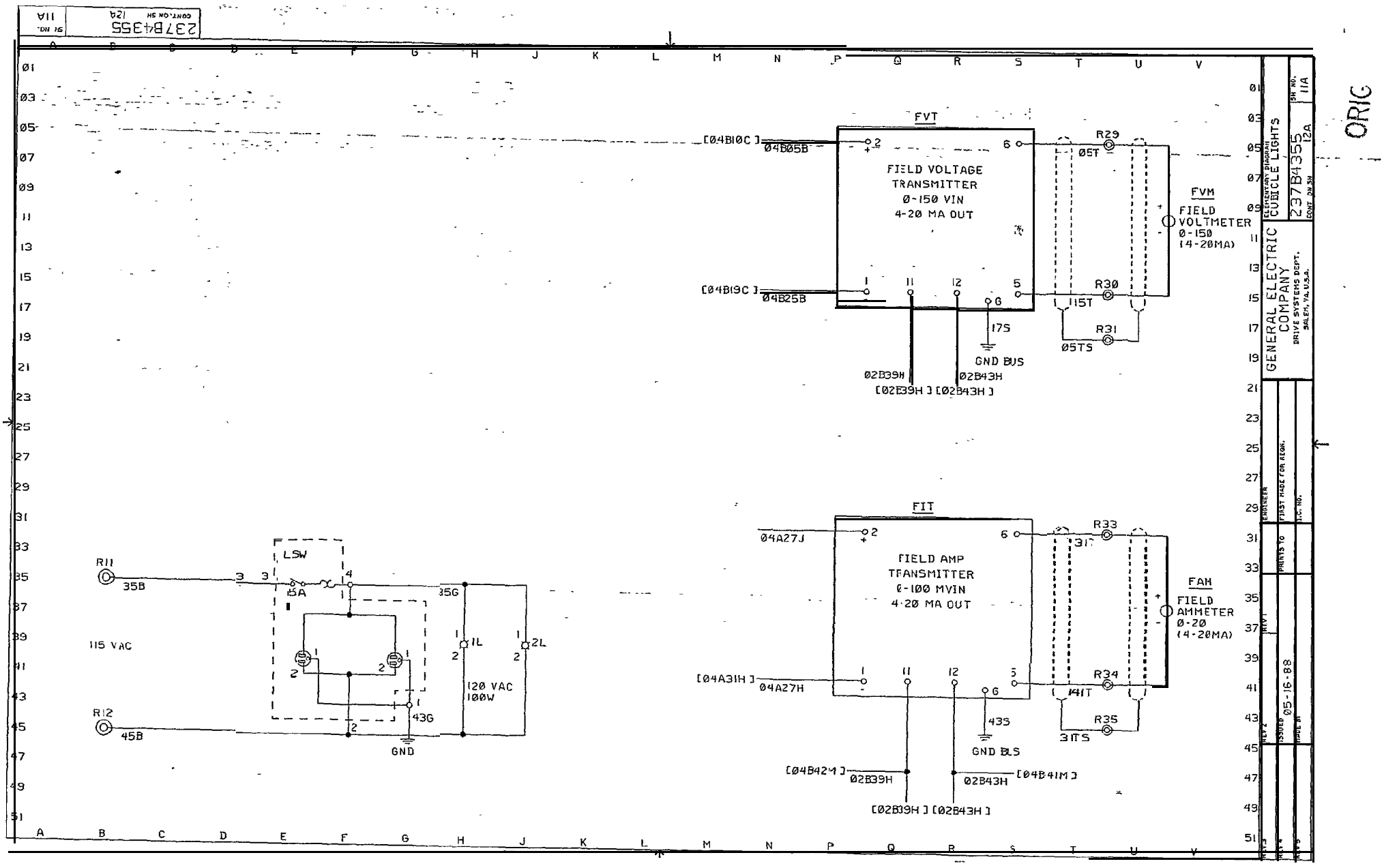

90P LIMIT SWITCH SETTING
LOWER LIMIT
RATED
UPPER LIMIT
LS1
LS2
LS3
LS4
LS5
LS6
LS1 - UPPER LIMIT
LS2 - RATED
LS3 - LOWER LIMIT
LS4 - SPARE
LS5 - LOWER
LS6 - RAISE
AC REG MOVA
3S793:HA115G
[08A49B]
08A16B
[08A49M]
08A16M
90CS
LOWER
90CS
RAISE
R39
R40
R41
R42
R43
07L
23L
26L
32L
38L
VARL
[05H43U]
VAR/PF
OPTION
VARR
[05H39U]
[05H35U]
125VDC
75VDC
[05A03]
90P
3600Ω
UPPER
RATED
LOWER
[09B05B]
[09B05Q]
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA, U.S.A.
90P MOVA OPTION
237B4355
09A
09B
05-16-88

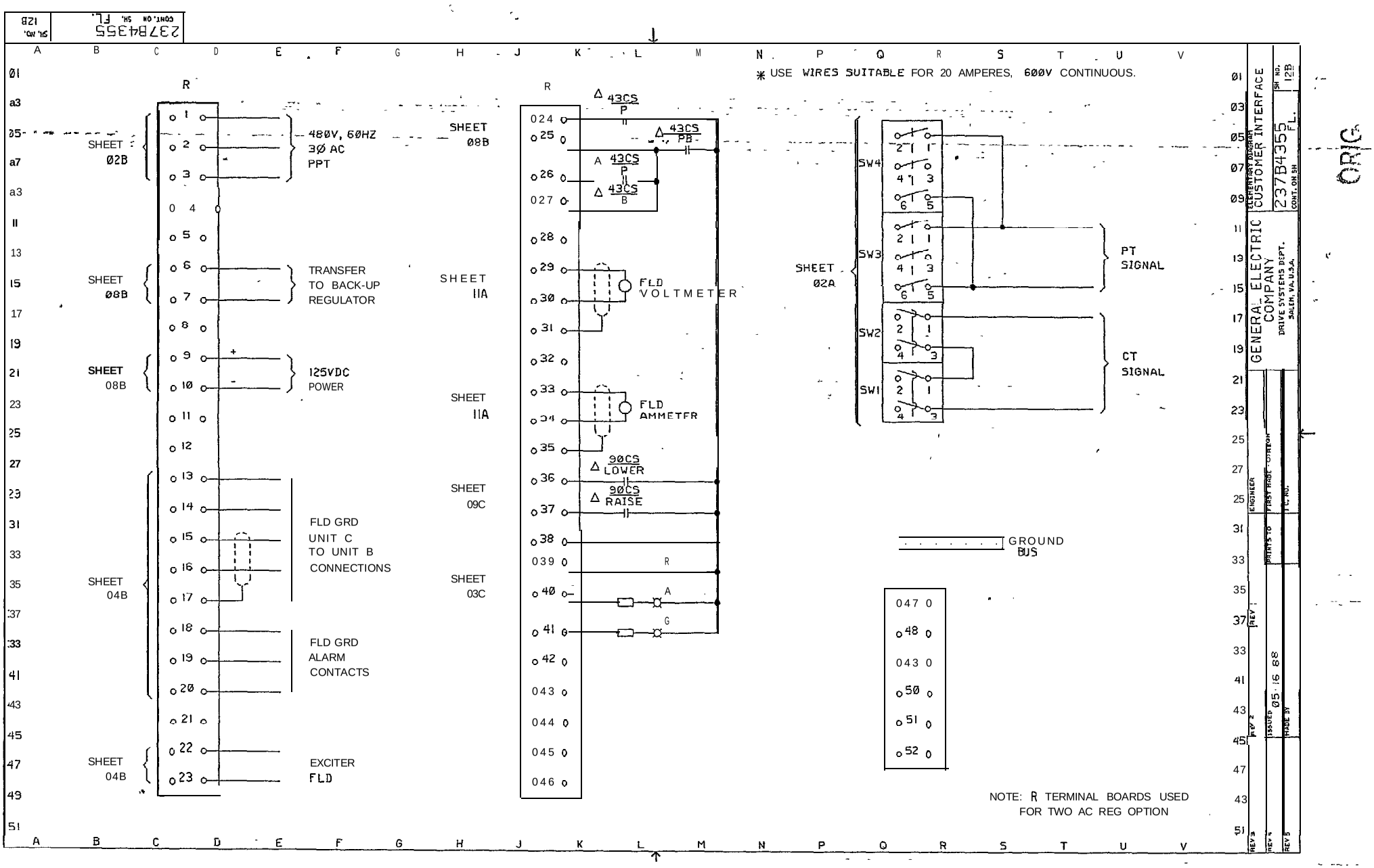

237B4355 OSA
CONT ON SH
SH NO 08B
[09C05Q ]
R9
1FU
(+) 09D
10A
09E
[03B05E ]
4
43CS
P
4C
R24
20F
59ETD
TRANS
6
12
59E
14
21G
125VDC
3
43CS
P,B
3C
R25
21L
1
43CS
P
1C
R26
27K
A
43P
B
31K
3
59ETD
TRANS
5
7
59ETD
TRANS
11
2
43CS
B
2C
23M
R27
A
43B
B
L1
59ETD
TRANS
L2
[03B03A
(-)
R10
37D
2FU
10A
37E
[09C05B ]
R6
41D
8
59ETD
TRANS
12
R7
45D
(TWO AC REGULATOR OPTION)
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA, U.S.A
237B4355
09A
SH NO 08B
CONT ON SH
ENGINEER
PRINTS TO
DRAWN BY
ISSUED 05-16-88
REV 1
REV 2
REV 3
REV 4
REV 5
ORIG

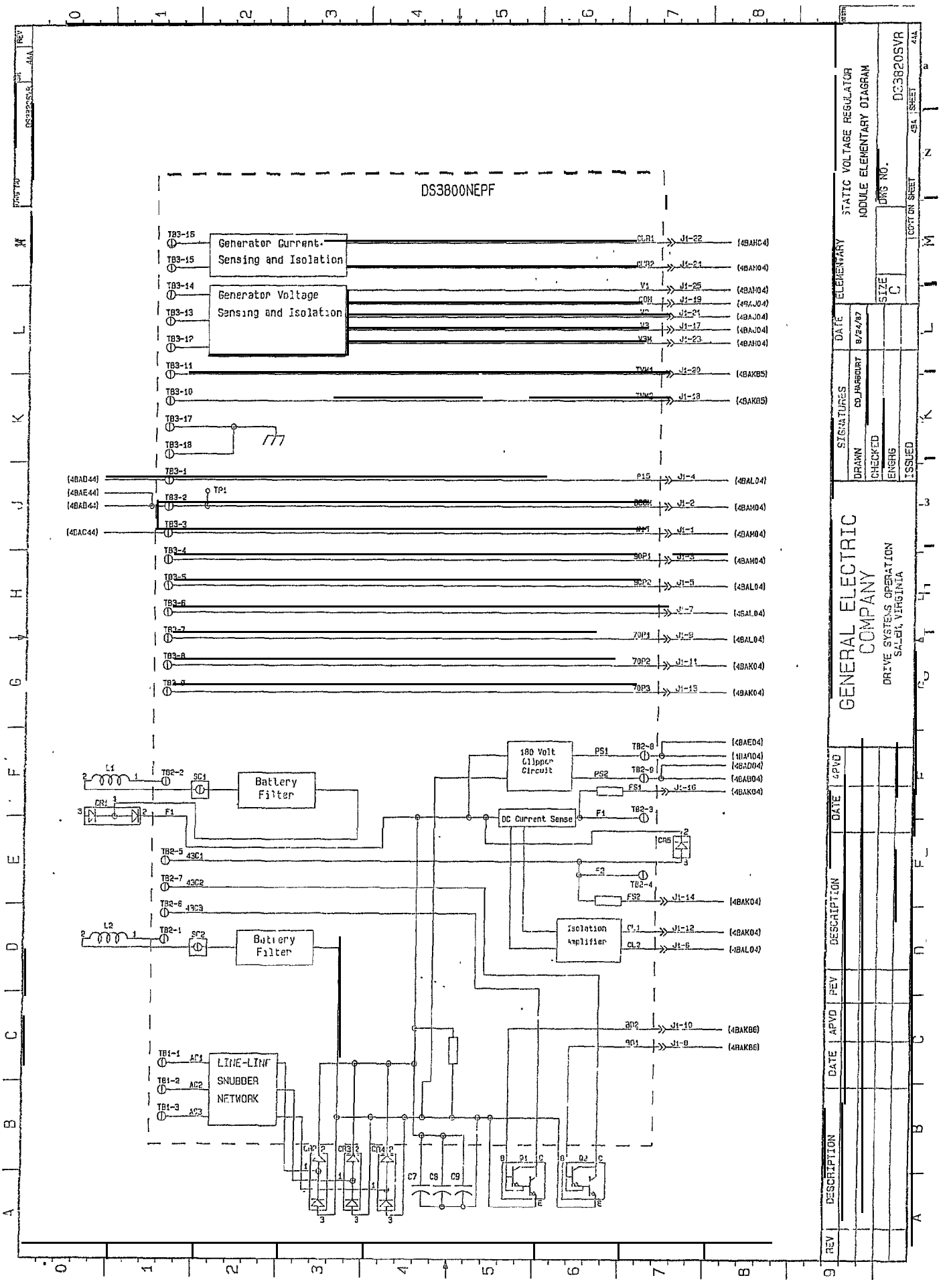

[09A48B]
08A16B
(-)
ITB4
K
11
10
ITB1
LS5
HM2
R34
70CS
LOWER
[09A47Q]
08A16M
(+)
07L
2
K
ITB5
K
6
4
K
1
3
3
HM1
IR
ITB3
ITB6
3600Ω
CCW
CW
ARM
75VDC
2K
70P
[05A50]
K
6
58
K
9
9
K
7
DC REG MOVA
3S7932HA115G
ITB2
LS6
HM3
R35
70CS
RAISE
23L
HM5
LS1
HM6
R36
UPPER
26L
R
HM4
HM8
LS2
HM7
R37
RATED
32L
A
HM9
HM10
LS3
HM12
R38
LOWER
38L
G
HM11
HM13
LS4
HM14
HM15
70P LIMIT SWITCH SETTING
LOWER LIMIT
RATED
UPPER LIMIT
LS1
LS2
LS3
LS4
LS5
LS6
LS1 - UPPER LIMIT
LS2 - RATED
LS3 - LOWER LIMIT
LS4 - SPARE
LS5 - LOWER
LS6 - RAISE
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA.U.S.A.
ELEMENTARY DIAGRAM
70P MOVA;OPTION
237B4355
SH NO 09B
CONT. ON SH 09C
ENGINEER
PRINTS TO
J.O. NO
REV 1
REV 2
ISSUED 05-16-88
MADE BY
REV 3
REV 4
REV 5
ORIG

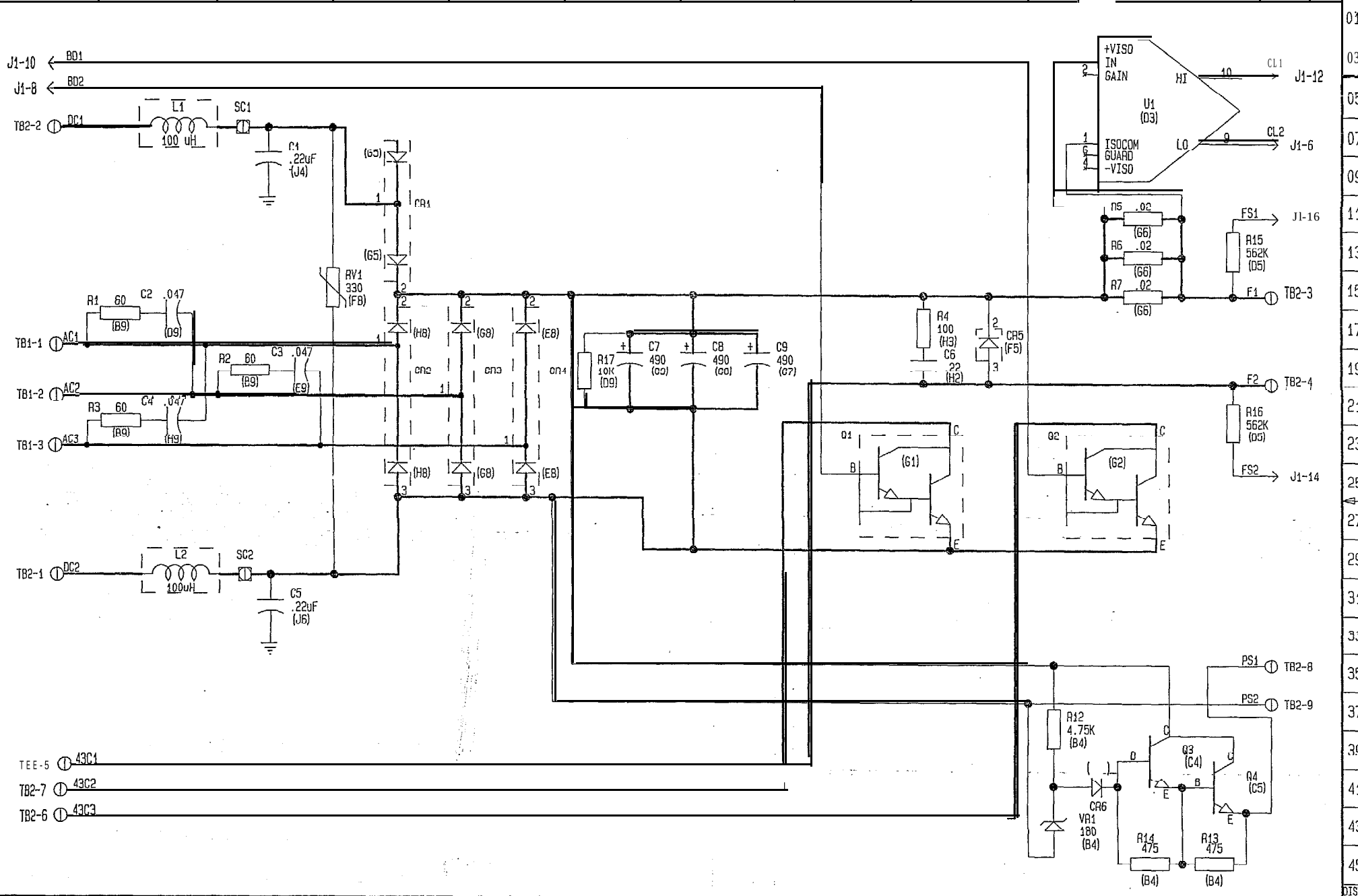

237B4355 11A
CONT ON SH
09C
[08B39F]
08B37E
(-)
K
ITB4
11
10
ITB1
LS5
HA2
07L
R36
90CS
LOWER
2
2C
[08B05F]
08B09E
R38
(+)
2
K
K 6 4 K 3
IK
90PA
[03A47]
ITB5
VARL
[05H43U]
HA1
IR
3600Ω
ITB6
ITB3
CCW
CW
ARM
75VDC
IK
90PB
[03B47]
VAR/PF
OPTION
[05H35U]
+125VDC
K 6 58 K 9
VARR
[05H39U]
AC REG MOVA
3S7932HA115G
9
7
K
ITB2
LS6
HA3
23L
R37
90CS
RAISE
1
1C
HA5
LS1
HA6
26L
R39
UPPER
R
HA4
HA8
LS2
HA7
32L
R40
RATED
A
HA9
HA10
LS3
HA12
38L
R42
LOWER
G
HA11
30P LIMIT SWITCH SETTING
LOWER LIMIT
RATED
UPPER LIMIT
LS1
LS2
LS3
LS4
LS5
LS6
LS1 • UPPER LIMIT
LS2 • RATED
LS3 • LOWER LIMIT
LS4 • SPARE
LS5 • LOWER
LS6 • RAISE
(TWO AC REGULATOR OPTION)
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM
90P MOVA OPTION
237B4355
11A
09C
CONT ON SH.
ENGINEER
DWG. NO.
REV.1
REV.2
REV.3
REV.4
REV.5
ISSUED 05-16-88
MADE BY


SHEET 02A
PPT.
SHEET 08A
ALARMS AND TRIPS
SHEET 04A
REGULATOR OUTPUT
SHEET 05A
TVM 10-0-10
FAM
FVM
SHEET 03A
125VDC POWER
SHEET 11A
120VAC POWER
SHEET 05A OR 09B
70P 70CS
SHEET 04A
FIELD GROUND DETECTOR
SHEET 05A OR 09A
90P 90CS
FIELD GND DET ALARM
SHEET 03A OR 08A
43cs RUN/STOP
526
43PF
VAR/PF ENABLE
SHEET 05H
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM CUSTOMER INTERFACE
237B4355 12B
CONT. ON SH.
SH. NO 12A
ISSUED 05-16-88
REV 2
REV 3
REV 4
REV 5


237B4355 CONT. ON SH 124 SH NO. 11A
FVT
[04B10C]
04B05B
FIELD VOLTAGE TRANSMITTER 0-150 VIN 4-20 MA OUT
[04B19C]
04B25B
02B39H
[02B39H]
02B43H
[02B43H]
17S
GND BUS
R29
05T
R30
15T
R31
05TS
FVM
FIELD VOLTMETER 0-150 (4-20MA)
FIT
04A27J
FIELD AMP TRANSMITTER 0-100 MVIN 4-20 MA OUT
[04A31H]
04A27H
43S
GND BUS
R33
31T
R34
41T
R35
31TS
FAM
FIELD AMMETER 0-20 (4-20MA)
[04B42M]
02B39H
[02B39H]
02B43H
[02B43H]
[04B41M]
LSW
R11
35B
15A
35G
115 VAC
1L
2L
120 VAC 100W
R12
45B
43G
GND
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA, U.S.A.
CUBICLE LIGHTS
237B4355
SH NO. 11A
CONT ON SH 12A
ISSUED 05-16-88
ORIG


* USE WIRES SUITABLE FOR 20 AMPERES, 600V CONTINUOUS.
R
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
480V, 60HZ 3Ø AC PPT
SHEET 08B
SHEET 02B
TRANSFER TO BACK-UP REGULATOR
SHEET 08B
SHEET 11A
125VDC POWER
SHEET 08B
SHEET 11A
FLD GRD UNIT C TO UNIT B CONNECTIONS
SHEET 09C
SHEET 04B
SHEET 03C
FLD GRD ALARM CONTACTS
EXCITER FLD
SHEET 04B
R
024
025
026
027
028
029
030
031
032
033
034
035
036
037
038
039
040
041
042
043
044
045
046
Δ 43CS P
Δ 43CS PB
A 43CS P
Δ 43CS B
FLD VOLTMETER
FLD AMMETER
Δ 90CS LOWER
Δ 90CS RAISE
R
A
G
SHEET 02A
SW4
SW3
SW2
SW1
PT SIGNAL
CT SIGNAL
GROUND BUS
047
048
043
050
051
052
NOTE: R TERMINAL BOARDS USED FOR TWO AC REG OPTION
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT. SALEM, VA, U.S.A.
ELEMENTARY DIAGRAM
CUSTOMER INTERFACE
237B4355
SH. NO. 12B
CONT. ON SH.
FL.
ENGINEER
FIRST PROD. OPTION
P.T.C. NO.
REV.
ISSUED 05-16-88
MADE BY
REV 2
REV 3
REV 4
REV 5
237B4355 FL.
CONT. ON SH.
SH. NO. 12B
ORIG


DS3800NEPF
Generator Current Sensing and Isolation
Generator Voltage Sensing and Isolation
Battery Filter
Battery Filter
180 Volt Clipper Circuit
DC Current Sense
Isolation Amplifier
LINE-LINE SNUBBER NETWORK
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS OPERATION
SALEM, VIRGINIA
STATIC VOLTAGE REGULATOR
MODULE ELEMENTARY DIAGRAM
DS3820SVR




| REV | DESCRIPTION | DATE | APPR | REV | DESCRIPTION | DATE | APPR |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS OPERATION
SALEM, VIRGINIA

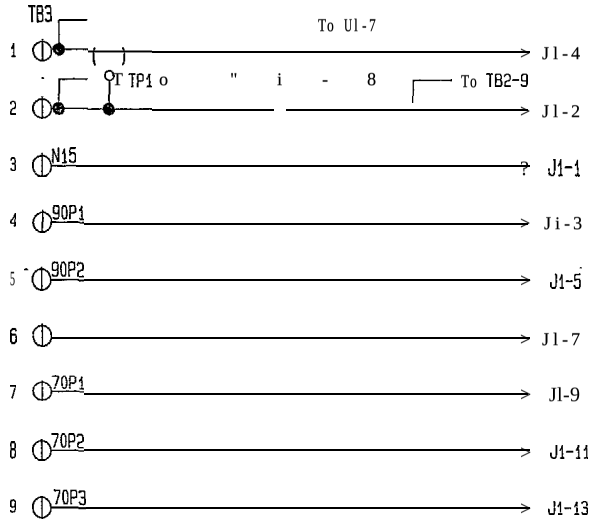
| SIGNATURES | DATE |
|---|---|
| DRAWN | 9/11/87 |
| CHECKED | 9/11/87 |
| ENGRG | 9/11/87 |
| ISSUED | 9/15/87 |

ELEMENTARY
STATIC VOLTAGE REGULATOR
EXCITATION POWER BOARD

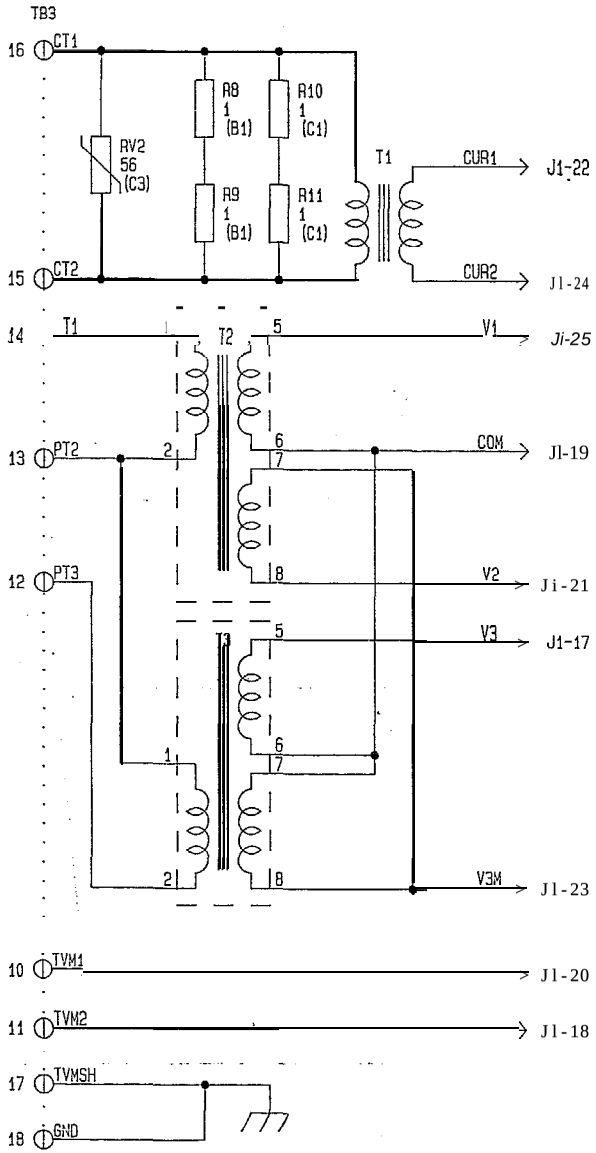
SIZE C | DWG NO DS3800NEPF

CONT ON SHEET 48A | SHEET 4AA

DISTR DL104




When single phase sensing is used, short TB3-12 and TB3-13 (PT2 to PT3).




| REV | DESCRIPTION | DATE | APPR | REV | DESCRIPTION | DATE | APPR |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS OPERATION
SALEM, VIRGINIA

| SIGNATURES | DATE |
|---|---|
| DRAWN M Reid | 9/16/87 |
| CHECKED M Reid | 9/16/87 |
| ENGRG Gyan Upadhyay | 9/11/87 |
| ISSUED M Reid | 9/16/87 |

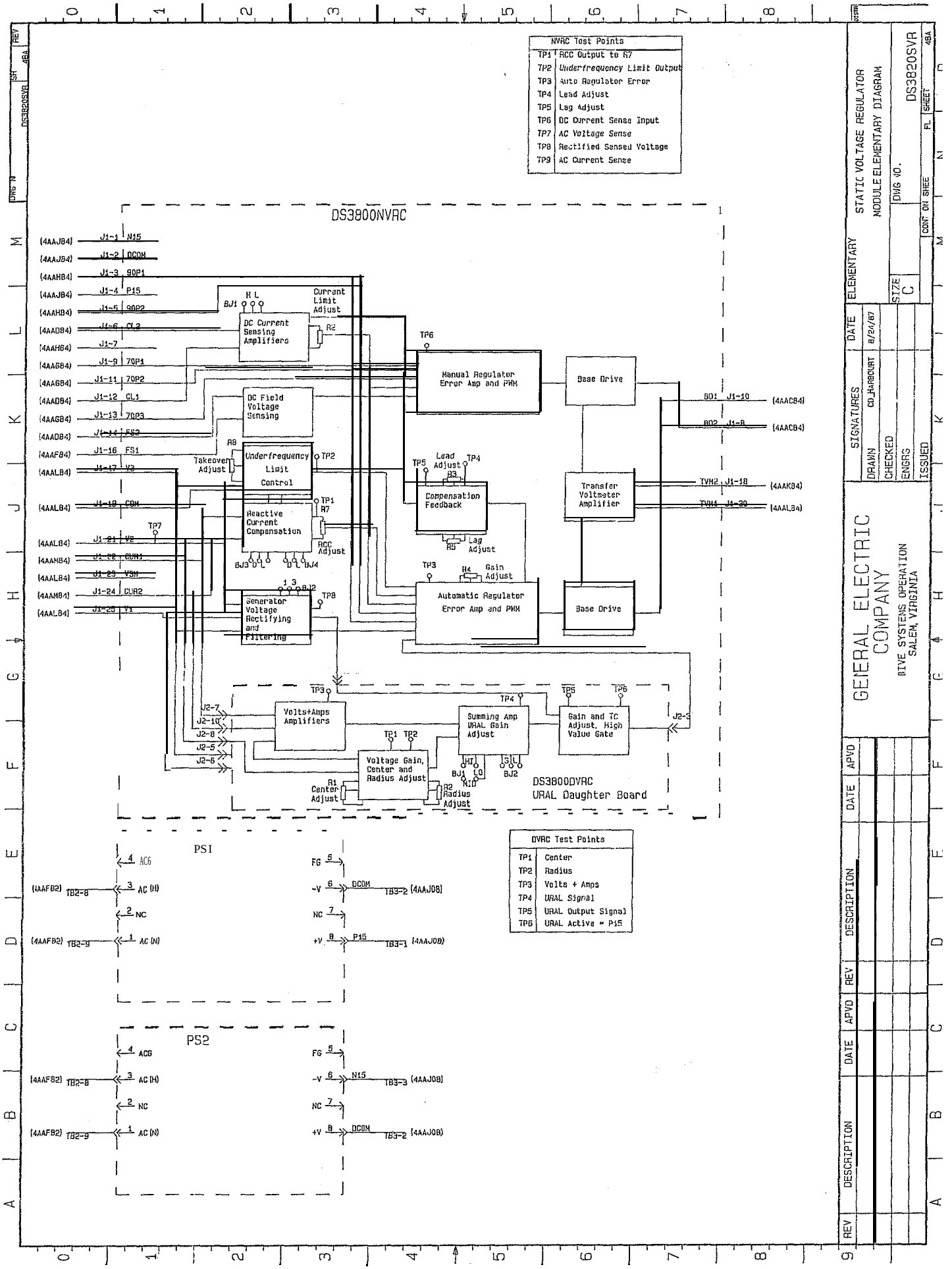
ELEMENTARY

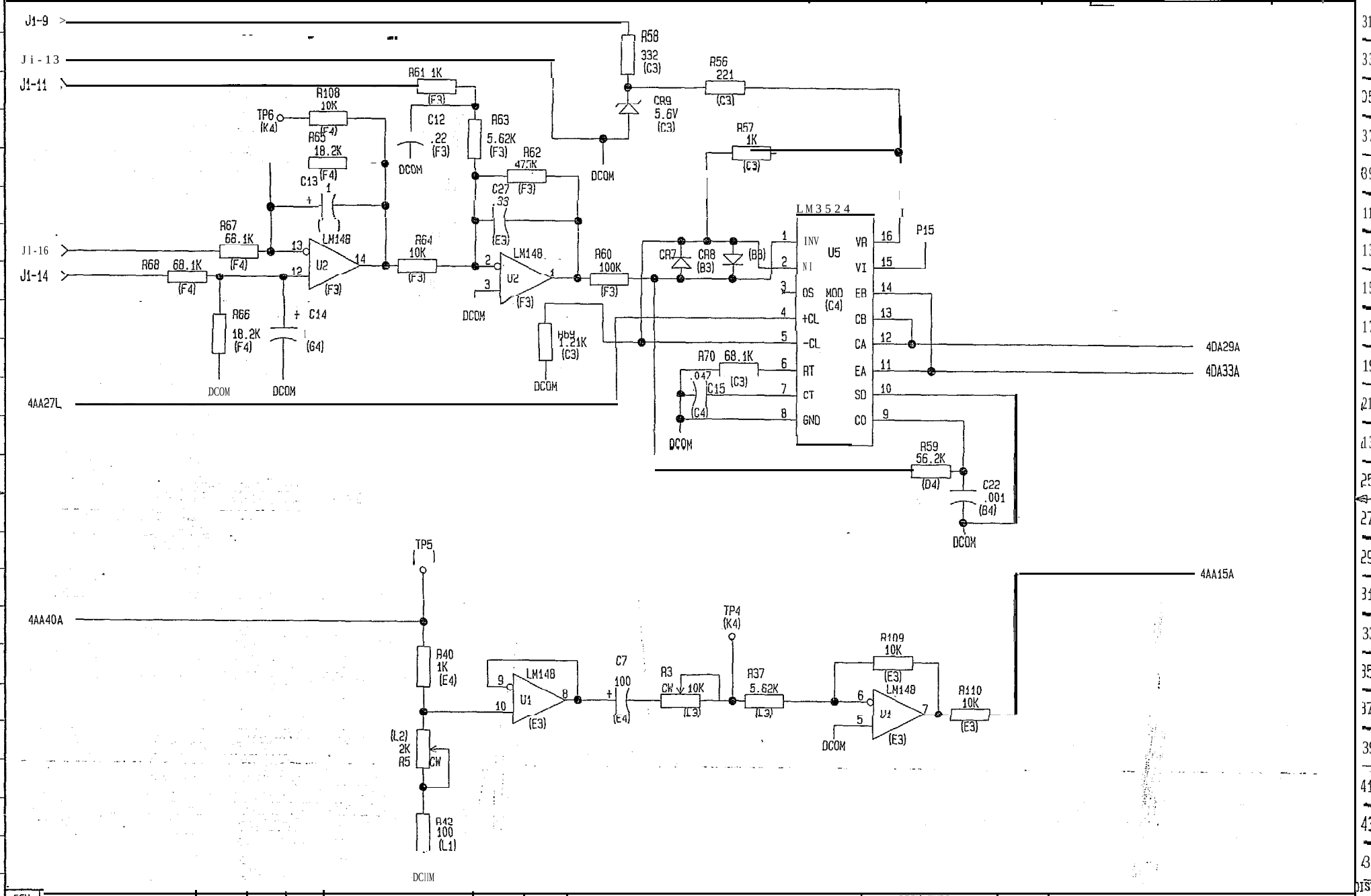
STATIC VOLTAGE REGULATOR
EXCITATION POWER BOARD

SIZE C | DWG NO DS3800NEPF | CONT ON SHEET 4CA | SHEET 4BA

DISTR

DL404


NVRC Test Points
TP1 ACC Output to R7
TP2 Underfrequency Limit Output
TP3 Auto Regulator Error
TP4 Lead Adjust
TP5 Lag Adjust
TP6 DC Current Sense Input
TP7 AC Voltage Sense
TP8 Rectified Sensed Voltage
TP9 AC Current Sense
DS3800NVRC
DC Current Sensing Amplifiers
Current Limit Adjust
DC Field Voltage Sensing
Underfrequency Limit Control
Takeover Adjust
Reactive Current Compensation
ACC Adjust
Generator Voltage Rectifying and Filtering
Manual Regulator Error Amp and PWM
Base Drive
Compensation Feedback
Lead Adjust
Lag Adjust
Transfer Voltmeter Amplifier
Automatic Regulator Error Amp and PWM
Gain Adjust
Base Drive
Volts+Amps Amplifiers
Summing Amp URAL Gain Adjust
Gain and TC Adjust, High Value Gate
Voltage Gain, Center and Radius Adjust
Center Adjust
Radius Adjust
DS3800DVRC
URAL Daughter Board
PSI
PS2
DVRC Test Points
TP1 Center
TP2 Radius
TP3 Volts + Amps
TP4 URAL Signal
TP5 URAL Output Signal
TP6 URAL Active = P15
STATIC VOLTAGE REGULATOR
MODULE ELEMENTARY DIAGRAM
DS3820SVR
ELEMENTARY
GENERAL ELECTRIC COMPANY
RIVE SYSTEMS OPERATION
SALEM, VIRGINIA




DCIIM

| REV | DESCRIPTION | DATE | APPR | REV | DESCRIPTION | DATE | APPR |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

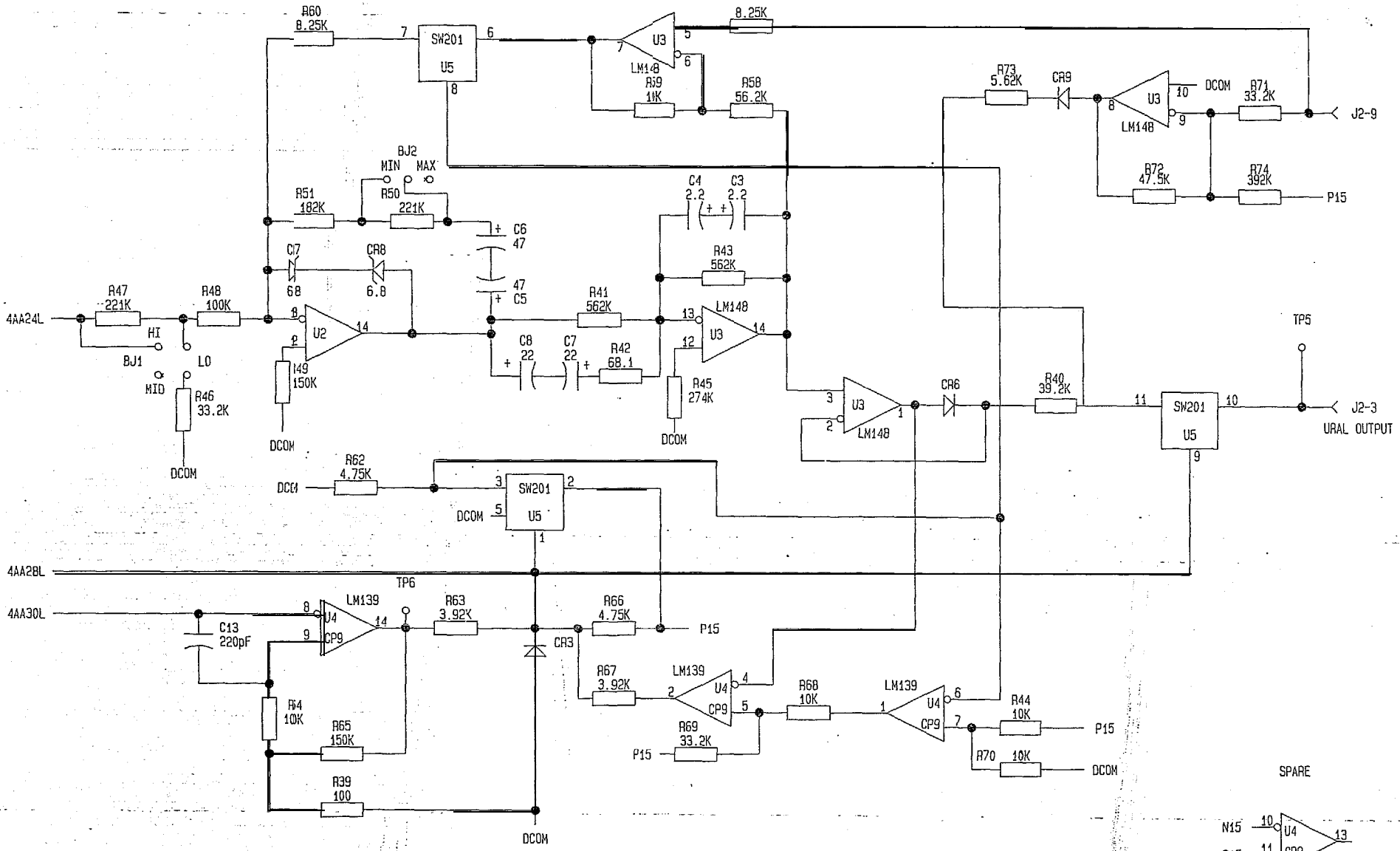
**GENERAL ELECTRIC COMPANY**
DRIVE SYSTEMS OPERATION
SALEM, VIRGINIA

| SIGNATURES | DATE |
|---|---|
| DRAWN | 4/16/87 |
| CHECKED | 4/16/87 |
| ENGRG | 4/16/87 |
| ISSUED | 4/16/87 |

ELEMENTARY — STATIC VOLTAGE REGULATOR
AUTO/MANUAL REGULATOR ELEMENTARY

SIZE C — DWG NO DS3800NVRC

CONT ON SHEET 4CA — SHEET 4BA




| REV | DESCRIPTION | DATE | APPR | REV | DESCRIPTION | DATE | APPR |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS OPERATION
SALEM, VIRGINIA

| SIGNATURES | | DATE |
|---|---|---|
| DRAWN | | 9/16/87 |
| CHECKED | | 9/16/87 |
| ENGRG | Lynn Hubbard | 9/16/87 |
| ISSUED | | 9/16/87 |

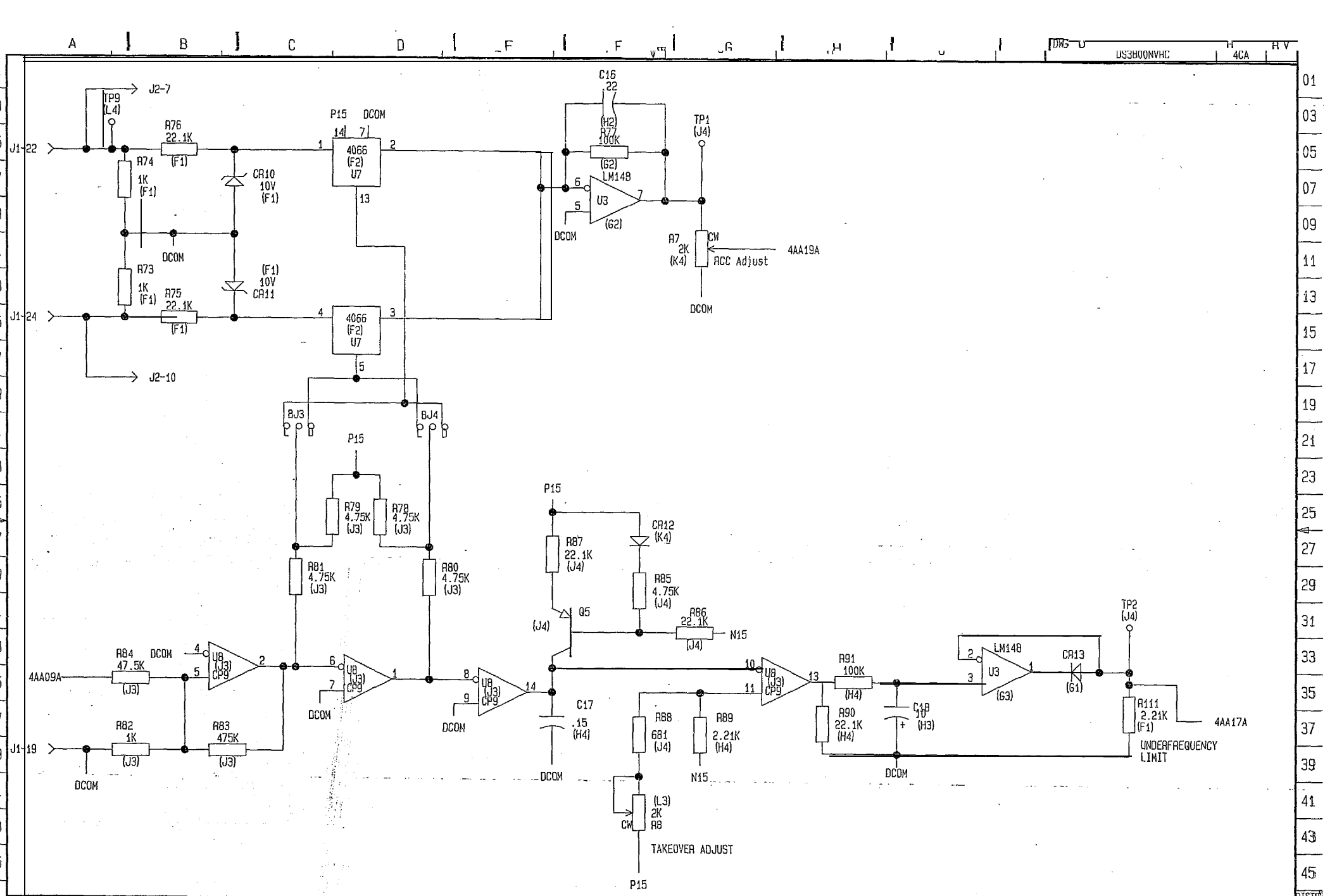
ELEMENTARY — STATIC VOLTAGE REGULATOR
URAL ELEMENTARY

SIZE C | DWG NO DS3800DVRC

CONT ON SHEET FL | SHEET 4BA

DISTR

OL104




| REV | DESCRIPTION | DATE | APPR | REV | DESCRIPTION | DATE | APPR |
|---|---|---|---|---|---|---|---|
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GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS OPERATION
SALEM, VIRGINIA

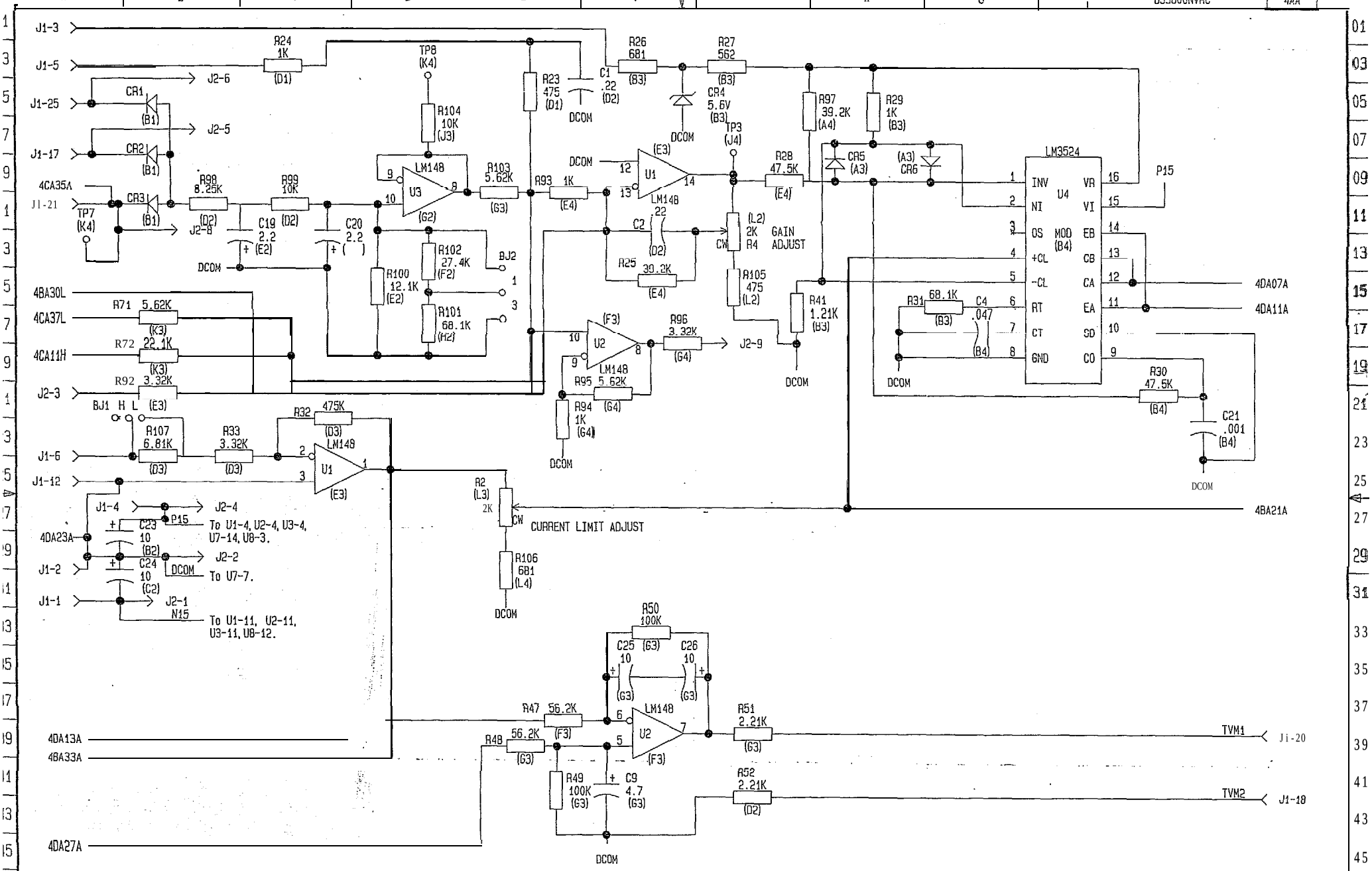
| SIGNATURES | DATE |
|---|---|
| DRAWN | 9/16/91 |
| CHECKED | 9/16/91 |
| ENGRG | 9/11/91 |
| ISSUED | 9/11/91 |

ELEMENTARY
STATIC VOLTAGE REGULATOR
AUTO/MANUAL REGULATOR ELEMENTARY

SIZE C | DWG NO DS3800NVRC

CONT ON SHEET 4DA | SHEET 4CA

DIST 0104




| REV | DESCRIPTION | DATE | APPR | REV | DESCRIPTION | DATE | APPR |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
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| | | | | | | | |

**GENERAL ELECTRIC COMPANY**
DRIVE SYSTEMS OPERATION
SALEM, VIRGINIA

| SIGNATURES | DATE |
|---|---|
| DRAWN | 9/16/87 |
| CHECKED | 9/16/87 |
| ENGRG | |
| ISSUED | 9/16/87 |

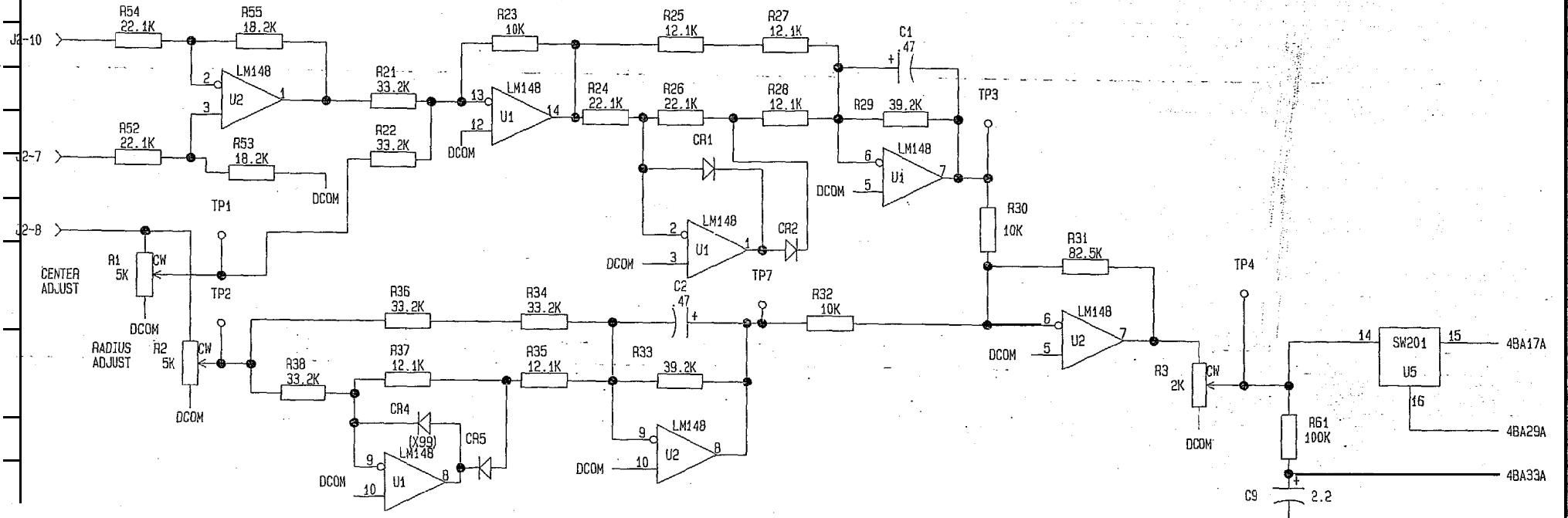
ELEMENTARY

STATIC VOLTAGE REGULATOR
AUTO/MANUAL REGULATOR ELEMENTARY

SIZE C-1

DWG NO DS3800NVRC

CONT ON SHEET 4BA (SHEET 4AA)




| REV | DESCRIPTION | DATE | APPR | REV | DESCRIPTION | DATE | APPR |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS OPERATION
SALEM, VIRGINIA

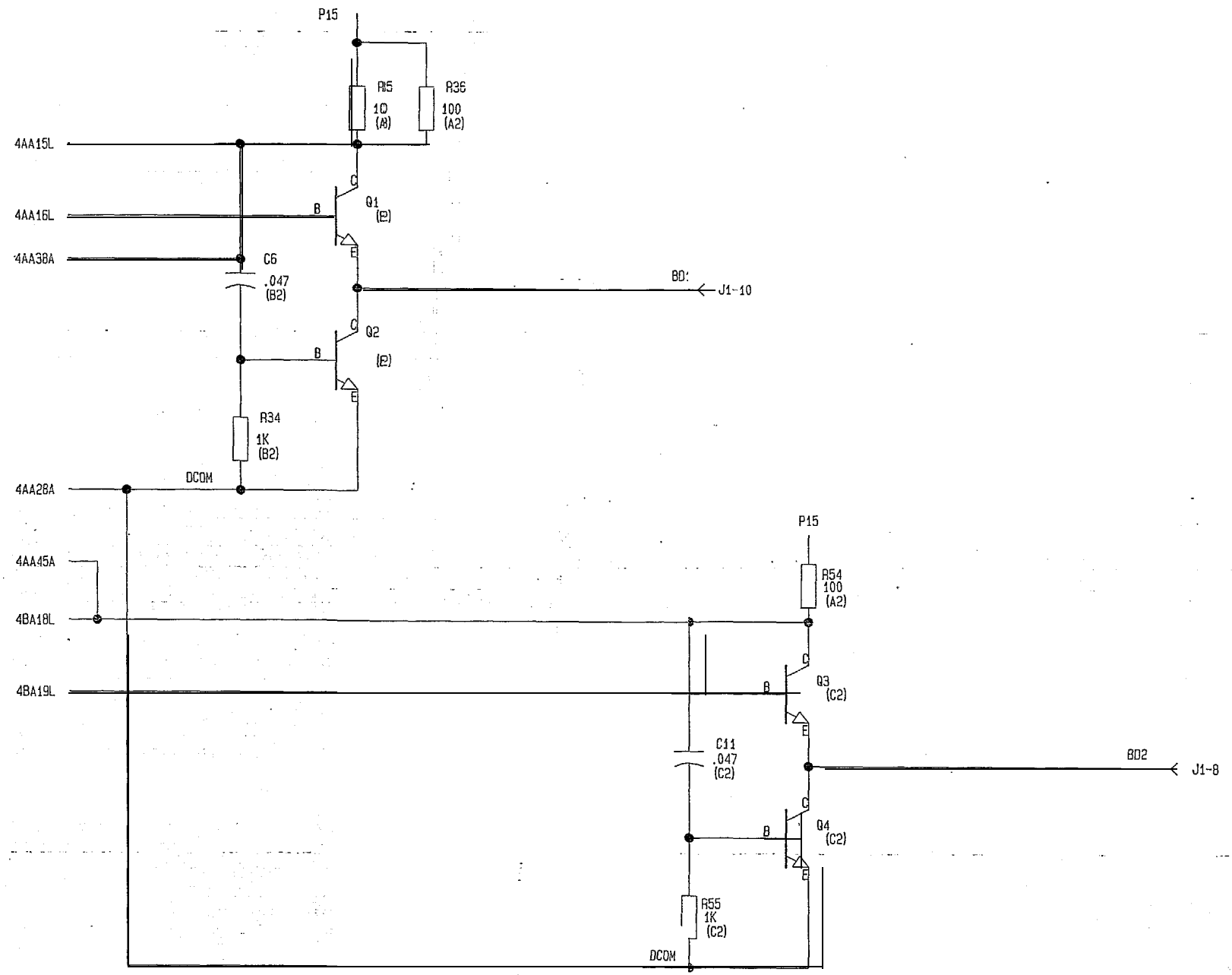
| SIGNATURES | DATE |
|---|---|
| DRAWN | 9/16/80 |
| CHECKED | 9/16/80 |
| ENGRG | 9/1/87 |
| ISSUED | 9/16/80 |

ELEMENTARY
STATIC VOLTAGE REGULATOR
URAL ELEMENTARY

SIZE C | DWG NO DS3800DVRC

CONT ON SHEET 4BA | SHEET 4AA

DL104




| REV | DESCRIPTION | DATE | APPR | REV | DESCRIPTION | DATE | APPR |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

GENERL ELECTRIC OMPANY
DRIVEYSTEMS OPERATION
SEM, VIRGINIA

| SIGNATURES | DATE |
|---|---|
| DRAWN M Reed | 9/16/87 |
| CHECKED M Reed | 9/16/87 |
| ENGRG Gary O'Halloran | 9/16/87 |
| ISSUED M Reed | 9/16/87 |

ELEMENTARY STATIC VOLTAGE REGULATOR
AUTO/MANUAL REGULATOR ELEMENTARY

SIZE C | DWG NO DS3800NVRC

CONT ON SHEET FL | SHEET 4DA

DISTR

0L104



**GE Industrial Systems**

*1-540-387-7000*
*www.ge.com*

***General Electric Company***
*1501 Roanoke Blvd.*
*Salem, VA 24153-6492 USA*